The spectrum of a compact internal space. I. Gauge structure and fermion content

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Abstract: This paper develops a spectral construction for gauge structure and fermion content. The basic datum is a discrete spectrum with finite multiplicities, from which a mass tower is formed. Physical quantities are defined by convergent spectral sums and closed analytic expressions. The construction starts from a disorder-dominated coarse-graining principle, formulated as a minimal unbiased spin-1/2 change. This specification fixes the primitive chiral carrier and its Clifford realisation. Its chiral structure selects $\mathbb{RP}^3 = S^3/\mathbb{Z}_2$ within the stated minimal quotient criterion. The associated Clifford module has dimension four, and the scale flow gives a $3+1$ decomposition. Condensation in the right sector reduces the isometry algebra $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$. An additional colour carrier, admitted as structural datum, is fixed by minimality to $\mathfrak{su}(3)$. The construction is conditional: the coarse-graining principle fixes the primitive chiral carrier, while the colour carrier, matter completion, and Higgs sector are stated as structural closures rather than absorbed into the principle. The gauge field contains a colour-singlet sector. The formulation specifies conditions for the spectral datum, physical quantities, state space, and mapping to field-theoretic structures. The free sector is recovered via Gaussian Osterwalder–Schrader reconstruction. The interacting sector is described by the spectral correspondence and spectral sums. Each ingredient is classified as theorem, definition, convention, or assumption.

Keywords: Spectral datum; Scale flow; Gauge structure; Positive scalar mass; Coarse-graining principle; Encoding; Osterwalder–Schrader reconstruction; Spectral sums

Contents

1 Introduction

Quantum field theory is ordinarily formulated in terms of fields defined on a prescribed spacetime. Its physical content, however, is often organised effectively through spectral data. The Källén–Lehmann representation expresses two-point functions in terms of a positive spectral measure [1, 2], and spectral geometry more generally encodes geometric information in the spectra of elliptic operators [3–6]. Compact internal spaces, of the kind introduced in Kaluza–Klein compactifications [7–9], provide discrete spectra with finite multiplicities. These facts motivate a concrete construction problem. The problem is to connect spectral data to the geometric, gauge, and field-theoretic structures used in quantum field theory.

A spectral formulation becomes a field-theoretic formulation only after it specifies how spectral data are encoded into fields, states, correlation functions, and interactions, together with the reconstruction rules relating the two descriptions. The best-developed instance of such an encoding is the spectral triple programme of noncommutative geometry [10], in which the Dirac operator and the associated Clifford structure replace the metric as the basic geometric datum. The spectral action principle [11] turns this datum into a classical action. The present paper studies a different organisation, based on a compact internal spectrum. Physical quantities are defined by absolutely convergent spectral sums and by closed analytic expressions constructed from the spectral datum and its derived parameters.

The present formulation is not the spectral action principle of noncommutative geometry [10, 12]. In that programme, as reviewed in [13], a spectral triple supplies the geometry and the action functional generates the dynamics; the Standard Model structure follows from a finite noncommutative space [12]. The two approaches share the language of spectra, but the present one defines physical quantities directly as spectral sums over a compact internal manifold, without passing through an action functional or a measure. The spectral sum is the definition, not a regularization. The free sector is recovered by the usual Osterwalder–Schrader construction; the interacting sector is carried by the spectral sums themselves. The relation to the standard formulation, and the boundary set by the triviality of four-dimensional ϕ^4 theory, is taken up in Section 10. Other recent work on spectral methods in related directions may be found in [14, 15].

The constructive starting point is the existence of a minimal, unbiased change with spin-1/2 as its quantum unit. The corresponding carrier is a two-dimensional complex spinor space with a chiral Weyl structure. The spinor structure on a manifold is subject to topological obstructions [16, 17]; the chiral pair selects a particular spin structure. The unit sphere of the carrier is S^3, and requiring the chiral pair to be compatible with a minimal admissible quotient selects the antipodal identification and gives

$$RP^3 = S^3/\mathbb{Z}_2.$$

The same chiral carrier determines the relevant Clifford module. The semispinor dimension relation selects Cl(4) and a four-dimensional real carrier. The scale field then separates this carrier into three level-set directions and one flow direction, giving the $3+1$ decomposition used in the formulation.

The internal geometry also organises the gauge structure. The isometry algebra of RP^3 is

$$\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R.$$

The condensation direction acts in the right sector and leaves its maximal torus unbroken,

$$\mathfrak{su}(2)_R \longrightarrow \mathfrak{u}(1)_R,$$

while the left factor remains intact. With the integer normalisation of the surviving torus generator, its representations carry an integral weight lattice. The associated geometric charges are consequently discrete. The identification of the surviving torus with the electroweak hypercharge factor uses the normalised electroweak embedding specified in the matter sector.

A separate colour carrier supplies the colour sector. Applying the minimality principle to this carrier selects the smallest simple compact algebra distinct from the electroweak factors,

$$\mathfrak{g}_{\rm col} = \mathfrak{su}(3).$$

The resulting gauge algebra is

$$\mathfrak{g}_{\rm SM} = \mathfrak{su}(3) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y.$$

The electroweak and colour factors are thus organised by the compact carrier structure and by the minimality criterion applied to the corresponding symmetry sectors. This algebraic organisation parallels the gauge structure obtained in noncommutative standard model constructions [12], but is derived here from a different set of spectral and statistical assumptions.

The chiral matter sector is represented by the minimal compatibility class consistent with the Weyl structure, the colour action, the Yukawa couplings, and gauge consistency. It contains a coloured weak doublet, an uncoloured weak doublet, the corresponding coloured singlets, and the charged-lepton singlet. In the usual notation, the one-generation content is

$$Q_L = (3,2)_{Y_Q}, \qquad u_R = (3,1)_{Y_u}, \qquad d_R = (3,1)_{Y_d},$$
$$L_L = (1,2)_{Y_L}, \qquad e_R = (1,1)_{Y_e}.$$

The anomaly equations [18–20] express the consistency conditions of the chiral gauge structure, while the Yukawa relations connect the representations through the Higgs doublet. The anomaly freedom of this representation class is verified. With the standard electroweak normalisation $Y_H = 1/2$, the hypercharge relations give

$$Y_Q = \tfrac{1}{6}, \qquad Y_u = \tfrac{2}{3}, \qquad Y_d = -\tfrac{1}{3}, \qquad Y_L = -\tfrac{1}{2}, \qquad Y_e = -1.$$

The charge assignment is therefore fixed by the consistency equations and the Yukawa structure once the electroweak normalisation has been specified.

The compact internal space also carries a discrete Dirac spectrum. The spin structure selected by the chiral construction determines the allowed spinor modes on RP^3. Coarse graining along the scale flow introduces an emergence window, and the spectral capacity of this window retains a finite set of modes from the Dirac ladder. The retained modes form the generation ladder used in the Standard Model identification. Their ordering along the scale flow gives a kinematic ordering of the generations within the spectral description.

The same spectral data enter the construction of the gauge field and the scalar sector. The gauge field is obtained from the internal connection and its mode expansion. In the colour-singlet sector, condensation produces a positive scalar mass at a condensed endpoint. The mass tower, correlation functions, thermal quantities, and related observables are expressed through spectral sums over the internal modes. In the free sector, the Gaussian structure leads to the usual Osterwalder–Schrader reconstruction [21, 22]. The interacting sector is represented by the spectral correspondence specified by the construction, with its physical quantities defined directly through the corresponding spectral sums. The statistical selection underlying the Gaussian structure follows the maximum-entropy principle [23].

The construction can be summarised as

$$\begin{gathered}\text{minimal unbiased change} \longrightarrow \text{chiral carrier} \longrightarrow RP^3 \longrightarrow \text{Clifford and gauge structure} \\ \longrightarrow \text{compatible matter content} \longrightarrow \text{spectral generations and mass scales.}\end{gathered}$$

The disorder-dominated coarse-graining principle supplies the primitive chiral unit. Compact geometry supplies the internal symmetry and the charge lattice. Minimality selects the gauge factors, given the additional colour carrier. The matter sector is organised by chirality, colour compatibility, anomaly freedom, and Yukawa structure under the stated matter closure. The emergence window determines the mode content used for generations and mass formation.

The Standard Model gauge group and fermion multiplets are not taken as primitive inputs. The starting point is instead the disorder-dominated coarse-graining principle, together with the structural closures stated explicitly below. Within these conditions, and with the spectral, topological, and representation-theoretic consistency requirements of the compact internal space, the admissible low-energy gauge and fermion structures become highly constrained. The resulting configuration gives the Standard Model one-generation pattern as a closed solution of the stated construction.

The remainder of the paper develops these ingredients in order. Section 2 reviews the Wightman framework, the maximum-entropy principle, and the relevant representation theory. Section 3 states the disorder-dominated coarse-graining principle and the minimality principle. Section 4 obtains the internal geometry, the Clifford dimension, and the gauge algebra, and derives the fermion content and the content symmetries. Section 5 introduces the spectral datum and the spectral-sum definition of physical quantities. Section 6 locates the standard quantum kinematics within the formulation, and Section 7 gives the mapping relations between spectral data and field-theoretic structures. Section 8 constructs the gauge field and the positive scalar mass. Section 9 develops the formal spectral correspondence for the interacting sector. Section 10 compares the spectral formulation with the standard measure-theoretic formulation. The appendices provide the spectral analysis, the product-space matching, and the free-sector reconstruction used in the main text.

2 Preliminaries

This section recalls some standard mathematical facts. It includes the Wightman axioms, the maximum entropy principle, and the relevant representation theory of $\mathrm{Spin}(4) \cong SU(2) \times SU(2)$.

2.1 Wightman axioms

Let H be a Hilbert space, let $U(a, \Lambda)$ be a unitary representation of the Poincaré group $\mathcal{P}_+^\uparrow$ on H, and let $\Omega \in H$ be a unit vector. Denote by $\mathcal{S}(\mathbb{R}^4)$ the Schwartz space.

Definition 2.1 (Wightman axioms [24]). *A family $\{\varphi_i\}_{i \in I}$ of operator-valued distributions on $\mathbb{R}^4$, with $\varphi_i : \mathcal{S}(\mathbb{R}^4) \to \mathcal{L}(H)$, is said to satisfy the Wightman axioms if the following conditions hold.*

(W1) There exists a strongly continuous unitary representation $U(a, \Lambda)$ of the Poincaré group on H that satisfies the spectral condition stated in (W3).

(W2) Each $\varphi_i(f)$ is a densely defined operator on H whose closure is a closed operator. Its domain contains a common dense subspace D such that $D \subset \mathrm{Dom}\, \varphi_i(f)$, $\varphi_i(f)D \subset D$, and the map $f \mapsto \langle \Psi, \varphi_i(f)\Phi \rangle$ is a tempered distribution on $\mathcal{S}(\mathbb{R}^4)$.

(W3) The joint spectrum of the energy–momentum operators P^μ, defined by $U(a, 1) = e^{ia_\mu P^\mu}$, is contained in the closed forward light cone $\bar{V}^+ = \{p : p^2 \geq 0,\ p^0 \geq 0\}$.

(W4) There exists a unique (up to a phase) U-invariant state Ω, i.e. $U(a, \Lambda)\Omega = \Omega$. The cyclicity of Ω is stated in (W6).

(W5) If the supports of f and g are spacelike separated, then $[\varphi_i(f), \varphi_j(g)]_\mp = 0$, where the minus sign applies to bosons and the plus sign to fermions (spin–statistics connection).

(W6) The set $\{\varphi_{i_1}(f_1) \cdots \varphi_{i_n}(f_n)\Omega\}$ spans a dense subspace of H.

Theorem 2.2 (Osterwalder–Schrader reconstruction [21, 22, 25]). *Let $\{\phi_i\}$ be a family of Euclidean fields whose Schwinger functions $\{S^{(n)}\}$ satisfy the OS axioms OS0 through OS4 (OS0 distribution property and regularity, OS1 Euclidean invariance, OS2 reflection positivity, OS3 symmetry, OS4 clustering). Then there exist quantum fields $\{\varphi_i\}$ satisfying Wightman axioms W1 through W6 such that their Wightman functions are the analytic continuations of the Schwinger functions. The axioms W1 through W3, W5 and W6 hold without OS4, while W4 follows from OS4.*

OS4 in Theorem 2.2 is the standard clustering axiom of the measure-theoretic formulation: the truncated correlation functions decay at large separation, equivalently the vacuum is unique. Its counterpart in the spectral-sum definition replaces that measure-theoretic decay condition by the exponential decay implied by the positive scalar mass $m_\delta > 0$, and is stated as Lemma 8.38 in Section 8. On the free sector the two coincide, with the same decay rate m_δ.

2.2 Maximum entropy principle

Let X be a random vector taking values in $\mathbb{R}^d$ with probability density ρ. The differential entropy is defined by

$$H(\rho) = -\int_{\mathbb{R}^d} \rho(x) \log \rho(x)\, d\mu_x.$$

Theorem 2.3 (Jaynes [23], maximum entropy principle)**.** *Under the constraints of fixed mean and covariance Σ, the unique maximiser of the differential entropy $H(\rho)$ is the Gaussian distribution with covariance Σ.*

A variational proof is given in [23]. For any ρ satisfying the same moment constraints one has $H(\rho) \le H(\rho_G)$, with equality if and only if $\rho = \rho_G$ almost everywhere.

Two facts about the Gaussian are used in this paper. (i) Under fixed mean and covariance, the Gaussian maximises the differential entropy (Theorem 2.3). (ii) Under the Fourier-conjugate entropy sum $H_x + H_p \ge 1 + \ln \pi$ (Beckner–Hirschman [26]), the Gaussian distributions are the unique saturators. For the self-dual two-point structure of Theorem 4.30, the Gaussian assigns equal spread to position and momentum, $H_x = H_p$, which is the equal-spread normalisation used at the condensation threshold (Proposition 8.27 of Section 8).

2.3 Representation theory preliminaries

Let $\mathrm{Spin}(4) \cong SU(2)_L \times SU(2)_R$. Its irreducible representations are labelled by (j_L, j_R) and have dimension $(2j_L + 1)(2j_R + 1)$.

Lemma 2.4 (Minimality)**.** *The minimal non-trivial irreducible representations of* $\mathrm{Spin}(4)$ *are precisely* $(1/2, 0)$ *and* $(0, 1/2)$*, both of dimension two. For every non-trivial irreducible representation ρ one has* $\dim \rho \ge 2$*, with equality if and only if* $\rho \in \{(1/2, 0), (0, 1/2)\}$*.*

Proof. The irreducible representations of $SU(2)$ are labelled by $j \in \frac{1}{2}\mathbb{N}$ and have dimension $2j + 1$. The condition $\dim \ge 2$ holds exactly when $j \ge 1/2$, and $\dim = 2$ exactly when $j = 1/2$. Irreducible representations of $\mathrm{Spin}(4)$ are of the form (j_L, j_R) with dimension $(2j_L + 1)(2j_R + 1)$. The minimal non-trivial dimension is two and occurs precisely for $(j_L, j_R) \in \{(1/2, 0), (0, 1/2)\}$. □

Lemma 2.5 (Faithfulness)**.** *The representations* $(1/2, 0)$ *and* $(0, 1/2)$ *are not faithful on* $\mathrm{Spin}(4)$*, since each is trivial on one of the two factors. The minimal faithful representation of* $\mathrm{Spin}(4)$ *is the Dirac representation* $(1/2, 0) \oplus (0, 1/2)$*, of dimension four.*

3 The Disorder-Dominated Coarse-Graining Principle

The word "principle" is used rather than "axiom" to emphasise that the present work establishes a conditional structural closure, not an unconditional ontological theorem.

This section states the coarse-graining principle and fixes the mathematical objects that follow from it. The aim is to give a precise formulation of a minimal and unbiased change, together with its symmetry group and statistical content.

A change is first understood as an invertible map of a set to itself. The coarse-graining principle then selects a particular kind of change by adding two requirements. The first fixes the quantum unit as spin-1/2. The second requires unbiasedness with respect to the symmetry group of the change.

Definition 3.1 (Change)**.** *Let X be a set. A* change *on X is a bijection $f : X \to X$. The identity transformation $f = \mathrm{id}_X$ is not regarded as a change.*

The use of a bijection means that every change remains within X and is reversible. The composition of all changes on a fixed set is again a change, and every change has an inverse, so the changes on X form a group. This is the natural symmetry group associated with the notion of change. The coarse-graining principle stated below imposes further conditions on this group.

Definition 3.2 (Unbiasedness)**.** *Let G be a group of transformations on X, and let μ be a probability measure on X. The measure μ is called* unbiased *with respect to G if it is G-invariant, that is,*

$$\mu(g^{-1}B) = \mu(B)$$

for all $g \in G$ and all measurable sets B.

Unbiasedness means that the statistical description does not favour any particular group element. When combined with the maximum entropy principle (Theorem 2.3), this requirement determines the statistical distribution of the change.

Definition 3.3 (Minimality)**.** *Let H be a finite-dimensional complex Hilbert space, and let $\rho : G \to GL(H)$ be a faithful continuous representation of a connected compact Lie group G. The pair (G, ρ) is called* minimal *if*

(i) ρ is irreducible, that is, H has no non-trivial G-invariant subspace, and

(ii) among all such pairs with irreducible ρ on the given carrier, $\dim G$ is minimal.

A change whose symmetry group acts on its carrier as a minimal pair is called a minimal change*. Condition (i) formalises indecomposability. An invariant subspace would describe a sub-change, so a reducible action factorises the change into independent parts. Condition (ii) is an economy criterion. It selects the smallest symmetry group consistent with (i). The carrier and the group are the data of the minimisation problem, and the order is the dimension of the group.*

Principle 3.4 (Disorder-dominated coarse-graining principle)**.** *There exists a minimal, unbiased, spin-1/2 change.*

To make its mathematical content explicit, the principle is unfolded into two clauses.

(A1) **Quantum unit.** The change takes spin-1/2 as its unit. The carrier of the change is a two-dimensional complex Hilbert space. The symmetry group of the change is minimal in the sense of Definition 3.3. The minimality principle selects $SU(2)$ with its two-dimensional fundamental representation. The carrier is equipped with a chiral structure, so that left- and right-handed components are distinguished.

(A2) **Unbiasedness.** The change is unbiased with respect to its symmetry group. This invariance fixes the mean to zero. Scale invariance fixes the covariance. With these moment constraints, the maximum entropy principle selects the Gaussian distribution.

Principle 3.4 is stated in a single sentence. The specification of spin-1/2 in clause (A1) is not a simple condition. It is a compact, information-dense statement. The spin quantum number 1/2 is an input of the principle. The input is minimal while its consequences are extensive. The concision of the principle is therefore a compression of information, not an omission of it.

Lemma 3.5 (Symmetry group of the change)**.** *Let $\rho : G \to GL(2, \mathbb{C})$ be a faithful, continuous, irreducible representation of a connected compact Lie group G on the two-dimensional carrier of the change. Then G is isomorphic to $SU(2)$ or to $U(2)$. Among these two candidates satisfying the stated conditions, the group of minimal dimension in the sense of Definition 3.3 is $SU(2) \cong \mathrm{Spin}(3)$, acting by its two-dimensional fundamental representation.*

Proof. By clause (A1) the carrier is a two-dimensional complex Hilbert space, and the change preserves its unitary structure. Hence $G \subseteq U(2)$, a connected compact Lie group. Irreducibility excludes every abelian group. An abelian compact group has only one-dimensional irreducible complex representations, so a two-dimensional representation of an abelian group is necessarily reducible. In particular $U(1)$ is excluded, despite having the faithful two-dimensional representation $z \mapsto \mathrm{diag}(1, z)$. The failing property is reducibility, not faithfulness.

For a non-abelian connected compact group with a two-dimensional faithful irreducible representation, the decomposition

$$G = (T^k \times G_1 \times \cdots \times G_r)/Z$$

into a central torus and simple factors forces $r = 1$. The irreducible representation is a tensor product of irreducible representations of the factors, whose dimensions multiply to 2. Exactly one simple factor is therefore non-trivial, and it must admit a two-dimensional faithful irreducible representation. The only simple Lie group with a two-dimensional faithful irreducible representation is $SU(2)$. Its fundamental representation is faithful and irreducible. The group $SO(3)$ has only odd-dimensional irreducible representations ($2j+1$, $j \in \mathbb{Z}_{\geq 0}$), and every simple group of rank at least 2 has minimal representation dimension at least 3. The central torus contributes only scalars, extending $SU(2)$ to $U(2) \cong (U(1) \times SU(2))/\mathbb{Z}_2$ without reducing the dimension. The candidates are therefore $SU(2)$, with

$\dim G = 3$, and $U(2)$, with $\dim G = 4$. By condition (ii) of Definition 3.3 the minimal group is $SU(2) \cong \mathrm{Spin}(3)$, whose fundamental representation has dimension 2. $\square$

The two clauses of Principle 3.4 are mutually independent. The quantum unit does not imply a statistical description. Unbiasedness does not imply an algebraic structure of the change. The spin quantum number $1/2$ in clause (A1) is an input of the principle. It is realised by the fundamental representation of $SU(2)$, which has spin label $1/2$ and is faithful.

The fundamental representation of $SU(2)$ is pseudoreal. It is equivalent to its complex conjugate, the equivalence being realised by the antisymmetric tensor of $SU(2)$. A left-right chiral distinction therefore cannot arise within $SU(2)$ alone. The chiral structure of clause (A1) is realised at the level of the full symmetry group of the carrier space. The carrier sphere S^3 has isometry group $O(4)$ with identity component $\mathrm{Spin}(4) \cong SU(2)_L \times SU(2)_R$. The two factors carry the two inequivalent minimal representations $(1/2, 0)$ and $(0, 1/2)$. This is the mathematical reason for introducing Spin(4) in Section 4. The passage to Spin(4) is the minimal faithful geometric realisation of the left-right Weyl pair associated with the two-dimensional carrier.

Theorem 3.6 (Quantum unit theorem)**.** *Under the coarse-graining principle, the symmetry group of the change is $SU(2)$, and the carrier is its two-dimensional fundamental representation. The spin label $1/2$ of clause (A1) is the spin quantum number of this representation.*

Proof. By Lemma 3.5 the minimal symmetry group of the two-dimensional carrier is $SU(2) \cong \mathrm{Spin}(3)$, acting by its fundamental representation. The irreducible representations of $SU(2)$ are labelled by $j \in \frac{1}{2}\mathbb{N}$ and have dimension $2j+1$. The fundamental representation has $j = 1/2$, dimension 2, and is faithful (its kernel is $\{e\}$). The unit $1/2$ of clause (A1) is therefore realised as the spin label of the minimal representation. It is an input of the principle, not a derived quantity. The angular momentum operators on the carrier space satisfy the $\mathfrak{su}(2)$ commutation relations, and their smallest absolute eigenvalue is $1/2$. $\square$

Lemma 3.7 (Clifford semispinor dimensions)**.** *Let $n \in \mathbb{N}$. The complexification of the Euclidean Clifford algebra* $\mathrm{Cl}(n)$ *satisfies*

$$\mathrm{Cl}(n) \otimes_{\mathbb{R}} \mathbb{C} \cong \begin{cases} M_{2^{\lfloor n/2 \rfloor}}(\mathbb{C}) & \text{if } n \text{ is even,} \\ M_{2^{\lfloor n/2 \rfloor}}(\mathbb{C}) \oplus M_{2^{\lfloor n/2 \rfloor}}(\mathbb{C}) & \text{if } n \text{ is odd.} \end{cases}$$

The spinor module has complex dimension $2^{\lfloor n/2 \rfloor}$. A chiral operator $\Gamma = \gamma_1 \cdots \gamma_n$ exists if and only if n is even, $n = 2m$, and then the spinor module decomposes into two semispinor modules, each of complex dimension 2^{m-1}. For odd n no chiral decomposition exists.

Proof. This is the standard classification of complexified Euclidean Clifford algebras [4]: for even n the complexification is a full matrix algebra and for odd n a direct sum of two matrix algebras, with the stated spinor-module dimensions. The chiral operator Γ and the semispinor decomposition exist exactly in the even case $n = 2m$. $\square$

The consistency of the two clauses of the principle can be seen from an explicit model. Take $X = S^3$, the unit sphere in the two-dimensional complex carrier space. Take $G = SU(2)$ acting on X by rotations. Take μ the round maximum-entropy measure. Then (A1) holds because the minimal faithful representation of $SU(2)$ on the two-dimensional carrier has dimension 2. (A2) holds because μ is G-invariant and is the maximum-entropy measure on the compact carrier. The coarse-graining principle is consistent with ZFC.

The maximum entropy principle (Theorem 2.3) requires fixing the mean and the covariance. The mean is fixed to zero by unbiasedness. A non-zero mean vector would select a preferred direction and a preferred location, while the measure is unbiased with respect to the symmetry group. The covariance is fixed by the scale-invariant two-point structure at the conformal fixed point, Theorem 4.30 of Section 4.7, which gives $C(x, y) = c/|x-y|^2$. Its spectral representation is defined in Section 4.7. The spectral density $\rho(p^2)$ encodes the full two-point correlation information. The statements of the paper do not depend on its detailed form beyond temperedness.

4 Emergent Structures

4.1 The minimal internal space

Clause (A1) fixes the carrier of the change as a two-dimensional complex spinor space. Its unit sphere is S^3. The chiral structure of (A1) distinguishes left-handed and right-handed components. That distinction has to be realised geometrically. The change therefore acts on a definite internal space, not on the carrier sphere alone. The passage to Spin(4) is the minimal faithful geometric realisation of the left-right Weyl pair (Section 3).

The action of Spin(4) on S^3 is not faithful. Its kernel is the central subgroup generated by $(-1_L, -1_R)$. The antipodal map $z \mapsto -z$ corresponds to the central element $(-1_L, +1_R)$. This element acts as -1 on the left-handed Weyl spinor and as $+1$ on the right-handed one. Thus the chiral structure of (A1) is realised precisely by the antipodal identification.

Lemma 4.1 (Antipodal realisation of chirality)**.** *The central element* $(-1_L, +1_R) \in \mathrm{Spin}(4)$ *is the unique element that acts as* -1 *on the left-handed Weyl spinor* $(1/2, 0)$ *and as* $+1$ *on the right-handed one* $(0, 1/2)$*. Consequently the chiral Weyl pair* $(1/2, 0) \oplus (0, 1/2)$ *descends to a twisted/untwisted pair of spinor sections on a free quotient* S^3/Γ *if and only if* Γ *contains the antipodal element.*

Proof. Write $g = (g_L, g_R) \in SU(2)_L \times SU(2)_R = \mathrm{Spin}(4)$. The element g acts on $(1/2, 0)$ through g_L and on $(0, 1/2)$ through g_R. It acts as -1 on $(1/2, 0)$ exactly when $g_L = -1_L$, and as $+1$ on $(0, 1/2)$ exactly when $g_R = +1_R$; hence $g = (-1_L, +1_R)$, the antipodal map, and this element is unique. For the second assertion, a section of the spinor bundle over S^3 descends to S^3/Γ only if it is invariant under the Γ-action lifted to the bundle; the twisted (resp. untwisted) character is the statement that the left-handed (resp. right-handed) component is odd (resp. even) under the antipodal element. If Γ does not contain the antipodal element, both components descend to the same untwisted bundle and no chiral distinction survives; if it does, the distinction is exactly the one stated. $\square$

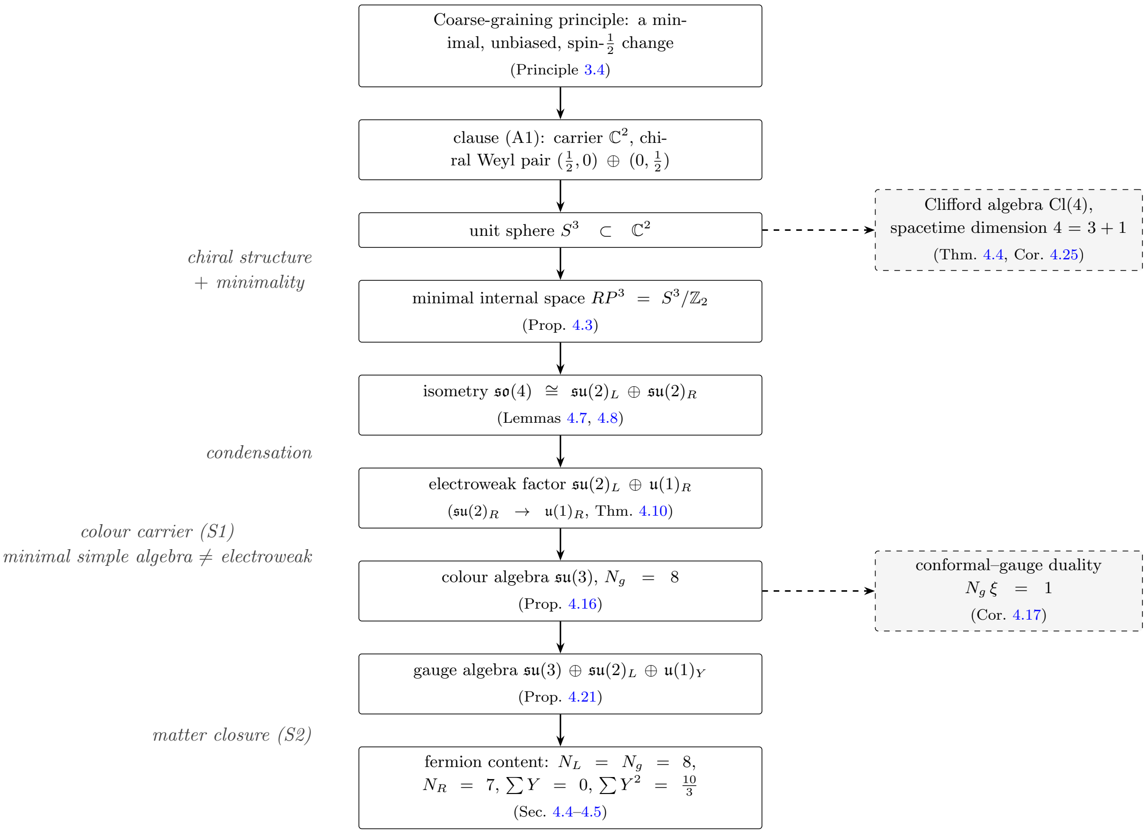

Figure 1. The emergence chain. Solid arrows carry the selections and constructions of Section 4. Dashed arrows mark derived consequences. A single minimality criterion is applied throughout: to the admissible quotient (giving RP^3), to the symmetry group (giving $SU(2)$), to the colour algebra (giving $\mathfrak{su}(3)$), to the matter content (giving the one-generation completion), and to the Higgs sector (the single doublet). The colour step is conditional on the additional colour carrier (S1) of Definition 4.13, and the fermion content is derived within the matter closure (S2) of that definition, not selected by the coarse-graining principle; the first two steps are selections within the coarse-graining principle.

Definition 4.2 (Minimal internal space)**.** *Let S^3 be the unit sphere in the two-dimensional complex carrier space. A free quotient S^3/Γ is called* admissible *if the chiral Weyl pair $(1/2, 0) \oplus (0, 1/2)$ is compatible with the quotient and Γ contains the antipodal element, so that the antipodal identification realises the chiral structure of clause (A1). Compatibility means that the two Weyl components define sections on the quotient: the left-handed component, odd under the antipodal map, defines a section of the twisted spinor bundle, and the right-handed component, even, defines a section of the untwisted one. Among all admissible free quotients, the one with the smallest order $|\Gamma|$ is called the* minimal internal space*.*

Proposition 4.3. *Under the admissibility conditions of Definition 4.2, the minimal internal space is $RP^3 = S^3/\mathbb{Z}_2$.*

Proof. First, $\mathbb{Z}_2$ is admissible. The antipodal map $x \mapsto -x$ acts as the Spin(4) element

$(-1_L, +1_R)$. Under this element, the left-handed spinor changes sign and the right-handed spinor does not. Hence the chiral pair is compatible with the quotient $S^3/\mathbb{Z}_2$: the left-handed component acquires the antipodal character and defines a section of the twisted spinor bundle, the right-handed component is even and defines a section of the untwisted bundle. Conversely, admissibility requires Γ to contain the antipodal element, which is what distinguishes the two Weyl components. Any admissible Γ therefore contains $\mathbb{Z}_2$. The smallest possible admissible quotient is accordingly $S^3/\mathbb{Z}_2 = RP^3$. The remaining symmetry statements follow from the standard action of Spin(4) on S^3. □

4.2 Clifford algebra and dimension

Theorem 4.4 (Dimension)**.** *Under the coarse-graining principle, the Clifford algebra carrying the chiral two-dimensional representation of clause (A1) is* Cl(4)*. Its generators span a real space V of dimension four.*

Proof. Clause (A1) fixes the representation to be the two-dimensional fundamental representation of $SU(2)$ with chiral structure. By Lemma 3.7, a chiral decomposition exists only in even dimensions $n = 2m$. Hence the representation is a semispinor module of some $\mathrm{Cl}(2m)$. The semispinor dimension equation then gives $2^{m-1} = 2$, from which $m = 2$ and $n = 4$. The generators $\gamma_1, \ldots, \gamma_4$, satisfying $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$, therefore span the four-dimensional real space

$$V = \mathrm{span}\{\gamma_1, \ldots, \gamma_4\}.$$

The relation $\{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}$ identifies the metric as the square of the Clifford generators. In this formulation, the metric structure used below is encoded through the Clifford relation rather than introduced as an independent Riemannian datum. The spin operators

$$S_{\mu\nu} = \frac{i}{4}[\gamma_\mu, \gamma_\nu] \in \mathrm{Cl}(4)$$

generate Spin(4). Its vector representation on V gives $SO(4)$. The quadratic form defined by the Clifford relation is the Euclidean metric on V. □

Corollary 4.5 (Euclidean dimension and Lorentzian reconstruction)**.** *Theorem 4.4 fixes the* Euclidean *real dimension of the carrier: the Clifford generators span the Euclidean space $V \cong \mathbb{R}^4$. The Lorentzian signature, the Poincaré covariance, and the causal (spectral) structure are supplied by the Osterwalder–Schrader reconstruction of Theorem 2.2 applied to a reflection-positive Euclidean measure on V (Appendix C), which yields a relativistic quantum field theory in $3+1$ Minkowski spacetime satisfying the Wightman axioms W1–W6. The Clifford argument by itself does not select a Lorentzian signature; the $3+1$ Minkowski structure is the standard content of the reconstruction theorem, not an additional consequence of the coarse-graining principle.*

Remark 4.6 (Status of the Clifford realisation). The identification of the two-dimensional chiral carrier of clause (A1) with a semispinor module of a Euclidean Clifford algebra is an encoding step of the chiral clause, not a further consequence of the coarse-graining principle. Given this identification, the dimension four of Theorem 4.4 is forced by the semispinor dimension equation alone. The principle fixes a two-dimensional chiral carrier; the Clifford module is the minimal algebraic realisation of its chirality, in the same way that the passage to Spin(4) in Section 3 is a geometric realisation.

4.3 The gauge algebra

The abstract symmetry group $SU(2)$ of clause (A1) acquires gauge meaning through the $SU(2)_L$ factor in the isometry algebra of RP^3. The isometry-sector breaking direction and the colour sector are fixed as follows. The R factor breaks to $\mathfrak{u}(1)_R$ via the condensation mechanism of Theorem 4.10, while $SU(2)_L$ remains unbroken. The minimal colour algebra is selected in Proposition 4.16.

Lemma 4.7. *The isometry algebra of RP^3 is $\mathfrak{so}(4)$.*

Proof. The isometry group of S^3 is $O(4)$. Since the antipodal map lies in the centre of $O(4)$, the isometry group of $RP^3 = S^3/\mathbb{Z}_2$ is $O(4)/\{\pm 1\}$. Its identity component has Lie algebra $\mathfrak{so}(4)$. □

Lemma 4.8. $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$.

Proof. The Hodge star on $\Lambda^2(\mathbb{R}^4)^* \cong \mathfrak{so}(4)$ decomposes the space into the ± 1 eigenspaces Λ^2_+ and Λ^2_-, each of dimension three. Equipped with the cross-product bracket, each summand is isomorphic to $\mathfrak{su}(2)$. □

Definition 4.9. *The gauge algebra $\mathfrak{g}_{\text{ch}}$ is the subalgebra of $\mathfrak{so}(4)$ retained as gauge symmetry. According to Theorem 4.10, $\mathfrak{su}(2)_L$ survives completely, while $\mathfrak{su}(2)_R$ breaks to its maximal torus $\mathfrak{u}(1)_R$. Thus*

$$\mathfrak{g}_{\text{ch}} = \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R.$$

Theorem 4.10 (Isometry-sector breaking direction)**.** *The condensation mechanism of Section 8.6 acts on the R sector. The long-root mode of this sector, the $J = 2$ squash deformation of the R-sector connection, carries charge 2 under the unbroken torus (Appendix A). Below the critical curvature R_c it becomes tachyonic. Its condensation is the squash of the internal geometry and breaks $\mathfrak{su}(2)_R$ to its maximal torus $\mathfrak{u}(1)_R$. The maximal torus theorem ensures uniqueness up to conjugation. The same tachyonic direction supplies the negative quadratic term that triggers the condensation of the colour-singlet scalar of Section 8.6 (Theorem 8.26); the two condensations act on different sectors. The surviving torus is the geometric $\mathfrak{u}(1)_R$ factor of the construction. Its identification with the SM weak hypercharge is fixed by the hypercharge assignment derived in Section 4.4.*

Proof. The long-root mode is the mode of the R-sector connection in the representation $(2, 1)$ of Spin(4), described in Appendix A. Its squared mass equals

$$m^2_{\text{long}} = \frac{8}{3}(R - R_c),$$

which is negative when $R < R_c$. Condensation of this mode breaks $\mathfrak{su}(2)_R$. The surviving subgroup is the stabiliser of the condensed direction. For a simple rank-one factor this stabiliser is a maximal torus, unique up to conjugation. The identification of the surviving $\mathfrak{u}(1)_R$ generator with weak hypercharge is fixed by the hypercharge assignment derived in Section 4.4: the geometric torus charge is $q = 2m_R$, an integer, and the SM hypercharge normalisation is the one carried by that assignment. The condensation mechanism of the colour-singlet scalar is established in Section 8.6, while the present theorem only identifies the breaking direction of the isometry sector. □

Lemma 4.11 (Weight-lattice quantisation from the compact electroweak factor)**.** *Let $G_R = SU(2)_R$ and let $T^3_R = \mathrm{diag}(1/2, -1/2)$ be its Cartan generator in the standard normalisation. For a state with $SU(2)_R$ weight m_R, the integer-normalised torus charge is*

$$q = 2m_R \in \mathbb{Z}.$$

Thus compactness fixes the charge weight lattice, $m_R \in \frac{1}{2}\mathbb{Z}$. If the surviving $U(1)_Y$ is identified with a specified normalised embedding of this torus, the hypercharge ratios are determined by the representation weights; the remaining hypercharge relations follow from the Yukawa structure and anomaly cancellation. The conventional value $Y_H = 1/2$ follows only after the $U(1)_Y$ embedding, the electromagnetic generator $Q = T^3_L + Y$, and the neutral Higgs vacuum condition have been specified.

Proof. The Cartan generator T^3_R has eigenvalues $m_R = -j_R, -j_R+1, \ldots, j_R$ in the spin-j_R representation, and the integer normalisation $q = 2T^3_R$ gives $q = 2m_R$. The compactness chain is explicit: RP^3 is compact, hence its isometry $SO(4)$ is compact, hence its factor $SU(2)_R$ is compact, hence the weight lattice is discrete, $m_R \in \frac{1}{2}\mathbb{Z}$. Therefore $q = 2m_R \in \mathbb{Z}$. The hypercharge statements follow from the definitions of the embedding, the charge relation, and the neutral vacuum. □

Remark 4.12 (Discreteness is internal, normalisation is a convention)*.* Lemma 4.11 makes the discreteness of the charge a consequence of the compactness of the internal space. Charge quantisation and the discreteness of the scalar spectrum $\{\varepsilon_l\}$ are two projections of one geometric property, the compactness of RP^3. The seed of both is the minimal non-trivial spin of clause (A1), $j_R = 1/2$, whose fundamental weights $m_R = \pm 1/2$ carry the unit charge $q = \pm 1$. What compactness does not fix is the normalisation, namely the embedding of $U(1)_Y$ in the torus, the charge relation $Q = T^3_L + Y$, and the generator rescaling $Y \to aY$, $g_Y \to g_Y/a$. These are stated conventions, not consequences of compactness. The dimensional normalisation used in the construction is the scale M_P of Theorem 8.39. The discrete structures are internal, while their normalisations are conventions.

Definition 4.13 (Structural closure assumptions)**.** *Beyond the coarse-graining principle, the construction rests on three structural closures, stated explicitly and not absorbed into the principle.*

(S1) Colour sector. *An independent colour carrier is introduced as a structural datum. It consists of a simple Lie algebra distinct from the electroweak factor $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R$, realised on the Kaluza–Klein zero modes of a connection on RP^3. This datum is not derived from the coarse-graining principle. Its algebraic form is fixed by the minimality principle.*

(S2) Matter completion. *The matter sector is the minimal anomaly-free Yukawa-complete chiral completion of the gauge algebra, the class of chiral fermion contents satisfying: (i) a single $SU(2)$ Higgs doublet; (ii) Yukawa couplings for u, d, e; (iii) all local and global gauge anomalies vanish; (iv) colour is vector-like; (v) no right-handed neutrino; (vi) no additional vector-like or sterile matter; (vii) minimal total representation dimension. Within this class the one-generation content is unique (Section 4.4).*

(S3) Higgs sector. *There is a single Higgs $SU(2)_L$ doublet, and its vacuum preserves the electromagnetic $U(1)$, so that the charge relation $Q = T_3 + Y$ holds with a neutral vacuum.*

These closures are conditions of the construction, not consequences of the coarse-graining principle, but they add no new selection principle: each is the same economy criterion of Definition 3.3 applied to a different carrier — (S1) to the colour algebra once the carrier is admitted, (S2) to the matter representation content, and (S3) to the Higgs sector. The one genuinely additional datum is the existence *of the colour carrier (S1). Conditional on these closures, the constructions of this section are unique. The hypercharge normalisation $Y_H = 1/2$ is not a further closure: it follows from Lemma 4.11 once the $U(1)_Y$ embedding, the charge relation $Q = T_3 + Y$, and the neutral-Higgs vacuum of (S3) are specified.*

Definition 4.14 (Colour carrier)**.** *Let $E_{\rm col}$ be the space of Kaluza–Klein zero modes of a connection on RP^3 with values in a simple Lie algebra $\mathfrak{g}_{\rm col}$ distinct from the electroweak factor $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R$. The natural representation of $\mathfrak{g}_{\rm col}$ on $E_{\rm col}$ is the adjoint representation induced by the zero-mode expansion.*

Remark 4.15 (Cross-checks on the colour assumption). The existence of a colour factor distinct from the electroweak one is the colour closure of Definition 4.13(S1). The coarse-graining principle fixes the electroweak factor through the isometry algebra of RP^3, and it does not by itself force a second simple factor. The colour carrier is therefore an assumed structural datum. However, this assumption is not an unconstrained input. Its presence is compatible with Witten anomaly cancellation. The chiral electroweak content contains an odd number of $SU(2)_L$ doublets, and a colour factor with odd N_c restores the even doublet count required for a well defined fermion determinant. This is a natural and minimal way to satisfy the condition, though not the only one. The content identities point to the same conclusion. The content symmetries of Section 4.5 and the conformal-gauge duality of Corollary 4.17 both give $N_g = 8$, which matches the dimension of the gauge algebra when $\mathfrak{su}(3)$ is added to the electroweak sector. Proposition 4.16 already shows that minimality among simple algebras distinct from the electroweak factor singles out $\mathfrak{su}(3)$. These observations do not prove necessity, but they indicate that the colour datum is tightly constrained by consistency, content closure, and minimality.

Proposition 4.16 (Colour algebra)**.** *Under the minimality principle of Definition 3.3 applied to the colour carrier of Definition 4.14, the colour algebra is fixed to $\mathfrak{su}(3)$.*

Proof. The candidates are the simple compact Lie algebras distinct from the electroweak factor $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R$, acting on the colour carrier through the natural adjoint representation of Definition 4.14. The adjoint representation of a simple algebra is faithful and irreducible, so condition (i) of Definition 3.3 holds for every candidate. Condition (ii), minimality of $\dim G$, then selects the smallest simple algebra distinct from the electroweak factor. A further constraint, independent of minimality, comes from the Witten global anomaly [27]: for the fermion determinant to be well defined the number of $SU(2)_L$ doublets must be even. In one generation the colour fundamental of dimension N_c contributes N_c coloured doublets $Q_L = (N_c, 2)$ and the lepton sector contributes one colourless doublet $L_L = (1, 2)$, so the total number of doublets is $N_c + 1$, and evenness forces N_c to be odd. The quarks sit in the fundamental (smallest non-trivial) representation of the colour algebra, of dimension N_c; the fundamental dimensions of the simple compact algebras begin with $\mathfrak{su}(2)$ ($N_c = 2$), $\mathfrak{su}(3)$ ($N_c = 3$), $\mathfrak{so}(5)$ ($N_c = 4$), $\mathfrak{su}(4)$ ($N_c = 4$), and G_2 ($N_c = 7$). The parity condition excludes the even-N_c algebras $\mathfrak{su}(2)$, $\mathfrak{so}(5)$ and $\mathfrak{su}(4)$; among the surviving odd-N_c algebras, minimality of $\dim G$ selects the smallest, namely $\mathfrak{su}(3)$ ($\dim 8$) over G_2 ($\dim 14$). Hence $\mathfrak{g}_{\mathrm{col}} = \mathfrak{su}(3)$, with

$$N_g = \dim \mathfrak{su}(3) = 8.$$

□

Corollary 4.17 (Conformal-gauge duality)**.** *The colour content and the conformal content of the emergent structure pair into the conserved unit*

$$N_g\, \xi = 1,$$

equivalently $N_g \Delta = 2(d-1)$*, where* $\Delta = (d-2)/2$ *is the scalar conformal weight and* $\xi = \Delta/[2(d-1)]$ *its Yamabe normalisation. The duality is a derived identity of the selections above.*

Proof. The internal space is the unit sphere S^3 of the two-dimensional carrier of clause (A1), so the internal dimension is $d = 3$ (Section 4.1). A scalar field on a curved space of dimension d is conformally invariant only when it couples to the scalar curvature through the term $\xi R \varphi^2$ with the Yamabe conformal coupling

$$\xi = \frac{d-2}{4(d-1)},$$

the coefficient of the Yamabe operator $\Delta + \xi R$, fixed uniquely by the conformal requirement and not by a convention of the present construction [4, 28]. At $d = 3$ this gives $\xi = 1/8$. Proposition 4.16 gives $N_g = 8$. Hence $N_g \xi = 8 \cdot 1/8 = 1$, and the equivalent form follows from $\xi = \Delta/[2(d-1)]$. The identity is structural rather than tautological. With $N_g = N_c^2 - 1$ and the Yamabe form of ξ, the pairing equation $N_g \xi = 1$ with $d = N_c$ is the polynomial $(N_c - 3)(N_c + 2) = 0$, whose unique positive solution is $N_c = d = 3$, the same two integers fixed independently here by the carrier and by minimality. □

Corollary 4.18 (Convergence of the colour determination)**.** *The colour number* $N_c = 3$ *is obtained consistently by three routes that agree on the same integer: (i) minimality among simple algebras distinct from the electroweak factor selects* $\mathfrak{su}(3)$*, hence* $N_g = N_c^2 - 1 =$

8 (Proposition 4.16); (ii) the conformal–gauge duality $N_g\xi = 1$ with the Yamabe value $\xi = 1/8$ fixes $N_g = 8$ (Corollary 4.17); (iii) the content symmetry $N_L = N_g$ with $N_L = 8$ (Section 4.5) fixes $N_g = 8$. The agreement is a consistency check on the construction rather than an independent derivation of each route.

Remark 4.19 (Status of the colour selection). The colour algebra is fixed by the economy criterion, the same minimality principle that selects $SU(2)$ over $U(2)$ in Lemma 3.5 and RP^3 over the larger lens spaces in Proposition 4.3. The three applications differ in status, however. For the symmetry group and the internal space the minimality criterion is applied to an object selected by the coarse-graining principle itself. For the colour algebra it is applied to the colour carrier of Definition 4.14, which is admitted as a structural datum (Remark 4.15). Only after this additional carrier is admitted does the economy criterion select $\mathfrak{su}(3)$. The generator count selected by minimality and the conformal coupling fixed by the carrier dimension are reciprocal. The identity records the coincidence of the two determinations, namely the carrier dimension $d = 3$ and the colour rank $N_c = 3$, in the single conserved unit $N_g\xi = 1$.

Definition 4.20 (Constructive data)**.** *The constructive data of the paper consist of the clauses (A1)–(A2) of the coarse-graining principle, including the chiral structure of clause (A1). The minimality principle of Definition 3.3 selects $SU(2)$ over $U(2)$ in Lemma 3.5, RP^3 over the larger lens spaces in Proposition 4.3, and the colour algebra $\mathfrak{su}(3)$ over the larger simple algebras in Proposition 4.16. The conformal-gauge duality $N_g\xi = 1$ of Corollary 4.17 is a derived identity of these selections. All further data are derived in Sections 4–8 and Appendix A.*

Proposition 4.21 (Gauge algebra, a selection under stated criteria)**.** *Under the stated criteria of this section, namely the coarse-graining principle, the minimality selections of Definition 4.20, and the structural datum of a colour carrier distinct from the electroweak factor (Remark 4.15), the gauge algebra is*

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y.$$

Proof. Lemmas 4.7 and 4.8 give $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$. Definition 4.9 fixes the electroweak part as $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R$. Proposition 4.16 selects the colour part $\mathfrak{su}(3)$. Taking the direct sum and identifying $\mathfrak{u}(1)_Y$ with the weak hypercharge generator yields the result. □

4.4 Fermion content

The chiral clause (A1) supplies the Weyl pair $(1/2, 0) \oplus (0, 1/2)$ of Spin(4), as recorded in Section 3. The left-handed component $(1/2, 0)$ is an $\mathfrak{su}(2)_L$ doublet. The colour algebra $\mathfrak{su}(3)$ of Proposition 4.16 acts on the fermions through its fundamental representation $\mathbf{3}$, the smallest non-trivial irreducible representation. Colour is vectorial, so the left-handed fermions carry $\mathbf{3}$ and the right-handed ones carry the conjugate $\bar{\mathbf{3}}$, which cancels the $\mathrm{SU}(3)^3$ anomaly.

The five representations are the minimal anomaly-free Yukawa-complete chiral completion of the gauge algebra, the matter closure of Definition 4.13(S2): the smallest chiral content on which all gauge and gravitational anomalies vanish and the Yukawa couplings for u, d, e exist, without a right-handed neutrino or additional vector-like matter. Within this stated class the content is unique; its “minimality” is part of the closure, not a separate selection principle of the coarse-graining principle. The left-handed doublet carries either colour, giving $Q_L = (3, 2)$, or no colour, giving $L_L = (1, 2)$. The right-handed singlets mirror the doublet members. The coloured members u_L, d_L give $u_R = (3, 1)$ and $d_R = (3, 1)$ in the conjugate representation, the uncoloured member e_L gives $e_R = (1, 1)$, and ν_L has no light right-handed partner. The five representations are

$$Q_L = (3, 2)_{Y_Q}, \quad u_R = (3, 1)_{Y_u}, \quad d_R = (3, 1)_{Y_d}, \quad L_L = (1, 2)_{Y_L}, \quad e_R = (1, 1)_{Y_e}.$$

The hypercharges are not free. The Higgs doublet of the closure (S3) is the minimal $SU(2)_L$ doublet whose vacuum preserves the electromagnetic $U(1)$, so that the charge relation $Q = T_3 + Y$ holds with a neutral vacuum; this fixes $Y_H = 1/2$ from $T_3(H^0) = -1/2$ and $Q(H^0) = 0$. By Lemma 4.11 the charge is quantised, and $Y_H = 1/2$ records the fundamental weights $m_R = \pm 1/2$ of clause (A1) in the hypercharge normalisation. The remaining hypercharges are then fixed by anomaly cancellation and the Yukawa structure. The charge convention $Q = T_3 + Y$ gives the Yukawa relations

$$Y_u = Y_Q + Y_H, \qquad Y_d = Y_Q - Y_H, \qquad Y_e = Y_L - Y_H.$$

The mixed $\mathrm{SU}(2)^2\mathrm{U}(1)$ anomaly vanishes when

$$3Y_Q + Y_L = 0,$$

and the mixed gravitational U(1) anomaly vanishes when

$$6Y_Q - 3Y_u - 3Y_d + 2Y_L - Y_e = 0,$$

where the right-handed fields enter with the opposite sign of their hypercharge, in the convention in which all fields are counted as left-handed. Substituting the Yukawa relations into these two equations gives $3Y_Q - Y_H = 0$, hence $Y_Q = 1/6$, and then $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$. The hypercharge assignment is therefore a consequence of anomaly cancellation, the Yukawa structure, and the neutral-Higgs vacuum condition of closure (S3).

Theorem 4.22 (Uniqueness of the matter completion)**.** *Within the class of Definition 4.13(S2) — a single $SU(2)$ Higgs doublet, Yukawa couplings for u, d, e, all local and global gauge anomalies vanishing, colour vector-like, no right-handed neutrino, no additional vector-like or sterile matter, and minimal total representation dimension — the one-generation content*

$$Q_L = (3, 2)_{1/6}, \quad u_R = (3, 1)_{2/3}, \quad d_R = (3, 1)_{-1/3}, \quad L_L = (1, 2)_{-1/2}, \quad e_R = (1, 1)_{-1}$$

is unique. Within this stated class the content is therefore fixed by the closure rather than over-determined by it.

Proof. The conditions fix the content in two steps. (i) *Representations.* Colour vector-like together with the absence of a right-handed neutrino forces the coloured matter into pairs $(3, R) \oplus (\bar{3}, R')$ of equal $SU(2)$ content, and the lepton sector into a doublet $L_L = (1, 2)$ and a charged singlet $e_R = (1, 1)$ with no neutral singlet. The Yukawa couplings for u, d, e require the coloured doublet $Q_L = (3, 2)$ and its two conjugate singlets u_R, d_R; minimality of the total representation dimension then excludes every further pair, leaving exactly the five representations of the statement. (ii) *Hypercharges.* With the five representations fixed, the Yukawa relations and the anomaly equations written above determine the hypercharges uniquely as $Y_Q = 1/6$, $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$. Steps (i) and (ii) leave no free parameter, so the content is unique. □

Proposition 4.23 (Global gauge group)**.** *The matter content of this section fixes the global gauge group to*

$$G_{\text{global}} \;=\; \big[SU(3) \times SU(2) \times U(1)_Y\big]/\mathbb{Z}_6,$$

where $\mathbb{Z}_6$ is the subgroup of the centre acting trivially on all five representations, generated by $(\omega, -1, 1)$ with $\omega = e^{2\pi i/3}$. The quotient is well defined precisely because the hypercharges are integer multiples of $1/6$; this is the global (topological) statement of charge quantisation.

Proof. In the all-left-handed Weyl convention the matter content is

$$Q_L = (3, 2)_{1/6}, \quad u^c_R = (\bar{3}, 1)_{-2/3}, \quad d^c_R = (\bar{3}, 1)_{1/3}, \quad L_L = (1, 2)_{-1/2}, \quad e^c_R = (1, 1)_1,$$

where u^c_R, d^c_R, e^c_R denote the conjugates of the right-handed singlets. An element $(a, b, e^{i\theta})$ of the centre $Z(SU(3)) \times Z(SU(2)) \times U(1)$, with $a = \omega^j$ $(j = 0, 1, 2)$ and $b = (-1)^k$ $(k = 0, 1)$, acts on the fields by the phases

$$Q_L:\ \omega^j(-1)^k e^{i\theta/6}, \quad u^c_R:\ \omega^{-j} e^{-2i\theta/3}, \quad d^c_R:\ \omega^{-j} e^{i\theta/3}, \quad L_L:\ (-1)^k e^{-i\theta/2}, \quad e^c_R:\ e^{i\theta}.$$

The kernel is the set of (j, k, θ) for which all five phases equal one. The fifth phase forces $\theta = 2\pi n$; the fourth forces $k+n$ even; the third forces $j \equiv n \pmod 3$; the first two then hold identically. The solutions are the six powers of $(\omega, -1, 1)$, a cyclic group of order six. The quotient by this kernel is the global gauge group, and the fact that the kernel is nontrivial — equivalently, that the fractional hypercharges are compatible with the $\mathbb{Z}_3 \times \mathbb{Z}_2$ centre of $SU(3) \times SU(2)$ — is exactly the global quantisation of charge recorded in Lemma 4.11. □

Remark 4.24 (Predictive boundary)*.* The predictive content of Sections 4.3 and 4.4 is delimited by three structural choices, and removing any of them changes the conclusion. (i) *Colour carrier.* Without the additional colour carrier of Definition 4.14, the construction retains only the electroweak factor $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R$; no $\mathfrak{su}(3)$ and no quark content would follow. (ii) *Minimality criterion.* If the economy criterion of Definition 3.3 were replaced by a different selection rule, a larger simple algebra would be admitted in Proposition 4.16, and $N_g = 8$ would not be fixed. (iii) *Higgs vacuum.* If the neutral-Higgs vacuum condition (S3) were relaxed, the hypercharges would remain determined only up to the overall electroweak normalisation. These three boundaries record where the construction selects and where it assumes.

4.5 Content symmetries

The five representations of Section 4.4 carry a counting symmetry. The left-handed doublets contribute Q_L with $3 \times 2 = 6$ components and L_L with $1 \times 2 = 2$ components, so the left-handed count per generation is

$$N_L = 6 + 2 = 8.$$

The right-handed singlets contribute u_R with 3, d_R with 3, and e_R with 1, so

$$N_R = 3 + 3 + 1 = 7.$$

The left-handed count equals the number of colour generators $N_g = 8$, and the right-handed count is one less. This content-gauge symmetry

$$N_L = N_g, \qquad N_R = N_g - 1$$

relates the fermion content to the gauge structure. The chiral asymmetry

$$N_L - N_R = 1$$

is odd. A compact oriented manifold admits a spin structure if and only if its second Stiefel–Whitney class vanishes, $w_2(M) = 0$, and the inequivalent spin structures are classified by $H^1(M, \mathbb{Z}_2)$. The internal space RP^3 admits two spin structures, classified by $H^1(RP^3, \mathbb{Z}_2) = \mathbb{Z}_2$. The chiral structure of clause (A1), whose antipodal action distinguishes the two Weyl components (Appendix A), selects the non-trivial spin structure; its Dirac spectrum is the half-integer sector recorded in Appendix A. The hypercharges of Section 4.4 satisfy

$$\sum Y = 0,$$

which is the vanishing of the mixed gravitational anomaly, and their first non-zero moment is

$$\sum Y^2 = \frac{10}{3}.$$

These content identities are exact consequences of the representation content of Section 4.4.

4.6 Spacetime dimension $4 = 3 + 1$

Corollary 4.25 (Local real dimension $4 = 3 + 1$)**.** *The spacetime V of Theorem 4.4 has local real dimension four: at each regular point of the scale field, the three-dimensional level sets of σ and the one-dimensional scale-flow direction span V. The internal space RP^3 of Proposition 4.3 is the compact internal factor of the product-space construction of Appendix B.*

Remark 4.26 (Scope of the dimension statement). Corollary 4.25 is a local statement about the real dimension of V and the $3 + 1$ split provided by the level sets of the scale field. Theorem 4.4 supplies the four-dimensional carrier V as the real space spanned by the Clifford generators, while Corollary 4.25 supplies the local $3 + 1$ split of that carrier by the level sets of the scale field; the two statements are related but distinct. It does not by itself fix a Lorentzian signature, a causal structure, or a global foliation by singularity-free three-dimensional slices. These are not used in the constructions of this paper and are left to a separate treatment; the four-dimensional factor entering the reconstruction (Appendix C) is Euclidean.

Proof. At every regular point of σ, the level set Σ_σ is a smooth submanifold of V of codimension one, and its normal direction is spanned by $\nabla\sigma$. Hence at such a point,

$$V = T_x\Sigma_\sigma \oplus \mathbb{R}\,\nabla\sigma(x),$$

with $\dim T_x\Sigma_\sigma = 3$ and $\dim \mathbb{R}\,\nabla\sigma = 1$. Therefore $\dim V = 4$, consistent with Theorem 4.4. The three spatial dimensions are the level sets of the scale field in V. The internal space RP^3 is an additional compact factor: its spectrum is carried by the ensemble geometry of the level sets (Theorems 4.34 and 4.36), and the product-space construction on $V \times RP^3$ with the decoupling of the massive modes is recorded in Appendix B. ☐

4.7 Scale field, equi-σ surfaces, and measure

The previous subsections fixed the algebraic structure of the theory. They determined the symmetry group, the internal space, the gauge algebra, and the spacetime dimension. None of this fixes the dynamical content, nor does it determine how the change acts at each point and at each scale. The remaining clause of the principle, unbiasedness (A2), supplies this content statistically. The statistical distribution of the change is the Gaussian fixed by the maximum entropy principle (Theorem 2.3). Realised on the emergent spacetime V, this statistical content becomes a random field. On the compact carrier sphere S^3 the invariant measure selected by unbiasedness is the uniform measure (Section 3); the Gaussian distribution is the realisation of this statistical content on the non-compact emergent spacetime V, where the fixed covariance of Theorem 4.30 turns it into a Euclidean Gaussian random field. The change sweeps out the scale direction, which is the fourth dimension of Corollary 4.25. The random field is therefore the scale field $\sigma(x)$. The coherence scale of the scale field is denoted L; it is both the radius of the internal space RP^3 and the Gaussian integral width of σ. Denoting the Gaussian width by b, the two are related by $L = \sqrt{2\pi}\,b$ at the condensation threshold (Proposition 8.27). At a point x, the value $\sigma(x)$ is the local value of the scale parameter. The statistics of σ are described by the unbiased Gaussian structure of (A2). The equi-σ surfaces are the sets $\Sigma_\sigma = \{x : \sigma(x) = \sigma\}$. They are the loci of constant local scale.

Theorem 4.27 (Gaussian measure of the scale field)**.** *A Gaussian probability measure $d\mu_{0,\sigma}$ exists on the configuration space $\mathcal{E} = \mathcal{S}'(V)$ with mean zero and covariance Σ_σ.*

Proof. By (A2) and Definition 3.2 the distribution is G-invariant. With moment constraints, Theorem 2.3 gives the Gaussian distribution. Minlos' theorem [29] extends the Gaussian cylinder measure to $\mathcal{E}$ when Σ_σ is a tempered kernel. ☐

Lemma 4.28 (Covariance structure)**.** *The covariance kernel of the σ field is radial, $C(x, y) = G(|x - y|)$.*

Proof. The measure is $E(4)$-invariant by (A2). Translation invariance gives $C(x,y) = G(x-y)$. Rotation invariance gives $G(x-y) = G(R(x-y))$ for every $R \in SO(4)$. Thus G depends only on $|x-y|$. $\square$

Definition 4.29 (Spectral representation of the Gaussian structure)**.** *The covariance kernel $C(x,y) = G(|x-y|)$ admits, by the Bochner–Schwartz theorem [30], the spectral representation*

$$C(x,y) = \int e^{ip\cdot(x-y)}\,\rho(p^2)\,\frac{d^4p}{(2\pi)^4},$$

where $\rho(p^2) \geq 0$ is a tempered spectral density. Rotational invariance makes ρ a function of p^2 alone. Its explicit form is fixed up to a normalisation constant by the scale-invariant fixed point of Theorem 4.30.

Theorem 4.30 (Conformal two-point structure at a fixed point)**.** *At a scale-invariant fixed point the two-point function takes the conformal form*

$$C(x,y) = \frac{c}{|x-y|^2},$$

with $c > 0$. Equivalently, the momentum-space covariance is proportional to p^{-2}. In the Källén–Lehmann spectral representation the spectral measure is $c\,\delta(\mu^2)$, describing a single massless mode. The constant c cancels in all ratios of physical quantities.

Proof. At a scale-invariant fixed point the two-point function is scale covariant, $C(\lambda x, \lambda y) = \lambda^{-2\Delta} C(x,y)$, where Δ is the scaling dimension of the free field. For the free Gaussian field associated with the Clifford structure of Theorem 4.4, the dimension is $\Delta = 1$. Scale and rotation invariance then fix the two-point function up to a constant, giving $C(x,y) = c\,|x-y|^{-2}$. The Fourier transform of $|x-y|^{-2}$ in four dimensions is proportional to p^{-2}. The mass spectral measure is therefore $c\,\delta(\mu^2)$. The constant c multiplies the covariance linearly and drops out of all ratios defined by the spectral sums of Definition 8.1. $\square$

Lemma 4.31 (Level set dimension)**.** *At a regular point of σ, the level set Σ_σ is a three-dimensional smooth submanifold of V.*

Proof. By the implicit function theorem, $\Sigma_\sigma = \sigma^{-1}(\sigma)$ has codimension one. Since $\dim V = 4$ by Theorem 4.4, $\dim \Sigma_\sigma = 3$. $\square$

The internal space is fixed twice in this paper, and the two routes meet at the same object. The first route is algebraic. In Section 4.1 the chiral structure of clause (A1) required the internal space to be an admissible free quotient of S^3. The minimal admissible quotient is $S^3/\mathbb{Z}_2$. The second route is statistical. The unbiased statistics of the scale field fixes the effective geometry to be a compact constant-curvature space S^3/Γ. The chiral structure requires Γ to contain the antipodal element. The same economy criterion then selects the smallest admissible Γ, namely $\mathbb{Z}_2$. Thus both routes lead to RP^3. They are not two independent assumptions. They are two applications of the same economy principle, one to the chiral quotient and one to the effective geometry.

Definition 4.32 (Ensemble geometry)**.** *The ensemble geometry of* $\{\Sigma_\sigma\}$ *is the spectral geometry of the family. It is the three-manifold whose Laplace spectrum is fixed by the correlation structure of the* σ *field on* Σ_σ*. The relevant sector is selected by the chiral structure of (A1). This gives the spectrum* $\varepsilon_l = l(l+2)/L^2$ *with multiplicities* $d_l = (l+1)^2$ *for even* l*. The assignment of this spectrum to the correlation structure is part of the constructive data of Definition 4.20. The ensemble geometry is a statistical object. The individual level sets of the Gaussian field are irregular random surfaces in general. No statement is made about the geometry of any single realisation.*

Proposition 4.33 (Maximal symmetry)**.** *The ensemble geometry is maximally symmetric. Its spectral data are the Laplace spectrum of a constant-curvature space.*

Proof. The spectral data $\varepsilon_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ are the Laplace spectrum of the round sphere S^3 of radius L. The ensemble data are the sector of this spectrum selected by the quotient structure. Every quotient of the round sphere has constant curvature and is therefore maximally symmetric. The specific quotient is selected by the chiral structure and verified by the spectral data. The maximal symmetry is a property of the ensemble geometry, not of the individual realisations. □

Theorem 4.34 (Effective geometry of the level sets at the flow endpoint)**.** *Let* σ *be the Gaussian scale field of (A2),* $E(4)$*-invariant with coherence scale* L*. Let the coarse-graining flow erase short-distance fluctuations of* Σ_σ *down to the scale* L*. Then the effective geometry of the level set* Σ_σ *at the flow endpoint is isotropic and homogeneous. It has constant curvature, with curvature scale* $R \sim 1/L^2$*. Its spectral data are those of the round sphere* S^3 *(Proposition 4.33), so the curvature is positive. The effective geometry is therefore a connected compact maximally symmetric space* S^3/Γ *[31].*

Proof. The $E(4)$-invariance of the measure fixes the distributional symmetry of the scale field. After integrating out fluctuations below the coherence scale L, the effective geometry inherits translational and rotational invariance. Hence it is homogeneous and isotropic. A homogeneous and isotropic Riemannian metric has constant curvature. The curvature scale is set by the coherence scale L, so by dimensional analysis $R = c/L^2$. The internal space is compact. A compact constant-curvature three-manifold is spherical, flat, or hyperbolic. The spectral data are those of the round sphere S^3 (Proposition 4.33), so the curvature is positive, which excludes the flat and hyperbolic cases. The effective geometry is therefore a spherical quotient S^3/Γ [31]. □

The chain is now explicit. The effective geometry of the level sets is determined by the statistics of the scale field. The spectral data are the Laplace spectrum of this geometry. The chain is

$$\text{statistics of the scale field} \;\to\; \text{effective geometry} \;\to\; \text{Laplace spectrum} \;\to\; \text{spectral data.}$$

The spectral data are the spectrum of the effective geometry fixed by the statistics.

Lemma 4.35 (Compactness of the internal space)**.** *The internal space is compact.*

Proof. In the spectral-sum definition of Definition 8.1, physical quantities are defined by absolutely convergent spectral sums over the spectrum of the internal Laplacian. This requires the spectrum to be purely discrete. A complete non-compact Riemannian manifold has non-empty essential spectrum. Hence a purely discrete spectrum implies compactness. Compactness also follows directly from Proposition 4.3, since $RP^3 = S^3/\mathbb{Z}_2$ is compact. The present argument shows consistency of the two routes. □

Theorem 4.36 (Classification)**.** *The ensemble geometry is a three-dimensional connected compact maximally symmetric space. It is therefore isomorphic to S^3/Γ, where $\Gamma \subset SO(4)$ is a finite subgroup acting freely. The quotient is selected by the chiral structure and the economy criterion, giving $RP^3 = S^3/\mathbb{Z}_2$. Its spectrum is the even-l sector of the sphere spectrum.*

Proof. Dimension three is Lemma 4.31. Compactness is Lemma 4.35. Maximal symmetry is Proposition 4.33. A connected compact maximally symmetric three-manifold has constant curvature and is of the form S^3/Γ [31]. The chiral structure requires Γ to contain the antipodal element. The economy criterion of Definition 4.2 selects the minimal quotient $\Gamma = \mathbb{Z}_2$. The resulting space is RP^3. Its spectrum is the even-l sector of the sphere spectrum. □

The ensemble geometry is defined by the spectral data, not by the geometry of individual level sets. The level sets of a stationary isotropic Gaussian field are irregular in general. The maximal symmetry of Proposition 4.33 is a property of the ensemble geometry, not of the realisations. The construction uses only the spectral data of the ensemble. These data are fixed by the $E(4)$-invariance of the measure through the correlation structure.

Proposition 4.37 (Chiral selection)**.** *The chirality of (A1) together with the economy criterion selects $\Gamma = \mathbb{Z}_2$, so the ensemble geometry is $RP^3 = S^3/\mathbb{Z}_2$.*

Proof. The chiral Weyl pair $(1/2, 0) \oplus (0, 1/2)$ decomposes under the antipodal map on S^3 as ψ_L odd and ψ_R even. This parity distinguishes left from right. For S^3/Γ to carry this chiral pair, the antipodal map must act trivially on the quotient. Hence Γ must contain the antipodal element. Among the free quotients S^3/Γ containing the antipodal identification, the economy criterion of Definition 4.2 selects the smallest order $|\Gamma|$. This gives $\Gamma = \mathbb{Z}_2$. Larger quotients such as the lens spaces $S^3/\mathbb{Z}_{2m}$ with $m \geq 2$ also contain the antipodal element but are excluded by the economy criterion. The resulting space RP^3 has the even-l sector of the sphere spectrum as its Laplace spectrum. □

Remark 4.38 (Status of the chiral quotient selection). The selection of $\Gamma = \mathbb{Z}_2$ uses the same economy criterion that selects $SU(2)$ over $U(2)$ in Lemma 3.5. In the present case the criterion is applied to the finite quotient, as stated in Definition 4.2. The statistics alone determine a constant-curvature compact space S^3/Γ. The reduction to RP^3 is driven by clause (A1), which requires Γ to contain the antipodal element, and by the economy criterion, which selects the smallest admissible $|\Gamma|$. The chirality requirement comes from (A1), and the minimality step is the economy criterion, part of the constructive data of Definition 4.20.

Corollary 4.39. *The equi-σ surface ensemble geometry is RP^3.*

Proposition 4.3 and Corollary 4.39 both give RP^3. This internal space determines the gauge algebra of Section 4.3 and supplies the spectral data used in Section 8.6.

4.8 Status of the principal ingredients

Table 1 records the logical status of the principal ingredients, so that the boundary between what is assumed, what is selected, what is stated as structural datum, and what is derived can be read off in one place. The classifications follow Definition 4.13 and Definition 4.20.

Table 1. Status of the principal ingredients

Ingredient	Status in the construction	First location	Consequence
coarse-graining principle	principle	Principle 3.4	statistical content (A1)–(A2)
spectral datum	definitionally specified datum	Definitions 5.1–5.8	spectral-sum-definition objects
RP^3 geometry	derived proposition	Prop. 4.3, Prop. 4.37	internal space, discrete spectrum
electroweak factor	structural theorem	Thm. 4.10	$\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_R$
colour carrier	stated structural datum	Def. 4.13(S1)	second simple factor exists
$\mathfrak{su}(3)$ colour algebra	minimality selection	Prop. 4.16	$N_g = 8$
matter closure (class)	stated closure	Def. 4.13(S2)	admissible fermion class
fermion content and hypercharges	derived (given the closures)	Sec. 4.4–4.5	$N_L = N_g = 8$, $N_R = 7$, $\sum Y = 0$, $\sum Y^2 = \frac{10}{3}$
window criterion	derived proposition	Prop. 8.14	$N_{\mathrm{win}}(c) = \#\{n \in 2\mathbb{N}_0 : n + \frac{3}{2} < c\}$
condensed endpoint	endpoint condition	Sec. 8.6, $R(\sigma_C) < R_c$	scale-flow endpoint
positive scalar mass	conditional theorem	Thm. 8.32	$m_\delta^2 > 0$
free-sector reconstruction	standard reconstruction theorem	Thm. 10.6	free scalar on $\mathbb{R}^4 \times RP^3$

The two rows that most affect how the construction is read are the colour carrier and the matter closure: both are stated structural data of the construction (Definition 4.13), not consequences of the coarse-graining principle. Conditional on them, the colour algebra $\mathfrak{su}(3)$ is selected by minimality (Proposition 4.16) and the fermion content with its hypercharges is derived (Section 4.4–4.5); neither is a free choice.

5 The spectral formulation

This section collects the conditions of the formulation in a self-contained form and states the status of each. Three conditions are definitional. They specify the primitive data, the rule for physical quantities, and the state space. Four further conditions describe the kinematic structure. They locate the standard rules of quantum mechanics within the formulation. These rules are probability, dynamics, composition, and locality. On the free sector each of them reduces to a standard theorem for free fields. On the interacting sector each of them is a conditional statement within the spectral-sum definition. The final condition is the mapping condition. It defines the relation between the spectral datum and the physical structure.

5.1 The spectral datum

Throughout, M denotes a compact three-dimensional Riemannian manifold without boundary, and Δ_M the non-negative scalar Laplacian on M. By the standard spectral theory of elliptic operators on compact manifolds, Δ_M has a discrete spectrum $\{\varepsilon_l\}_{l\geq 0}$ with finite multiplicities $\{d_l\}_{l\geq 0}$, accumulating only at infinity, and the counting function $N(\lambda) = \#\{l : \varepsilon_l \leq \lambda\}$ satisfies the Weyl asymptotics $N(\lambda) \sim C\,\lambda^{3/2}$ as $\lambda \to \infty$.

Definition 5.1 (Spectral datum)**.** *The primitive object of the formulation is the spectral datum*

$$S = \big(M,\ \{\varepsilon_l\},\ \{d_l\}\big), \tag{5.1}$$

consisting of the compact three-dimensional Riemannian manifold M and the spectrum $\{\varepsilon_l\}$ of its scalar Laplacian with multiplicities $\{d_l\}$. No further parameters are introduced by hand.

From the spectrum and the derived squared-mass parameter m_δ^2 the mass tower is defined as

$$M_l^2 = \varepsilon_l + m_\delta^2. \tag{5.2}$$

The spectral projector kernels $K_l(y, y')$ of the l-th eigenspace of Δ_M are used throughout. The derivation of m_δ^2 from the spectral datum is given in Theorem 8.32. It does not require any additional input.

The positive scalar mass m_δ, the quartic coupling λ, and the condensate $\langle E\rangle$ are not independent inputs. They are constructed from the datum in Sections 7 and 8, where their existence and positivity are established. Throughout, m_δ denotes the mass and m_δ^2 its square, which is the derived squared-mass parameter entering the mass tower (5.2); the bare symbol m_δ always denotes the mass, and a squared mass is always written with the exponent.

A concrete instance of the datum is obtained by taking $M = RP^3$, the real projective three-space of radius L. The scalar Laplacian has spectrum $\varepsilon_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ for even l [32]. The mass tower is then $M_l^2 = l(l+2)/L^2 + m_\delta^2$, with m_δ^2 derived as in Section 8. This example is used throughout the paper to illustrate the general definitions. Nothing in the conditions requires it. The reconstruction theorem of Appendix C holds for any datum satisfying the stated conditions.

Definition 5.2 (Physical quantities). *A physical quantity is a function of the spectral datum of Definition 5.1 of the form*

$$Q = \sum_l d_l \, f(\varepsilon_l), \tag{5.3}$$

where f is such that the series converges absolutely, or a closed analytic expression in the datum. A closed analytic expression is one obtained from the spectrum and the derived parameters of the datum by finitely many elementary operations and known special functions. Physical quantities are defined by these spectral sums. They are not defined as expectations with respect to a measure. This rule is the constitutive definition of the formulation. It specifies the meaning of the term physical quantity within the formulation.

For the mass-tower resolvent sums that appear below, the Weyl law gives $\sum_l d_l\,(\varepsilon_l + m^2)^{-p} \sim \int (\lambda + m^2)^{-p}\,\lambda^{1/2}\,d\lambda$, so absolute convergence requires $p > 3/2$. This is the Weyl exponent of the three-dimensional factor M. The exponent $3/2 = \frac{1}{2}\dim M$ is the source of the convergence threshold.

Definition 5.3 (State space). *The states of the theory are the normalised vectors and density operators of the physical Hilbert space $\mathcal{H}$, the reflection-positive reconstruction space of the Gaussian structure supplied by the datum (Appendix C). On the free sector the reconstruction is the Fock construction $\mathcal{H} = \mathcal{F}(h)$ over the mass tower $\{M_l\}$. This holds unconditionally. On the interacting sector the standard Osterwalder–Schrader reconstruction is conditional. The formulation defines the kinematic carrier of the interacting sector through the effective Fock space of Theorem 7.8 and the formal correspondence of Proposition 9.2.*

Definition 5.4 (Probability). *A measurement of a self-adjoint observable is a projection-valued measure. The probability of an outcome is the Born rule $p = \langle\psi|P\psi\rangle$. This is the kinematic rule of quantum mechanics on a Hilbert space. It is the unique probability assignment on the projection lattice by Gleason's theorem [33]. It is not derived from the Gaussian structure alone.*

Definition 5.5 (Dynamics). *Time evolution is the unitary group supplied by the reflection-positive reconstruction and Stone's theorem, $U(t) = e^{-\mathrm{i}Ht}$ with H self-adjoint. The states satisfy the Schrödinger equation. On the free sector $H = d\Gamma(\omega)$ with $\omega = \sqrt{L_V + M_l^2}$ on each mode of the mass tower, where $L_V = -\sum_{\mu=1}^4 \partial_\mu^2 \geq 0$ is the non-negative Laplacian on the emergent spacetime $V = \mathbb{R}^4$. This holds unconditionally. The scale parameter of the formulation and the emergent time t are independent directions.*

Definition 5.6 (Composition). *The composite of two systems is the tensor product of their physical Hilbert spaces. The independence of preparations is expressed by product initial states. On the free sector the tensor product is the Fock-space tensor product $\mathcal{F}(h_1 \oplus h_2) \cong \mathcal{F}(h_1) \otimes \mathcal{F}(h_2)$. This holds unconditionally.*

Definition 5.7 (Locality). *The field algebra satisfies the spacetime locality conditions. Fields at spacelike separation commute for bosons and anticommute for fermions. The fermion anticommutation relations are supplied by the spin–statistics connection.*

Definition 5.8 (Mapping)**.** *The physical structure of the formulation is defined to be the reconstruction of the spectral datum. This includes the Hilbert space, the field algebra, the dynamics, and the vacuum. The free-sector reconstruction is a theorem (Appendix C). The interacting-sector reconstruction is stipulated by the spectral-sum definition. It is defined by the formal correspondence of Section 9 and the spectral-sum definition of physical quantities. It is not claimed as a theorem in the standard measure-theoretic sense.*

Remark 5.9 (Status of the conditions). The conditions fall into three classes. Definitions 5.1, 5.2 and 5.3 are definitional. They specify the primitive data, the rule for physical quantities, and the state space. Definitions 5.4–5.7 are kinematic structure. On the free sector they reduce to standard theorems for free fields. They are not new results of this paper. On the interacting sector they are conditional statements within the spectral-sum definition. Definition 5.8 is the mapping condition. It is a definitional statement specifying the reconstruction relation between the spectral datum and the physical structure. The free-sector instance is proved. The interacting instance is defined within the spectral-sum definition.

5.2 Physical quantities and closure criteria

Definition 5.10 (Formulation criterion)**.** *A system of conditions constitutes a formulation of quantum field theory if it determines the following physical structures as well-defined mathematical objects.*

(i) State space, a Hilbert space with a vacuum.

(ii) Observables, an algebra of self-adjoint operators.

(iii) Probability, the Born rule of projective measurement.

(iv) Dynamics, unitary time evolution with a self-adjoint Hamiltonian.

(v) Composite systems, the tensor-product structure.

(vi) Locality, spacelike commutation or anticommutation.

(vii) Spectrum, the mass spectrum and the spectral gap.

(viii) Correlation functions, the n-point spectral functions.

(ix) Scattering, the scattering amplitudes.

(x) Thermal states, the KMS states.

(xi) Measurement, the system–apparatus coupling and decoherence.

Theorem 5.11 (Status of the physical structures)**.** *The conditions of Section 5.1 locate the structures required by Definition 5.10 as follows. Each entry is marked as derived from the spectral datum, defined within the spectral-sum definition, or standard kinematics.*

(i) State space. *Derived on the free sector. Defined on the interacting sector by the effective Fock space of Theorem 7.8 and the formal correspondence of Proposition 9.2.*

(ii) Observables. *Derived on the free sector from the Fock construction. Defined on the interacting sector by the mapping dictionary of Proposition 9.2.*

(iii) Probability. *Standard kinematics. The Born rule is part of the kinematic conditions and follows from Gleason's theorem.*

(iv) Dynamics. *Standard kinematics. Unitary evolution follows from the kinematic conditions and Stone's theorem applied to the reconstructed Hamiltonian.*

(v) Composite systems. *Standard kinematics. The tensor product is the standard Fock-space tensor product on the free sector.*

(vi) Locality. *Derived on the free sector from the commutator of free fields. On the effective sector it follows from Proposition 7.9.*

(vii) Spectrum. *Derived from the spectral datum. The mass tower* $\{M_l\}$ *and the mass parameter* m_δ *are obtained from the spectrum through the constructions of Sections 7 and 8.*

(viii) Correlation functions. *Derived on the free sector from the spectral kernel. On the interacting sector they are spectral-sum defined.*

(ix) Scattering. *Defined within the spectral-sum formulation by spectral-sum amplitudes. The particle content is identified by the spectral poles.*

(x) Thermal states. *Derived on the free and effective sectors from the KMS construction.*

(xi) Measurement. *Standard kinematics. The system–apparatus coupling and decoherence are standard.*

On the free sector each structure reduces to a standard free-field statement. On the interacting sector the structures are defined within the spectral-sum definition. No external physical parameters are used beyond the spectral datum.

The correspondence of these conditions to the standard Wightman, Osterwalder–Schrader and Haag–Kastler [34] formulations is recorded in Section 10, with the free-sector instance in Appendix C. The item-by-item status is as follows. The table records where the formulation obtains the standard counterpart and where it deliberately does not follow the standard route: the interacting content has no pointwise fields, so the locality entry ends with the field structures that exist.

5.3 Gaussian measure and spacetime dimension within the formulation

The conditions of Section 5.1 are stated for an arbitrary compact three-manifold M. The four-dimensional factor $\mathbb{R}^4$ of the reconstruction enters through the Osterwalder–Schrader construction of Appendix C. Two standard mathematical facts then fix the Gaussian form

Table 2. Correspondence between the standard items and their spectral-sum-formulation counterparts

standard item	counterpart in the spectral-sum formulation	status
Hilbert space and vacuum	Definition 5.3; the reconstruction of Definition 6.2	defined; free sector derived (Fock space)
Field operators, locality	Definition 5.7; Theorem 8.41; Proposition 7.9	derived on the free and effective sectors; the interacting observables are global spectral sums (Remark 7.10)
Spectral condition	Theorems 8.32 and 8.37: $M_l \geq m_\delta > 0$	derived
Covariance	Theorem 8.18; Theorem 8.23 ($SO(4)$ at the level of spectral data)	derived
Clustering	Lemma 8.38	derived; needs no measure

of the measure and the dimension of spacetime. Both facts follow from the conditions themselves.

The first fact is the maximum entropy principle, Theorem 2.3 of Section 2. The mean is zero as a direct consequence of unbiasedness (A2): a non-zero mean would single out a preferred configuration, while the constraints carry no such preference. The covariance, by contrast, is not fixed by unbiasedness alone; it is fixed by the scale-invariant two-point structure at the conformal fixed point, Theorem 4.30 of Section 4.7, which gives $C(x, y) = c/|x - y|^2$. With the mean and covariance so fixed, the maximum entropy theorem selects the Gaussian measure as the least-information distribution. The maximum entropy theorem is a standard mathematical result used inside clause (A2) and introduces no additional physical parameter.

The second fact is the Clifford algebra dimension. Each component $(1/2, 0)$ and $(0, 1/2)$ of the chiral Weyl pair is a semispinor module of complex dimension two (Lemma 3.7 of Section 3).

By Lemma 3.7 a chiral decomposition exists only in even dimensions $n = 2m$, and the semispinor dimension equation $2^{m-1} = 2$ gives $m = 2$, hence $n = 4$. This is the content of Theorem 4.4 of Section 4: the Clifford algebra carrying the chiral pair is Cl(4), whose generators span a four-dimensional real vector space, so $\dim \mathbb{R}^4 = 4 = 3 + 1$. The identification of the carrier as a semispinor module is part of the kinematic content of the chiral clause (A1). The Gaussian form and the ambient dimension four therefore follow from the conditions together with the standard facts of Sections 2 and 3, without additional input.

6 Kinematics

This section locates the standard kinematic rules of quantum mechanics within the formulation of Section 5. The rules treated are probability, the state space, the Born rule, non-commutativity, spin and statistics, unitary time evolution, energy quantisation, and the measurement process. Each rule is marked as standard kinematics or as derived from the spectral datum. On the free sector the Fock space of the reconstruction theorem is unconditional. On the interacting sector the corresponding statements are addressed by the mapping relations of Section 7.

6.1 Probability and the Born rule

Definition 6.1 (Probability measure)**.** *Let $d\mu$ be the Gaussian measure on the configuration space $\mathcal{E} = \mathcal{S}'(\mathbb{R}^4)$. Its covariance is the two-point kernel determined by the spectral datum through the mass tower (5.2) of Definition 5.1. For every measurable set $B \subset \mathcal{E}$, the number $P(B) = \mu(B)$ is the probability that the field configuration lies in B.*

The Gaussian measure is the reconstruction input. The physical probabilities are those computed in the reconstructed Hilbert space by the Born rule. On the free sector the two descriptions agree.

Among all distributions consistent with a fixed mean and covariance, the unique maximum-entropy distribution is the Gaussian (Theorem 2.3 of Section 2). The mean vanishes by unbiasedness (A2), since a non-zero mean would single out a preferred configuration. The covariance is fixed by the spectral datum through the mass tower (5.2). Hence the Gaussian measure of Definition 6.1 is the unique probability measure consistent with the datum.

6.2 State space and superposition

Definition 6.2 (Euclidean and physical Hilbert spaces)**.** *Let $\mathcal{E} = \mathcal{S}'(\mathbb{R}^4)$ be the configuration space with the Gaussian measure $d\mu$ of Definition 6.1. The Euclidean Hilbert space is $\mathcal{H}_E = L^2(\mathcal{E}, d\mu)$, the space of square-integrable functionals on $\mathcal{E}$. The physical Hilbert space $\mathcal{H}$ is the reflection-positive reconstruction space of the Osterwalder–Schrader construction [21, 22]. It is the completion of the positive-time functionals modulo the null space of the reflection-positive sesquilinear form. On the free sector this gives the bosonic Fock space $\mathcal{H} = \mathcal{F}(h)$ over the one-particle space h of the mass tower $\{M_l\}$. This holds unconditionally. On the interacting sector the standard reconstruction is conditional. The formulation defines the kinematic carrier through the effective Fock space of Theorem 7.8 and the formal correspondence of Proposition 9.2.*

The physical Hilbert space $\mathcal{H}$ is not the Euclidean space $\mathcal{H}_E$. It is a quotient and completion of the positive-time subspace. It carries the vacuum and the relativistic dynamics that the Euclidean space does not. The states of the theory are the normalised vectors and density operators of $\mathcal{H}$.

Theorem 6.3 (Born rule, standard kinematic rule)**.** *Let ψ, ϕ be normalised states of $\mathcal{H}$. The probability that a system prepared in the state ψ is found in the state ϕ is*

$$p([\psi], [\phi]) = |\langle\psi|\phi\rangle|^2. \tag{6.1}$$

This rule is standard in quantum mechanics. For completeness an elementary uniqueness proof is included on finite-dimensional subspaces. The full infinite-dimensional statement is provided by Gleason's theorem [33].

Proof. It suffices to prove the statement on an arbitrary finite-dimensional subspace, since any pair of unit rays lies in such a subspace. The unitary group acts transitively on pairs of unit rays with a fixed value of $|\langle\psi|\phi\rangle|$. A unitary-invariant transition probability therefore depends only on this value. Write $p([\psi],[\phi]) = f(|\langle\psi|\phi\rangle|)$ with f continuous. The resolution-of-unity condition fixes f. A basis containing $\phi = \psi$ gives $f(1) = 1$ and $f(0) = 0$. Restricting to a two-dimensional subspace, write $\psi = a\phi_1 + b\phi_2$ with $|a|^2 + |b|^2 = 1$. This gives $f(|a|) + f(|b|) = 1$. In terms of $g(t) = f(\sqrt{t})$, this is $g(t) + g(1-t) = 1$. Restricting to a three-dimensional subspace with $|a|^2 = |b|^2 = t$ gives $2g(t) + g(1-2t) = 1$. The two-dimensional relation gives $g(1-2t) = 1 - g(2t)$. Combining these yields $g(2t) = 2g(t)$. By continuity $g(t) = ct$ on $[0,1]$, and $g(1) = 1$ fixes $c = 1$. Hence $g(t) = t$ and $p([\psi],[\phi]) = |\langle\psi|\phi\rangle|^2$. □

Theorem 6.4 (Superposition)**.** *The state space $\mathcal{H}$ is a complex Hilbert space. Every complex linear combination of two non-zero vectors in $\mathcal{H}$ is a legitimate state of the theory.*

Proof. The complex structure of $\mathcal{H}$ follows from the complex-valued nature of the square-integrable functionals. The algebraic closure of a complex vector space guarantees that every complex linear combination of two state vectors belongs to $\mathcal{H}$. Such a combination defines a consistent probability assignment through the transition probability of Theorem 6.3. It is therefore a legitimate state of the theory. □

Remark 6.5 (Status of the Born rule)*.* The Born rule is the standard kinematic rule of quantum mechanics on a Hilbert space. The preceding proof shows uniqueness on finite-dimensional subspaces under unitary invariance, continuity, and resolution of unity. The full infinite-dimensional statement is Gleason's theorem [33]. The Born rule is not derived from the Gaussian measure alone. It belongs to the Hilbert-space structure of the physical space $\mathcal{H}$.

6.3 Composite systems and the tensor product

Definition 6.6 (Entangled state)**.** *A state $\Psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$ of a bipartite system is called entangled if it cannot be written as a product state $\psi_1 \otimes \psi_2$ with $\psi_1 \in \mathcal{H}_1$ and $\psi_2 \in \mathcal{H}_2$.*

Two layers of the composite-system structure are distinguished. At the level of Euclidean configuration spaces, the factorisation identity

$$L^2(\mathcal{E}_1 \times \mathcal{E}_2, d\mu_1 \otimes d\mu_2) \cong L^2(\mathcal{E}_1, d\mu_1) \otimes L^2(\mathcal{E}_2, d\mu_2) \tag{6.2}$$

holds by the standard factorisation theorem for independent measures. The independence of the two systems is expressed by the product measure. At the level of physical Hilbert spaces, the tensor product is supplied by the reconstruction. On the free sector the Fock-space tensor product $\mathcal{F}(h_1 \oplus h_2) \cong \mathcal{F}(h_1) \otimes \mathcal{F}(h_2)$ is the standard multi-particle structure and holds unconditionally. On the interacting sector the tensor product is not derived from a standard reconstruction. It is stipulated as part of the spectral-sum definition through the effective construction of Theorem 7.8 and the formal correspondence of Proposition 9.2. The composite-system rule is therefore the independence of preparations together with the reconstruction on the free sector, and a stipulation of the spectral-sum definition on the interacting sector.

6.4 Field operators and non-commutativity

Definition 6.7 (Field operators)**.** *Let $\phi(f)$ be the Wightman field operators supplied by the Osterwalder–Schrader reconstruction theorem [21, 22, 35]. They are defined on a dense subspace of the physical Hilbert space $\mathcal{H}$.*

Theorem 6.8 (Non-commutativity of the fermionic sector)**.** *The chiral fermion fields of spin $1/2$ satisfy canonical anticommutation relations, so the fermionic sector of the field algebra is non-commutative. The non-Abelian gauge structure contributes the classical Lie-algebraic bracket $[A_\mu, A_\nu] \neq 0$. Quantum commutators of the gauge-field operators are not asserted here.*

Proof. The field operators of Definition 6.7 are operator-valued distributions acting on the physical Hilbert space $\mathcal{H}$ of Definition 6.2. Non-commutativity is not automatic for operators. A commutative algebra of multiplication operators exists on L^2. In the present formulation the non-commutativity arises from the fermionic sector and the gauge sector. The chiral Weyl fields of spin $1/2$ satisfy canonical anticommutation relations. For suitable test functions f, g with overlapping supports,

$$\psi(f)\psi^\dagger(g) + \psi^\dagger(g)\psi(f) = \langle f, g\rangle,$$

and the same pair satisfies $\psi(f)\psi^\dagger(g) \neq \psi^\dagger(g)\psi(f)$. Therefore the algebra generated by the fermionic fields is not commutative in the ordinary sense. Scalar fields satisfy commutation relations. The non-Abelian gauge interaction enters through the field strength

$$F = dA + g[A, A].$$

The bracket $[A_\mu, A_\nu] \neq 0$ is a direct consequence of the non-vanishing structure constants $f^{abc} \neq 0$ of the gauge algebra. This is the Lie-algebraic self-interaction content of the gauge field. Quantum commutators of gauge-field operators are not asserted here. The non-commutativity of the field algebra rests on the fermionic anticommutators and the classical non-Abelian structure. □

The reconstructed field operators are self-adjoint on a dense domain. This is a standard property of the Osterwalder–Schrader reconstruction [21]. Physical observables built from the fields are therefore self-adjoint operators, and their spectra are real by the spectral theorem.

6.5 Spin and statistics

Theorem 6.9 (Spin–statistics, standard theorem)**.** *The quantum fields corresponding to the chiral Weyl pair $(1/2, 0) \oplus (0, 1/2)$ satisfy Fermi anticommutation relations. Scalar fields satisfy Bose commutation relations.*

Proof. This is the standard spin–statistics theorem for relativistic quantum fields [24]. It states that fields of half-odd-integer spin satisfy anticommutation relations and fields of integer spin satisfy commutation relations. The theorem requires the standard hypotheses of relativistic quantum field theory, including relativistic covariance, locality, the spectral condition, and positivity of the Hilbert-space structure. The spin value is one ingredient, not the sole input. The theorem is used here as a standard result. It is not re-derived from the spin value alone. □

6.6 Time evolution

Theorem 6.10 (Unitary time evolution)**.** *The Osterwalder–Schrader reconstruction supplies a strongly continuous unitary representation of the time-translation group on $\mathcal{H}$. By Stone's theorem this representation is generated by a self-adjoint Hamiltonian H,*

$$U(t) = e^{-\mathrm{i}Ht},$$

and the states satisfy the Schrödinger equation

$$i\partial_t \Psi = H\Psi.$$

Proof. The Euclidean measure satisfies the Osterwalder–Schrader axioms [21, 25]. In particular it is invariant under Euclidean time translations. The reconstruction therefore supplies a strongly continuous unitary representation of the time-translation group on the physical Hilbert space $\mathcal{H}$ of Definition 6.2. By Stone's theorem this representation is generated by a self-adjoint Hamiltonian H. On the free sector the Hamiltonian is the Fock operator $H = d\Gamma(\omega)$, where

$$\omega = \sqrt{L_V + M_l^2}$$

on each mode of the mass tower. This holds unconditionally. The scale parameters of the formulation and the emergent time t are independent quantities. The scale parameters belong to the spectral datum, while t is the time direction of the reconstructed spacetime. □

6.7 Energy quantisation

Theorem 6.11 (Energy quantum and Bohr frequency condition)**.** *A stationary state of the reconstructed theory has energy $E = \hbar\omega$, where ω is its angular frequency. A transition between two stationary states of energies E_i and E_f emits or absorbs radiation at frequency $\Delta E/h$, where $\Delta E = E_f - E_i$ and $h = 2\pi\hbar$.*

Proof. Theorem 6.10 supplies the Schrödinger equation. A stationary state of energy E evolves as

$$\Psi(t) = e^{-\mathrm{i}Et/\hbar}\psi.$$

Its phase rotates at angular frequency $\omega = E/\hbar$. Thus $E = \hbar\omega$. The transition rule follows from the spectral representation of the coupling to radiation. Let the system couple to a radiation field by

$$H_{\text{int}} = g\, O \otimes A,$$

where O is a system observable and A is the radiation field. In first-order time-dependent perturbation theory the transition amplitude from ψ_i to ψ_f with emission of a photon of energy $\hbar\omega_\gamma$ is proportional to

$$\int_0^t dt'\, e^{\mathrm{i}(E_f - E_i - \hbar\omega_\gamma)t'/\hbar} \langle\psi_f|O|\psi_i\rangle \langle\gamma_f|A|\gamma_i\rangle. \tag{6.3}$$

This integral is resonant at $\hbar\omega_\gamma = E_f - E_i$. Transitions therefore occur only when the emitted photon carries the energy difference. This is the Fermi golden rule. Hence the transition frequency satisfies $\Delta E = h\nu$ with $h = 2\pi\hbar$. □

Remark 6.12 (Status of $\hbar$). The Planck constant $\hbar$ is the conversion scale of the formulation's units. It converts dimensionless representation data into dimensionful observables, such as $J^2 = \hbar^2 j(j+1)$ and $E = \hbar\omega$. Its numerical value fixes the unit system. The present section only records the kinematic relation $E = \hbar\omega$ and the Bohr frequency condition derived from the radiation coupling. The discreteness of the energy spectrum is a separate property of the Hamiltonian spectrum.

6.8 The measurement process

The measurement process is described by the standard quantum-mechanical treatment. In the present formulation this treatment is consistent with the conditions of Section 5 and the kinematic Born rule of Theorem 6.3.

Definition 6.13 (Measurement apparatus)**.** *A measurement apparatus $\mathcal{A}$ is a quantum system with a macroscopically large number of degrees of freedom. Its Hilbert space $\mathcal{H}_{\mathcal{A}}$ is constructed in the same way as the physical Hilbert space of Definition 6.2. On the free sector this is the Fock construction. In general it is given by the spectral-sum-definition construction of the corresponding sector. The pointer states $|\phi_k\rangle$ are the macroscopically distinct configurations of the read-out. They form a mutually orthogonal basis to a good approximation. Their stability is guaranteed by a positive gap of the apparatus sector.*

Theorem 6.14 (Interaction establishes correlation)**.** *Let $\mathcal{S}$ be a quantum system with Hilbert space $\mathcal{H}_{\mathcal{S}}$, and let $\mathcal{A}$ be a measurement apparatus with Hilbert space $\mathcal{H}_{\mathcal{A}}$. Let the interaction between $\mathcal{S}$ and $\mathcal{A}$ be the von Neumann coupling [36]*

$$H_{\text{int}} = g\, O \otimes Q,$$

where O is the measured observable and Q is the pointer observable. Let the initial system state have non-zero components in at least two eigenspaces of O, and let the displaced pointer states

$$\phi_k(t) = e^{-\mathrm{i}g o_k Q t} \phi_{\mathcal{A}}$$

be pairwise distinct for distinct eigenvalues o_k. Then the initial product state evolves into an entangled state of the composite system.

Proof. The evolution is generated by the total Hamiltonian

$$H = H_{\mathcal{S}} + H_{\mathcal{A}} + H_{\text{int}}.$$

Writing the initial state in the eigenbasis of O,

$$\psi_{\mathcal{S}} = \sum_k c_k \hat{\psi}_k,$$

the evolution under H_{int} gives

$$\Psi(t) = e^{-\mathrm{i}H_{\text{int}}t}(\psi_{\mathcal{S}} \otimes \phi_{\mathcal{A}}) = \sum_k c_k\, \hat{\psi}_k \otimes \phi_k(t),$$

where $\phi_k(t)$ is the pointer state displaced by the eigenvalue o_k. The displaced pointer states are pairwise distinct by hypothesis, and the eigenbranches $\hat{\psi}_k$ are orthogonal. A product decomposition $\Psi(t) = \psi \otimes \phi$ would require a common apparatus factor. This is impossible when at least two distinct displaced pointer states occur. Hence $\Psi(t)$ is entangled. □

Theorem 6.15 (Decoherence from many internal modes, physical argument)**.** *Let $\mathcal{A}$ be a macroscopic apparatus whose pointer states $|\phi_k\rangle$ are the macroscopically distinct configurations of Definition 6.13. Assume that the internal mode frequencies and couplings are sufficiently regular, so that the phases of different modes are effectively uncorrelated. Then the off-diagonal elements of the reduced density matrix of the system, expressed in the pointer basis, decay exponentially. The dephasing rate is determined by the couplings of the system observable to the internal modes of the apparatus.*

Proof. After tracing out the apparatus, the reduced density matrix of the system is

$$\rho_{\mathcal{S}} = \mathrm{Tr}_{\mathcal{A}}\, |\Psi(t)\rangle\langle\Psi(t)|.$$

In the eigenbranch basis $\{\hat{\psi}_k\}$ its off-diagonal elements are

$$\rho_{kl}(t) = c_k \bar{c}_l\, f_{kl}(t),$$

where the suppression factor is

$$f_{kl}(t) = \langle\phi_l(t)|\phi_k(t)\rangle.$$

The interaction couples the observable O to a collective pointer observable

$$Q = \sum_m g_m Q_m,$$

the sum running over the N internal modes of the apparatus. Each mode contributes an independent phase factor to $f_{kl}(t)$. Under the regularity assumption on the internal modes, the phases are uncorrelated for large N, and the product is exponentially suppressed,

$$|f_{kl}(t)| \leq e^{-\Gamma t},$$

with a dephasing rate Γ determined by the couplings and fluctuations of the modes. The positive gap of the apparatus sector ensures stability of the pointer states. Decoherence is therefore a dynamical consequence of the macroscopic number of internal modes under the stated assumptions [37]; the exponential suppression, however, is a physical argument rather than a rigorous theorem without further hypotheses. □

Theorem 6.16 (Born rule as projective measurement)**.** *Let the apparatus $\mathcal{A}$ measure an observable O of the system $\mathcal{S}$ on the physical Hilbert space $\mathcal{H}_{\mathcal{S}}$. Let*

$$O = \sum_k o_k P_k$$

be the spectral decomposition, where P_k is the projector onto the eigenspace with eigenvalue o_k. The probability of obtaining the outcome o_k when the system is in the state $\psi_{\mathcal{S}}$ is

$$p_k = \|P_k \psi_{\mathcal{S}}\|^2.$$

Proof. On the physical Hilbert space, a measurement of a self-adjoint observable is a projection-valued measure. By the spectral theorem, $O = \sum_k o_k P_k$. The probability of the outcome o_k is the standard kinematic rule

$$p_k = \langle \psi_{\mathcal{S}} | P_k \psi_{\mathcal{S}} \rangle = \|P_k \psi_{\mathcal{S}}\|^2.$$

This is the transition probability of Theorem 6.3 between the initial state and the eigenbranch $\hat{\psi}_k = P_k \psi_{\mathcal{S}} / \|P_k \psi_{\mathcal{S}}\|$. The measurement interaction realises this kinematic rule dynamically. Before the measurement the system and apparatus are independent. The state of the composite system is $\Psi(0) = \psi_{\mathcal{S}} \otimes \phi_{\mathcal{A}}$. The von Neumann coupling [36] correlates each eigenbranch with a distinct pointer state. After the coupling the reduced density matrix of the apparatus is diagonal in the pointer basis, consistent with the kinematic rule $p_k = |c_k|^2 = \|P_k \psi_{\mathcal{S}}\|^2$. This is the Born rule. Its status is kinematic. □

6.9 Location of the quantum postulates

The standard postulates of quantum mechanics are located in the formulation as follows.

7 Mapping relations

The mapping condition of Section 5 states the intended relation between the spectral datum and the physical structure. This section collects the mapping statements. The free-sector instance is proved in Appendix C. The interacting-sector instance is defined within the spectral-sum definition by the analytic, structural, effective, and non-perturbative statements of the present section.

Theorem 7.1 (Mapping within the formulation)**.** *Let S be the spectral datum of Definition 5.1. The reconstruction of Definitions 5.3–5.7 yields a physical structure $(\mathcal{H}, \mathcal{A}, U, \Omega)$. On the free sector every physical quantity of Definition 5.2 is a spectral-data quantity of the reconstructed structure, as a trace of the internal Green operator of the reconstructed*

Table 3. Location of the quantum postulates within the spectral formulation

Standard postulate	Located in
State space is a complex Hilbert space	Physical Hilbert space of Definition 6.2. Fock space on the free sector
Superposition principle	Complex Hilbert-space structure, Theorem 6.4
Probability via the Born rule	Projective measurement, Theorem 6.16
Observables are Hermitian operators	Self-adjoint field operators, Section 6.4
Spin–statistics relation	Standard spin–statistics theorem, Theorem 6.9
Time evolution by the Schrödinger equation	Reconstruction and Stone's theorem, Theorem 6.10
Energy quantisation	Phase dynamics of stationary states, Theorem 6.11
Measurement is an interaction	System–apparatus coupling, Theorem 6.14
Decoherence suppresses interference	Many-mode dephasing of the apparatus, Theorem 6.15

propagator (Theorem 7.7). On the interacting sector every physical quantity is spectral-sum defined, with formal correspondence to the field algebra given by Proposition 9.2 and Theorem 9.5. The spectrum of the Hamiltonian contains the mass tower $\{M_l\}$ with lower bound m_δ. The free-sector statement is proved by Theorem C.2. The interacting-sector statement is defined within the spectral-sum definition. It is not claimed as a theorem in the standard measure-theoretic sense.

Remark 7.2 (Interacting mapping and the standard formulation). The interacting content of the formulation has no realisation as a standard four-dimensional elementary scalar field theory, because the continuum limits of four-dimensional lattice scalar models are Gaussian. The interacting mapping is therefore a statement about the formulation itself. The precise locus of this boundary in the standard formulation is discussed in Section 10.4.

7.1 Analytic mapping: analytic control, positivity, condensation, and gap

The analytic part of the mapping records the analytic properties of the spectral sums. The input is the spectral datum of Definition 5.1, namely the compact three-manifold M and its spectrum $\{\varepsilon_l\}$ with multiplicities $\{d_l\}$. The parameters m_δ, λ, and $\langle E\rangle$ are derived from this datum in Section 8.

Theorem 7.3 (Well-definedness and analytic control). *For $p > 3/2$ the spectral sum*

$$Q_p(m^2) = \sum_l d_l(\varepsilon_l + m^2)^{-p}$$

is absolutely convergent, positive, and analytic in m^2 on $(0, \infty)$, with the bound

$$Q_p(m^2) \leq C_p \,\mathrm{vol}(M)\, m^{3-2p}$$

for $m^2 \geq m_\delta^2$.

Proof. By the Weyl law for the scalar Laplacian on the compact three-manifold M, the counting function $N(\mu) = \#\{l : \varepsilon_l \le \mu\}$ satisfies $N(\mu) \sim C\,\mathrm{vol}(M)\,\mu^{3/2}$ as $\mu \to \infty$. The series is therefore summable if and only if the integral

$$\int (\mu + m^2)^{-p}\, dN(\mu) \sim \int (\mu + m^2)^{-p}\, \mu^{1/2}\, d\mu$$

converges at infinity. This requires $p > 3/2 = \frac{1}{2}\dim M$. The series converges absolutely and uniformly on compact subsets of $(0,\infty)$. Each term is positive and analytic in m^2. The sum is therefore positive and analytic. For the bound, split the sum at $\mu_* \sim m^2$. The low modes give

$$Q_p^{(\mathrm{low})} \le m^{-2p} \sum_{\varepsilon_l \le m^2} d_l \sim m^{-2p} N(m^2) \sim \mathrm{vol}(M)\, m^{3-2p}.$$

The high modes give

$$Q_p^{(\mathrm{high})} \le \sum_{\varepsilon_l > m^2} d_l\, \varepsilon_l^{-p} \sim \int_{m^2}^{\infty} \mu^{-p}\, \mu^{1/2}\, d\mu \sim m^{3-2p}.$$

Both parts are of the stated form. This is the analytic control of the physical quantities. Every physical quantity of Definition 5.2 is a well-defined analytic function of its parameters, with no infrared singularities for $m^2 \ge m_\delta^2 > 0$. □

Theorem 7.4 (Spectral positivity)**.** *For every non-negative function f, the physical quantity*

$$Q = \sum_l d_l\, f(\varepsilon_l)$$

of Definition 5.2 is non-negative. In particular, the two-point spectral quantities are positive, and the spectral measure of the mass tower

$$\rho(\mu^2) = \sum_l d_l\, \delta(\mu^2 - M_l^2)$$

is non-negative with support in $[m_\delta^2, \infty)$.

Proof. Every multiplicity $d_l \ge 1$ is positive, and every eigenvalue satisfies

$$M_l^2 = \varepsilon_l + m_\delta^2 \ge m_\delta^2 > 0.$$

Every term is therefore non-negative, and the support statement follows. The positivity of the two-point spectral quantities is a direct consequence of the spectral data. □

Theorem 7.5 (Condensation)**.** *Let λ and $\langle E\rangle$ be the derived parameters of Section 8. Define the long-root mass parameter by*

$$m_{\mathrm{long}}^2 = -\lambda \langle E\rangle^2 < 0. \tag{7.1}$$

Then the Landau potential

$$V(E) = \tfrac{1}{2} m_{\mathrm{long}}^2 E^2 + \tfrac{\lambda}{4} E^4 \tag{7.2}$$

has a unique non-trivial minimum at $E = \langle E\rangle$. The curvature at the minimum is

$$m_\delta^2 = V''(\langle E\rangle) = 2\lambda \langle E\rangle^2 = -2 m_{\mathrm{long}}^2 > 0. \tag{7.3}$$

Proof. The Landau potential (7.2) is even and has derivative

$$V'(E) = E\,(m_{\mathrm{long}}^2 + \lambda E^2).$$

With $m_{\mathrm{long}}^2 < 0$ (by (7.1)), the critical points are $E = 0$ and $E = \pm\sqrt{-m_{\mathrm{long}}^2/\lambda} = \pm|\langle E\rangle|$. The point $E = 0$ is a local maximum because $V''(0) = m_{\mathrm{long}}^2 < 0$. The non-zero critical points are the two minima. The condensate $\langle E\rangle \neq 0$ is the non-trivial minimum. The curvature there is

$$V''(\langle E\rangle) = m_{\mathrm{long}}^2 + 3\lambda\langle E\rangle^2 = -2m_{\mathrm{long}}^2 = 2\lambda\langle E\rangle^2 = m_\delta^2.$$

The condensation is therefore a consequence of the derived parameters λ and $\langle E\rangle$. □

Theorem 7.6 (Positive spectral lower bound)**.** *Let m_δ be the derived parameter of Theorem 7.5. Then every spectral mass satisfies*

$$M_l^2 = \varepsilon_l + m_\delta^2 \geq m_\delta^2 > 0.$$

No massless mode occurs in the spectrum. The physical structure of Definitions 5.3–5.7 carries the lower bound $m_\delta > 0$.

Proof. At the level of the spectral data, every spectral mass satisfies $M_l \geq m_\delta > 0$ by Theorem 7.4. No massless mode occurs in the spectrum. At the operator level the lower bound is explicit on the free sector through the Fock spectrum of the mass tower (Theorem C.2). On the interacting sector it is defined by the effective construction of Theorem 7.8. The parameter m_δ is derived from the spectral datum through Theorem 7.5. This theorem records the propagation of the lower bound from the spectral datum to the physical structure. □

Theorems 7.3–7.6 record the analytic part of the mapping. They establish the analytic control, the positivity, the condensation, and the positive lower bound of the physical quantities. Their input is the spectral datum of Definition 5.1. Together with the free-sector mapping of Theorem C.2, they constitute the body of the mapping statement of Theorem 7.1.

7.2 Structural mapping: two-point and composite quantities

The structural mapping records the free-sector instance of Definition 5.8. It shows that the spectral sums of Definition 5.2 are traces of the reconstructed propagator. The input is the free sector of the recovery theorem (Theorem C.2), namely the Fock space $\mathcal{H} = \mathcal{F}(h)$ over the mass tower $\{M_l\}$ with field operators $\varphi(f \otimes g)$, where $f \in \mathcal{S}(\mathbb{R}^4)$ and $g \in C^\infty(M)$.

Theorem 7.7 (Mapping of two-point and composite quantities)**.** *Let the free sector of the recovery theorem be given. Then the following hold.*

(a) The vacuum two-point function of the field operators is the full spectral kernel. For internal test functions g, g',

$$\langle \Omega, \varphi(f \otimes g)\,\varphi(f' \otimes g')\Omega\rangle = \sum_l (g, K_l g')_{L^2}\,\langle \Omega, \varphi_l(f)\,\varphi_l(f')\Omega\rangle,$$

where K_l is the spectral projector kernel of the l-th eigenspace and φ_l is the mode field of mass M_l.

(b) *The spectral sum*

$$Q_p(m^2) = \sum_l d_l(\varepsilon_l + m^2)^{-p}$$

with $p > 3/2$ is the trace of the p-th power of the internal Green operator,

$$Q_p(m^2) = \mathrm{Tr}_{L^2(M)}[(\Delta_M + m^2)^{-p}] = \sum_l \frac{d_l}{(\varepsilon_l + m^2)^p}.$$

For $p = 2$ the trace is the integral of the square of the internal Green function,

$$Q_2(m^2) = \iint_{M\times M} G_{m^2}(y, y')^2\, dy\, dy',$$

where

$$G_{m^2}(y, y') = \sum_l \frac{K_l(y, y')}{\varepsilon_l + m^2}.$$

This is the zero-momentum internal trace of the full propagator of the recovery theorem. It is not the vacuum expectation of the four-dimensional normal-ordered field $:\varphi^2:$, whose expectation involves the square of the full propagator over all momenta and is not the quantity Q_2.

(c) *Every physical quantity of Definition 5.2 with $f(\lambda) = (\lambda + m^2)^{-p}$ and $p > 3/2$ is therefore a trace of the spectral data. It is linked to the field structure of the free sector through the zero-momentum component of the full spectral kernel.*

Proof. (a) This is the standard Fock two-point function. The modes φ_l are independent free fields of mass M_l by Theorem C.2, and the internal projector kernels implement the orthogonality of the modes. (b) By the spectral decomposition of the internal Green function and the orthogonality of the projector kernels,

$$\int_M K_l K_{l'} = \delta_{ll'}\, d_l,$$

one obtains

$$\iint_{M\times M} G_{m^2}^2 = \sum_{l,l'} \frac{\int_M K_l K_{l'}}{(\varepsilon_l + m^2)(\varepsilon_{l'} + m^2)} = \sum_l \frac{d_l}{(\varepsilon_l + m^2)^2} = Q_2(m^2).$$

The general p is the trace of the p-th power,

$$\mathrm{Tr}[(\Delta_M + m^2)^{-p}] = \sum_l \frac{d_l}{(\varepsilon_l + m^2)^p}.$$

The link to the reconstructed field structure is through the zero-momentum component of the full spectral kernel. The full propagator of the recovery theorem is $G_{\mathrm{full}}((x, y), (x', y')) = \sum_l K_l(y, y')\, G_{M_l}(x - x')$ with $M_l^2 = \varepsilon_l + m_\delta^2$, and its zero-momentum component is

$$\int_{\mathbb{R}^4} G_{\mathrm{full}}((x, y), (0, y'))\, dx = \sum_l \frac{K_l(y, y')}{M_l^2} = G_{m_\delta^2}(y, y').$$

Thus the resolvent kernel G_{m^2} is the zero-momentum component of the reconstructed propagator at the physical value $m^2 = m_\delta^2$; for general m^2 it is the auxiliary analytic continuation used for the convergence control of Theorem 7.3. The quantity Q_p is the internal trace of the reconstructed propagator, and it is not a vacuum expectation of the four-dimensional fields. (c) Each Q_p is a spectral-data trace by (b), absolutely convergent by Theorem 7.3. The correspondence to the reconstructed field structure is through the zero-momentum component of the full spectral kernel. □

The correspondence between the spectral sums and the field-theoretic quantities is summarised as follows. The structural mapping of Definition 5.8 is thereby recorded on the

Table 4. Spectral-data realisation of the free-sector quantities

Quantity	Spectral-data realisation	Status
$Q_1(m^2) = \sum_l d_l(\varepsilon_l + m^2)^{-1}$	multiplicity diagonal of the two-point function (divergent; the well-defined object is the full spectral kernel)	Theorem C.2
$Q_2(m^2) = \sum_l d_l(\varepsilon_l + m^2)^{-2}$	$\mathrm{Tr}[(\Delta_M + m^2)^{-2}] = \iint G_{m^2}^2$ (internal Green trace)	Theorem 7.7
Q_p, $p > 3/2$	$\mathrm{Tr}[(\Delta_M + m^2)^{-p}]$, the p-th power of the internal Green operator	Theorem 7.7

free sector. The spectral sums of Definition 5.2 are spectral-data traces of the reconstructed propagator, specifically its zero-momentum internal components. They are not vacuum expectations of the four-dimensional fields.

7.3 Effective Hilbert space of the condensed sector

The free-sector mapping of Theorem 7.7 records the free content of Definition 5.8. The present subsection defines the kinematic carrier of the interacting content within the formulation. The condensed sector defines an effective theory whose parameters are closed functions of the spectral data. Its Hilbert space is the Fock space of the fluctuation field of mass m_δ. The construction is the effective-level instance of the mapping condition. It is not a standard-formulation continuum limit.

Theorem 7.8 (Effective Hilbert space of the condensed sector)**.** *Let λ, $\langle E\rangle$, and m_δ be the derived parameters of Section 8. Define the effective field*

$$e = E - \langle E\rangle.$$

Then the following hold.

(a) The condensate is non-vanishing, and the fluctuation mass is the curvature of the effective potential at its minimum,

$$m_\delta^2 = V''(\langle E\rangle) = 2\lambda\langle E\rangle^2.$$

Both are closed expressions in the spectral data.

(b) *The effective Hilbert space of the condensed sector is the Fock space $\mathcal{H}_{\mathit{eff}} = \mathcal{F}(h_\delta)$ of the fluctuation field e of mass m_δ. The single-particle space is*

$$h_\delta = L^2\left(\mathbb{R}^3, \frac{d^3p}{2\omega_\delta}\right),$$

with

$$\omega_\delta = \sqrt{p^2 + m_\delta^2}.$$

The free Hamiltonian is the Fock operator $d\Gamma(\omega_\delta)$, which is a strict operator. The interaction term λe^4 is an effective object of the formulation. Its status is given by the mapping dictionary and the spectral-sum amplitudes of Theorem 7.15, Proposition 9.2 and Theorem 9.5. It is not a strict Fock-space operator. The parameters of the effective theory are all closed functions of the spectral data.

(c) *The effective theory carries a positive lower bound $m_\delta > 0$. The spectrum of $d\Gamma(\omega_\delta)$ has no weight below m_δ above the vacuum.*

Proof. (a) The Landau form has its unique non-trivial minimum at

$$E_*^2 = -\frac{m_{\mathrm{long}}^2}{\lambda} = \langle E \rangle^2 > 0$$

by Theorem 7.5. The curvature there is m_δ^2 by (7.3). (b) The effective field e is a scalar of mass m_δ. Its Fock space is constructed by standard second quantisation from the spectral data. The single-particle space is determined by m_δ. The quartic term is the effective interaction with the spectral-data coupling λ. (c) The single-particle spectrum of $d\Gamma(\omega_\delta)$ starts at $m_\delta > 0$. The vacuum has no weight below m_δ. □

Proposition 7.9 (Locality of the effective sector)**.** *The fluctuation field e of the effective Hilbert space satisfies the locality condition of Definition 5.7. For test functions f, g with spacelike-separated supports,*

$$[e(f), e(g)] = 0.$$

Proof. The effective field e is a free scalar field of mass m_δ on its Fock space. The commutator of a free scalar field is

$$[e(x), e(y)] = i\Delta(x - y),$$

where Δ is the Pauli–Jordan function, which vanishes for spacelike separations. Hence the locality condition holds on the effective sector. □

Remark 7.10. The field structure of the free sector and the effective sector satisfies the locality condition. The spectral-sum quantities of Definition 5.2 are global functions of the spectral data. They are not local operators, so spacelike commutation is not defined for them. Locality is a property of the field structure, not of the numerical spectral sums. In particular, every field that carries a pointwise dependence on the reconstructed spacetime is either a free-sector field or the effective fluctuation field, and both satisfy the spacelike commutativity of Definition 5.7 (Theorem 6.8 and Proposition 7.9). The interacting content introduces no further pointwise fields, so no further locality condition arises; the boundary of the standard formulation for that content is the triviality theorem of Section 10.

7.4 Thermal states

The temperature β is an external physical condition. On the free sector the thermal states are the standard KMS states, which are Gaussian and determined by the temperature-parameterised spectral functions.

Theorem 7.11 (KMS states of the free sector)**.** *Let the free sector of the recovery theorem be given. For every inverse temperature $\beta > 0$ there exists a unique quasi-free KMS state ω_β in the Gaussian class on the field algebra. Its two-point function is the per-mode thermal propagator*

$$\langle \varphi(f \otimes g)\, \varphi(f' \otimes g') \rangle_{\omega_\beta} = \sum_l (g, K_l g') \iint dx\, dx'\, f(x)\, f'(x')\, G_\beta^{(M_l)}(x - x'),$$

where

$$G_\beta^{(M)}(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_M(p)} \Big[(1 + n_\beta(\omega_M(p)))e^{-i(\omega_M(p)x^0 - \mathbf{p}\cdot\mathbf{x})} + n_\beta(\omega_M(p))e^{+i(\omega_M(p)x^0 - \mathbf{p}\cdot\mathbf{x})}\Big], \quad (7.4)$$

with $\omega_M(p) = \sqrt{\mathbf{p}^2 + M^2}$ and the Bose occupation number

$$n_\beta(\omega) = (e^{\beta\omega} - 1)^{-1}.$$

This is the standard thermal two-point function. The state satisfies the KMS condition

$$\omega_\beta(A\, \varphi_t(B)) = \omega_\beta(\varphi_{t+i\beta}(B)\, A)$$

for all polynomials A, B in the fields. The thermal expectation of the composite quantities of Theorem 7.7 is the temperature-parameterised spectral sum

$$Q_p(\beta, m^2) = \sum_l d_l\, T_p(\beta, M_l),$$

where T_p is the thermal spectral function.

Proof. The Gaussian state with the thermal two-point function is the standard KMS state of the free field. The two-point function satisfies the KMS condition in the distributional sense. The higher correlations are Wick products of the two-point function. The KMS condition for polynomials follows from the two-point case. Uniqueness holds within the Gaussian class, whose two-point function is fixed by the KMS condition. The state is determined by the spectral data of the mass tower $\{M_l\}$ and the temperature β. □

Theorem 7.12 (KMS states of the effective sector)**.** *Let the effective sector of Theorem 7.8 be given. For every inverse temperature $\beta > 0$ there exists a unique quasi-free KMS state of the free part, whose two-point function is the thermal propagator $G_\beta^{(m_\delta)}$ of mass m_δ. The correction by the interaction is the perturbative series of the effective theory. The n-th order correction is a function of the spectral data λ and m_δ, computed with the thermal Feynman rules and the dispersion subtraction of Theorem 9.3.*

Proof. The free part of the effective Hamiltonian is the Fock operator of the field e of mass m_δ. By the construction of Theorem 7.11 there is a unique quasi-free KMS state with the thermal propagator $G_\beta^{(m_\delta)}$. The interaction correction is the perturbative thermal series of the effective theory. At each order the thermal amplitudes are integrals of thermal propagators and the vertex λ. The vertex λ is a spectral sum by Theorem 7.3. Each order is therefore a definite function of the spectral data. □

Theorem 7.13 (Asymptotic states of the effective sector)**.** *The asymptotic states of the effective sector are the Fock multi-particle states of the free field e. The scattering amplitudes of Theorem 9.3 are the transition amplitudes between these asymptotic states.*

Proof. The effective Hilbert space is the Fock space of the free field e, by Theorem 7.8. The asymptotic states are therefore the Fock multi-particle states. The scattering amplitudes of Theorem 9.3 are the transition amplitudes between these states, computed by the Feynman rules of the effective theory with the spectral-data parameters λ and m_δ. □

The definition of scattering within the formulation uses the spectral pole criterion together with the effective amplitudes. The isolated poles of the two-point spectral functions correspond to physical single-particle states within the formulation. In the Källén–Lehmann representation the atomic part of the spectral measure is the single-particle spectrum, as in Theorem 7.4. On the free sector the single-particle states are the Fock one-particle states of the mass tower, and the S-matrix of the free theory is trivial. The interacting S-matrix is defined by the amplitudes of the effective theory, which are explicit functions of the spectral data. The pole criterion identifies the particles of the theory from the spectral data.

7.5 Scattering

Scattering within the formulation is defined from the same spectral data as the other physical quantities. The particles are identified by the single-particle states of the spectral pole criterion. The scattering amplitudes are the perturbative amplitudes of the effective theory of Theorem 7.8. Their parameters, the mass m_δ and the coupling λ, are closed functions of the spectral data. The amplitudes are therefore explicit functions of the spectral data.

Theorem 7.14 (Particle spectrum from spectral poles)**.** *Let the two-point spectral function have an isolated pole at $p^2 = M_l^2$. Then the spectral measure has an atom at M_l, corresponding to a single-particle state of mass M_l within the formulation. On the free sector the single-particle states are the Fock one-particle states of the mass tower $\{M_l\}$ by Theorem C.2. On the effective sector the single particle is the fluctuation quantum of mass m_δ, corresponding to the pole of the e-field propagator at $p^2 = m_\delta^2$.*

Proof. In the Källén–Lehmann representation of the two-point function, the atomic part of the spectral measure is the single-particle spectrum. An isolated pole at $p^2 = M_l^2$ is an atom of the spectral measure at M_l. On the free sector this atom is the mass of the l-th Fock mode by Theorem C.2. On the effective sector the propagator of the free field e has a single pole at $p^2 = m_\delta^2$, which identifies the single particle of the condensed sector. □

Theorem 7.15 (Tree-level amplitude of the effective sector)**.** *Let the effective theory of Theorem 7.8 be written as*

$$\mathcal{L}_{\mathit{eff}} = \frac{1}{2}(\partial e)^2 + \frac{1}{2}m_\delta^2 e^2 + \frac{\lambda}{4}e^4.$$

The parameters m_δ and λ are closed functions of the spectral data by Theorems 7.3 and 7.6. The tree-level $2 \to 2$ scattering amplitude of the effective quanta is

$$\mathcal{M}(s,t) = -6\lambda.$$

The differential cross-section is

$$\frac{d\sigma}{d\Omega} = \frac{9\lambda^2}{16\pi^2 s},$$

where s is the Mandelstam variable.

Proof. The Feynman rules of the effective theory are standard. The propagator is

$$\frac{i}{p^2 - m_\delta^2 + i\varepsilon},$$

and the e^4 vertex is $-6i\lambda$ (the quartic coupling is normalised as in the Landau potential $V = (\lambda/4)E^4$ of Theorem 8.32, so the vertex carries the combinatorial factor 4! of the four fields). At tree level the $2 \to 2$ amplitude receives only the contact contribution $-6i\lambda$. There is no s-, t-, or u-channel exchange because the effective theory is stipulated quartic, with no e^3 term. Hence $\mathcal{M} = -6\lambda$. The cross-section for equal-mass $2 \to 2$ scattering with a contact interaction is

$$\frac{d\sigma}{d\Omega} = \frac{|\mathcal{M}|^2}{64\pi^2 s} = \frac{9\lambda^2}{16\pi^2 s},$$

which is angular-independent. The coupling λ is a convergent spectral sum of the spectral data by Proposition 8.31 and Theorem 7.3. The amplitude is therefore an explicit function of the spectral data. □

The full correspondence collected in this section is summarised as follows. The interface column records the three classes (C1)–(C3) of Section 8 through which each object is attached to the spectral datum.

8 Construction of the gauge field

Section 4 identified the gauge field A_μ as a connection on the four-dimensional spacetime V, realised by Kaluza–Klein zero modes. Its algebraic structure was given in Proposition 4.21. The present section develops the dynamical description of this gauge field within the spectral-sum definition.

The constructions of Section 4 produced the gauge algebra, the internal space, the spacetime dimension, and the Gaussian measure. The present section assembles these building blocks into a quantum field theory. The assembly is carried out within the spectral-sum definition of Definition 8.1. All subsequent statements are made within this definition.

Table 5. Summary of the spectral-to-field-theoretic correspondence

spectral datum object	physical structure	interface	scope
$\{\varepsilon_l, d_l, m_\delta\}$	mass tower $\{M_l\}$	spectral sums (Definition 5.2)	whole construction
$\lambda = \kappa \sum_l d_l(\varepsilon_l + \xi R)^{-2}$	quartic vertex	spectral sum; polygamma closed form (Theorem 8.39)	interacting sector
$\langle E\rangle^2 = \frac{8}{3}(R_c - R)/\lambda$, $m_\delta^2 = 2\lambda\langle E\rangle^2$	condensate, curvature, squared masses	closed forms (Theorem 8.32, Lemma 8.2)	interacting sector
full spectral kernel G_{full}	two-point function	Osterwalder–Schrader reconstruction (Appendix C, Theorem 10.5)	free sector
$Q_p(m^2) = \sum_l d_l(\varepsilon_l + m^2)^{-p}$	trace powers of the internal Green operator	spectral sums (Theorem 7.7)	whole construction
$\mathcal{M}(s,t) = -6\lambda$	tree $2 \to 2$ amplitude of the effective quanta	effective theory (Theorem 7.15)	effective sector

Definition 8.1 (Spectral-sum definition)**.** *The spectral-sum definition of this paper is the evaluation scheme defined as follows.*

(i) The basic datum is the spectral data of the emergent internal space RP^3, namely the scalar Laplacian spectrum $\varepsilon_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ for even l, together with the scale flow $\{\Phi_\sigma\}$ defined in Section 8.4 (Theorem 8.18).

(ii) A physical quantity is an expression of the form

$$Q = \sum_{l\ \text{even}} d_l \frac{f(\varepsilon_l)}{(\varepsilon_l + m^2)^p}, \qquad p > 3/2,$$

where f is determined by the quantity in question, or a closed analytic expression obtained from the scale flow.

(iii) The value of Q is its endpoint evaluation at σ_C. By Lemma 8.19 the series converges absolutely, and the closed forms of Theorem 8.39 are explicit.

(iv) The spectral-sum-definition analogue of the Osterwalder–Schrader axioms is defined as follows. Regularity corresponds to the absolute convergence of the spectral sums. Euclidean invariance corresponds to the $SO(4)$-invariance of the spectral data under the scale flow, together with the $E(4)$-invariance of the Gaussian part. Reflection positivity corresponds to the reflection positivity of the lattice-regularised measure, conditionally on the existence of the weak limit; the weak limit of the gauge-field measure is the open boundary recorded in Section 10. Symmetry corresponds to the Euclidean covariance of the action. Clustering corresponds to the exponential decay implied by the positive lower bound m_δ. The spectral-sum-definition analogue of the Wightman axioms is understood similarly.

(v) *The relation between the spectral-sum definition and the standard measure-theoretic formulation is analysed in Section 10. The free part has a standard-formulation construction, given there. For the interacting scalar content, the boundary of the standard continuum formulation is the triviality of the four-dimensional elementary scalar models, recorded in Section 10.*

These conditions are analogues of the standard axioms, not equivalent statements. On the free sector they are the standard Osterwalder–Schrader axioms, and the reconstruction of Appendix C yields a genuine quantum field theory; on the interacting sector they define the field-theoretic structure of the spectral-sum definition.

Lemma 8.2 (Closure of physical quantities)**.** *Let Q_1, Q_2 be spectral sums of the form of Definition 8.1(ii), convergent at the flow endpoint. Then every finite algebraic combination of Q_1, Q_2 built from addition, multiplication, division by nonzero quantities, and the standard elementary functions (polynomials, exponentials, logarithms, polygamma functions) is a physical quantity of the spectral-sum formulation, provided it is finite and real at the endpoint σ_C.*

Proof. Definition 5.2 and Definition 8.1(ii) take as physical quantities the convergent spectral sums and the closed analytic expressions obtained from the scale flow. The listed operations preserve closed analytic form and finiteness at σ_C; the lemma records this closure explicitly. The endpoint evaluation of Theorem 8.39 applies to the resulting combination. □

The term "physical quantity" of Definition 5.2 and Definition 8.1 falls into three classes, distinguished by how the quantity is obtained from the spectral datum.

(C1) *Spectral sums.* Direct convergent sums over $\{\varepsilon_l, d_l\}$: the quartic coupling $\lambda = \kappa \sum_l d_l(\varepsilon_l + \xi R)^{-2}$ of Theorem 8.39, and the trace powers $Q_p(m^2)$ of Theorem 7.7.

(C2) *Closed forms.* Finite algebraic combinations of spectral sums built from the operations of Lemma 8.2, obtained from the scale flow: the condensate $\langle E\rangle^2 = \frac{8}{3}(R_c - R)/\lambda$ and the squared masses $m_\delta^2 = 2\lambda\langle E\rangle^2$ of Theorem 8.32.

(C3) *Field-theoretic quantities.* Operators and correlation functions obtained by the free-sector reconstruction (Appendix C) and by the effective-sector construction (Theorem 7.8): the two-point function, and the tree amplitude $\mathcal{M}(s,t) = -6\lambda$ of Theorem 7.15.

Each class has its own interface to the spectral datum. A quantity is a physical quantity of the formulation in exactly one of these senses, and the sense in which it is used is always indicated.

Remark 8.3 (Terminology)*.* Throughout this paper, the term "spectral-sum definition" refers to the evaluation scheme in which physical quantities are defined as convergent spectral sums over a discrete spectrum, rather than as expectations with respect to a measure. This terminology is used only to name the rule stated in Definition 5.2; it does not assert an equivalence with any other formulation.

The objects appearing in this definition, namely the scale flow Φ_σ, the endpoint σ_C, and the spectral sums, are constructed in the following subsections.

8.1 Gauge field and scale semigroup

The scale field σ was introduced in Section 4.7 as a coarse-graining scale. Its statistics are governed by the unbiased Gaussian measure. A natural evolution of the scale parameter σ is generated by the heat-kernel smoothing semigroup. Let

$$L_V = -\sum_{\mu=1}^{4} \partial_\mu^2 \geq 0$$

be the non-negative Laplacian on the emergent spacetime $V = \mathbb{R}^4$ (the spacetime factor, to be distinguished from the internal Laplacian Δ_{RP^3} of Section 4), and let

$$t = \frac{\sigma^2}{2}$$

be the additive scale parameter. The heat kernel is then

$$C_t = e^{-tL_V},$$

and $\{C_t\}_{t>0}$ satisfies the additive semigroup law $C_{t_1}C_{t_2} = C_{t_1+t_2}$ in t (not in σ). This semigroup decomposes a field into momentum modes and defines the effective degrees of freedom at each scale.

Definition 8.4 (Slow mode)**.** *Let $\{C_t\}_{t>0}$ with $t = \sigma^2/2$ be the Gaussian heat semigroup of the scale flow, and let*

$$\Pi_\sigma = \mathbf{1}_{[0,\sigma^{-2}]}(L_V)$$

be the spectral projection onto $[0, \sigma^{-2}]$ of the non-negative Laplacian L_V. A field ϕ is called a slow mode *at scale σ if $\phi = \Pi_\sigma \phi$, that is, if its momentum support lies in the region $|k| \lesssim 1/\sigma$. The sharp projection Π_σ only designates the slow-mode sector; the actual smoothing is the exponential suppression of the heat semigroup C_t.*

The gauge field A_μ of Proposition 4.21 is reduced from the connection on RP^3 to the spacetime V by Kaluza–Klein expansion. Its internal space components are carried by zero modes whose momentum support is concentrated at the low end. The gauge field is therefore automatically a slow mode under the scale semigroup. The statistics of the gauge field are jointly determined by the unbiased Gaussian measure of clause (A2) and the spectral representation of the Gaussian structure.

8.2 Measure construction

As a quantum field, the statistical properties of the gauge field are described by a probability measure on the space of field configurations. Clause (A2) supplies a Gaussian measure for the free fields. The measure for the gauge fields must incorporate the F^2 nonlinear self-interaction, whose form is dictated by the emergent gauge algebra.

Theorem 8.5 (Interaction from the group structure)**.** *Let A_μ be the four-dimensional gauge field emerging from Principle 3.4 via Proposition 4.21. Its construction is given by the Kaluza–Klein mechanism. The connection on the principal $G_{SM} = SU(3) \times SU(2) \times U(1)$ bundle over RP^3 is expanded in zero modes. Their coefficients are the gauge fields $A_\mu(x)$ on the four-dimensional spacetime V. The colour part is carried by the eight zero modes of the colour algebra $\mathfrak{su}(3)$. The electroweak part is carried by the four isometric Killing zero modes of the retained gauge symmetries $SU(2)_L \times U(1)_R \subset SO(4)$. This gives twelve gauge bosons. The field strength of A_μ is*

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu,$$

where the f^{abc} are the structure constants of $\mathfrak{g}_{SM}$. The expansion of F^2 yields three-gluon vertices proportional to gf^{abc} and four-gluon vertices proportional to $g^2 f^{abe} f^{cde}$. The chiral fermions of clause (A1), forming the Weyl pair $(1/2, 0) \oplus (0, 1/2)$, couple to the gauge field through the covariant derivative

$$D_\mu \psi = (\partial_\mu + g A^a_\mu T^a)\psi.$$

The factor $SU(2)_L$ acts on the left-handed component, and $U(1)_Y$ acts with different weights on the two chiral components.

Proof. The curvature formula $F = dA + g[A, A]$ is a standard consequence of principal bundle geometry. The non-Abelian character $[A_\mu, A_\nu] \neq 0$ follows from the non-vanishing structure constants $f^{abc} \neq 0$ of the non-Abelian factors $\mathfrak{su}(3)$ and $\mathfrak{su}(2)$. The three-gluon and four-gluon vertices are algebraic consequences of the expansion of $\mathrm{Tr}\, F^2$. The fermion–gauge coupling is given by the minimal coupling through the covariant derivative, together with the chiral content (A1). □

Proposition 8.6 (Measure specification)**.** *Let A_μ be the gauge field emerging from Principle 3.4 via Proposition 4.21. Its Euclidean measure is defined after emergence and consists of two parts.*

For the matter fields, the base space is the continuous Euclidean space V. The matter fields are the chiral fermions of clause (A1). Their free measure $d\mu_{mat}$ is taken as a Grassmann Gaussian measure. The free covariance is given by the spectral representation of the Gaussian structure. The Berezin integral is well defined in finite volume, and the infinite-volume limit is realised through a quasi-free state construction.

For the gauge fields, the field strength F contains nonlinear self-interactions, and its measure is non-Gaussian. The total measure is defined by the gauge-invariant action

$$S = S_{\text{gauge}}[F] + S_{\text{ferm}}[\psi, A],$$

where S_{gauge} is the $\mathrm{Tr}\, F^2$ term and S_{ferm} is the minimal coupling term. The forms of these terms are determined by the requirement of gauge invariance and the minimal coupling structure. The scale dependence is governed by the scale semigroup. The gauge-field measure is here specified, not constructed: its non-Gaussian part is the open constructive problem recorded in Section 10, and the present proposition fixes its formal (action-level) form only.

Proof. The gauge group G_{SM} is given by Proposition 4.21. The field strength and the action are determined by gauge invariance and minimal coupling, as stated in Theorem 8.5. The measure for the matter fields is supplied by clause (A2) and the spectral representation of the Gaussian structure. $\square$

8.3 Technical prerequisites

Before entering the scale-flow dynamics, this section records the basic technical prerequisites for the measure of the gauge and matter fields. These points concern lattice regularisation, fermion doubling, the chiral determinant, and the $U(1)$ sector. They are independent of the dynamical analysis in the subsequent sections.

The gauge-field measure contains the nonlinear F^2 term. In continuous form it is a formal path integral. A standard way to define the gauge-field measure is lattice regularisation, with the lattice spacing a serving purely as a regularisation parameter. The emergent geometry is continuous, and no finite spacing acts as a physical minimal length. A hypercubic lattice breaks $SO(4)$ to its hyperoctahedral subgroup at finite spacing. The full $SO(4)$ invariance of the emergent geometry is recovered in the continuum limit.

The lattice action is taken as the standard plaquette action for the gauge theory with group G_{SM}, with link variables $U_p \in G_{\text{SM}}$. The plaquette expansion gives

$$S_{\text{plaq}} = \frac{a^4}{4g^2} \int \operatorname{Tr} F^2 + O(a^6)$$

for any compact Lie group G. At the formal level the continuum limit recovers the Lie-algebra structure $SU(3) \times SU(2) \times U(1)$ of Proposition 4.21. The lattice gauge group coincides with the emergent gauge group.

In the spectral-sum definition the free part of the measure is defined directly by the maximum-entropy principle of clause (A2) together with the spectral representation of the Gaussian structure. No independent regulator cutoff enters the free measure. The parameter σ is a physical coarse-graining length, while the sharp projection Π_σ only designates the slow-mode sector (Definition 8.4). The interaction is subsumed into the scale-flow evolution, and physical quantities are given by spectral sums at the flow endpoint. The lattice regularisation described above serves only as a technical device for establishing reflection positivity in the gauge sector. It does not belong to the definition of the measure itself.

Proposition 8.7 (No doubling)**.** *The formulation does not produce a fermion doubling problem.*

Proof. The Nielsen–Ninomiya theorem [38, 39] constrains local, translation-invariant, Hermitian, chiral fermion actions on a lattice. Such actions necessarily produce doublers. The matter-field measure in Proposition 8.6 is a continuous Grassmann Gaussian measure. It does not introduce a lattice fermion action. The hypothesis of the Nielsen–Ninomiya theorem is therefore not satisfied. The fermion content is fixed at the operator level by the representation theory of clause (A1) and the spectral data on RP^3. $\square$

Proposition 8.8 (Well-definedness of the chiral determinant)**.** *Let D_+ be the Weyl operator coupled to the gauge field A. Its determinant* $\det D_+$ *is well defined.*

Proof. One has

$$|\det D_+|^2 = \det(D_+^\dagger D_+),$$

where $D_+^\dagger D_+$ is a self-adjoint positive elliptic operator with a discrete non-negative spectrum. Its spectral zeta function

$$\zeta(s) = \sum_j \varepsilon_j^{-s}$$

admits a meromorphic continuation, and the determinant is defined by

$$\det = \exp(-\zeta'(0))$$

after regularisation. On the compact internal space this is the standard zeta-function construction. On the non-compact base V the determinant is regularised within the spectral-sum definition of Definition 8.1.

The phase of $\det D_+$ is well defined if and only if all gauge anomaly coefficients vanish. The fermion content used for this verification is the standard model content, one generation Q_L, u_R, d_R, L_L, e_R with the standard hypercharge assignment $Y = (1/6, 2/3, -1/3, -1/2, -1)$. This content is the minimal anomaly-free Yukawa-complete chiral completion of Definition 4.13(S2); the present proposition verifies its consistency term by term.

One verifies term by term. The $SU(3)^2U(1)$ anomaly vanishes, $2Y_Q - Y_u - Y_d = 0$. The $SU(2)^2U(1)$ anomaly vanishes, $3Y_Q + Y_L = 0$. The $U(1)^3$ anomaly vanishes, $\sum Y^3 = 0$. The gravitational $U(1)$ anomaly vanishes, $\sum Y = 0$. The $SU(3)^3$ anomaly vanishes because the colour representation is vector-like. The Witten $SU(2)$ anomaly is avoided because the number of doublet states is four, which is even.

In the sums $\sum Y^3$ and $\sum Y$ all fermion fields are counted as left-handed Weyl components. The right-handed fields enter with the opposite hypercharges. The cancellation holds per generation, independently of the total number of generations. All anomaly coefficients of the standard model content are zero. The Weyl determinant is therefore well defined for this content, and the chiral fermion measure exists. No assertion is made here about anomaly freedom for any additional fermion content. □

Remark 8.9 (Status of the fermion content). The hypercharges of Section 4.4 follow from anomaly cancellation and the Yukawa structure together with the neutral-Higgs vacuum condition of Definition 4.13(S3). The representation content is the minimal anomaly-free Yukawa-complete chiral completion of Definition 4.13(S2), compatible with the chiral clause (A1) and the colour vectoriality recorded there. Proposition 8.8 verifies the anomaly freedom of this content term by term.

Remark 8.10 (The $U(1)$ sector). Triviality of four-dimensional pure $U(1)$ gauge theory is at present only a Landau-pole conjecture. Rigorous triviality theorems exist only for ϕ_4^4 theory and do not apply to $U(1)$ gauge theory. In this paper the scale evolution is defined on a finite interval, and the $U(1)_Y$ coupling does not encounter a Landau pole within that interval. The statements of the paper concerning the positive lower bound in the colour-singlet sector and the spectral-sum-definition analogue of the Wightman axioms do not depend on the non-triviality of the $U(1)$ sector in the continuum limit.

8.4 Scale flow and consistent family

When the scale parameter σ varies continuously, the effective theory evolves accordingly. This evolution is generated by the scale flow. The present section develops the mathematical description of this evolution and records the consistency and analytic properties of the scale family.

Lemma 8.11 (β-function)**.** *Let V_{eff} be the effective potential and let $k = 1/\sigma$ be the scale parameter. Then*

$$V_{\text{eff}}(k) = -\frac{C}{2}\big(k^4 - M_P^4\big), \qquad \beta(\sigma) = -\frac{dV_{\text{eff}}}{d\sigma} = -2C\sigma^{-5}, \qquad C < 0.$$

The reference scale M_P is the Planck-scale boundary value of the scale parameter in the present formulation. Its numerical value is not used in this paper.

Proof. The renormalisation group source is a pure power law $\text{source}(k) = Ck^4$. Its form is fixed by dimensional analysis and homogeneity. The unbiased structure of clause (A2) forces the scale-dependent part of the effective potential to be a single power proportional to k^4. The ultraviolet divergence structure of the coefficient C is supplied by the heat-kernel expansion of the spectral sums on the internal space [3, 4], with leading Seeley–DeWitt coefficient $a_0 \propto \operatorname{Vol}\operatorname{Str}(1)$. Hence $C \propto a_0 \propto \operatorname{Str}(1)$. The scale derivative is $dV/dk = -2\,\text{source}(k)/k = -2Ck^3$. With the boundary condition $V_{\text{eff}}(M_P) = 0$, integration gives

$$V_{\text{eff}}(k) = -\frac{C}{2}\big(k^4 - M_P^4\big).$$

From $d\sigma/dk = -1/k^2$ one obtains

$$\beta(\sigma) = -\frac{dV}{dk}\frac{dk}{d\sigma} = -2C\sigma^{-5}.$$

The sign $C < 0$ follows from fermion dominance in the spectral representation. A sufficient condition is given in Proposition 8.12. The scope of this β-function is the free Gaussian part within the local potential approximation, as discussed in Remark 8.16. $\square$

Proposition 8.12 (Condition for $C < 0$)**.** *Let C be the source coefficient of Lemma 8.11,*

$$C \propto \operatorname{Str}(1) = \sum_b \operatorname{Tr} 1 - \sum_f \operatorname{Tr} 1.$$

For the emergent content of this paper, with n_g generations, 15 Weyl fermions per generation, 12 gauge bosons, the four-dimensional spin-2 sector, and n_s scalars, one has

$$C < 0 \iff 30n_g > 29 + n_s.$$

The condition holds for $n_g \geq 2$. The scalar content satisfies $n_s \lesssim 10$, so $C < 0$ is satisfied.

Proof. The homogeneity of the spectral representation implies $\text{source}(k) \propto k^4\operatorname{Str}(1)$. The fermionic contribution is $-15n_g\cdot 2 = -30n_g$, and the bosonic contribution is $12\cdot 2 + 5 + n_s = 29 + n_s$. The weights count physical field degrees of freedom in the heat-kernel trace.

Two transverse polarisations per gauge boson are counted, with the remaining components removed by the Faddeev–Popov procedure. The transverse-traceless spin-2 field contributes $10 - 4 - 1 = 5$ independent components. Each Weyl fermion contributes weight 2 relative to a real scalar, and each real scalar contributes weight 1. Thus $C < 0$ is equivalent to $\mathrm{Str}(1) < 0$, which is $30n_g > 29 + n_s$. □

Theorem 8.13 (Spin-2 sector)**.** *The spin-2 content of the* Str(1) *counting is the four-dimensional spin-2 sector. The conserved symmetric tensor of the emergent action of Proposition 8.6 has a transverse-traceless part with* $10-4-1 = 5$ *independent components. The weight 5 is therefore the component count of the four-dimensional spin-2 sector. It is distinct from the internal transverse-traceless modes of* RP^3*, which have no zero mode (Theorem 8.37).*

Proof. Translation invariance of the emergent action of Proposition 8.6 gives, by Noether's theorem, a conserved symmetric tensor $T_{\mu\nu}$. Its transverse-traceless part is the spin-2 sector of the formulation. The component count is standard, $10 - 4 - 1 = 5$. □

Proposition 8.14 (Window criterion for retained spinor modes)**.** *Let* $c = (kL)^2$ *denote the window capacity of the scale flow, where* k *is the flow momentum scale and* L *is the internal radius. The coarse graining retains the spinor modes whose internal momentum lies below the inverse radius* $1/L$*, that is, the modes whose dimensionless Dirac eigenvalue* $n + 3/2$ *lies below* c*:*

$$n + \frac{3}{2} < c, \qquad n = 0, 2, 4, \dots .$$

Consequently the number of retained modes is

$$N_{\mathrm{win}}(c) = \#\left\{n \in 2\mathbb{N}_0 : n + \tfrac{3}{2} < c\right\}.$$

The window capacity c *is fixed self-consistently by the spectral sums, not chosen by hand; its value determines how many spinor modes are retained.*

Proof. The Dirac spectrum of the selected spin structure of RP^3 is the tower $\pm(n+3/2)/L$ for $n = 0, 2, 4, \dots$, computed in Appendix A. The coarse graining of the scale flow retains the modes whose wavelength exceeds the internal radius L, which is the condition $n + 3/2 < (kL)^2$. The window capacity $(kL)^2$ is the value at which the spectral sums close. The extrusion order of the retained modes along the scale flow gives a kinematic ordering of the generations. □

The generation count follows from the window capacity rather than from an independent threshold. The window capacity $c = (kL)^2$ is fixed self-consistently by the spectral sums, not chosen by hand. The internal radius is the Gaussian integral width of the scale field, $L = \sqrt{2\pi}\, b$ (Section 4.7, Proposition 8.27); with $k = 1/\sigma$ at the emergence scale $\sigma = b$ this gives the Gaussian benchmark $kL = \sqrt{2\pi}$, hence $c = 2\pi$. The actual closure value of kL is fixed self-consistently by the spectral sums and need not coincide with this benchmark; the criterion $n+\frac{3}{2} < c$ retains exactly the modes $n = 0, 2, 4$ and excludes $n = 6$ for every capacity in the platform interval $(\frac{11}{2}, \frac{15}{2})$ of Proposition 8.15, which contains the benchmark 2π, so $N_{\text{win}} = 3$. The retained modes, ordered by their extrusion order along the scale flow, form the generation ladder used in the Standard Model identification.

Proposition 8.15 (Robustness of the generation count)**.** *The generation count $N_{\text{win}}(c)$ is a platform in c: the value $N_{\text{win}}(c) = 3$ holds for every window capacity in the open interval*

$$\frac{11}{2} < c < \frac{15}{2},$$

that is, for $(kL)^2$ ranging over an interval of width 2 centred at 6.5. The count $n_g = 3$ is therefore insensitive to the precise value of the window capacity: any $(kL)^2 \in (\frac{11}{2}, \frac{15}{2})$ retains exactly the three modes $n = 0, 2, 4$ and excludes $n = 6$.

Proof. The mode $n = 4$ is retained while $n = 6$ is excluded exactly when $4 + \frac{3}{2} < c < 6 + \frac{3}{2}$, that is $\frac{11}{2} < c < \frac{15}{2}$. On this interval the retained modes are $n = 0, 2, 4$ (with $n + \frac{3}{2} = \frac{3}{2}, \frac{7}{2}, \frac{11}{2}$, all below c), while $n = 6$ (with $n + \frac{3}{2} = \frac{15}{2}$) is excluded. Hence $N_{\text{win}}(c) = 3$ identically on an interval of width 2, so the count does not hinge on a finely tuned window capacity. □

Remark 8.16 (Status of the β-function). The β-function of Lemma 8.11 is the derivative of the effective potential along the scale flow, obtained within the local potential approximation. The derivation uses only the homogeneity of clause (A2) and the spectral representation of the Gaussian structure. For a Gaussian measure the heat-kernel smoothing $C_t = e^{-tL_V}$ (with $t = \sigma^2/2$) is an exact conditional expectation, and the scale dependence of the free part of the effective potential can be computed exactly from the spectral zeta function. The β-function therefore governs the evolution of the free Gaussian part of the effective potential. The non-Abelian interaction content of Theorem 8.5 contributes to the effective potential beyond the free part. Its back-reaction on the spectral data is controlled by Theorem 8.17, and its standard-formulation realisation is analysed in Section 10. All statements that use the β-function are statements about the spectral-sum definition (Definition 8.1).

Along the Gaussian heat flow $C_t = e^{-tL_V}$ the differential entropy is non-decreasing (de Bruijn's identity). This records the entropy increase of the effective coarse graining. It is neither the quantum time evolution of Section 6 (the emergent time and the scale parameter are independent directions) nor the maximum-entropy selection of clause (A2), which is a variational statement at fixed moments.

Theorem 8.17 (One-loop stability of the scalar mass)**.** *The interaction back-reaction enters the scale flow through the running of the quartic coupling λ. In the local potential approximation its one-loop β-function is non-negative for $\lambda \geq 0$, with $\beta_\lambda = c\lambda^2$ and $c > 0$. The analytic positivity $\lambda > 0$ of Proposition 8.31 is therefore preserved along the flow. The condensation condition $m^2_{\text{long}} < 0$ for $R < R_c$ and the lower bound $m^2_\delta = 2\lambda\langle E\rangle^2 > 0$ are stable under this back-reaction.*

Proof. Within the local potential approximation the running of the quartic coupling of a scalar with $\lambda \geq 0$ is $\beta_\lambda = c\lambda^2$ with $c > 0$. Hence λ is non-decreasing along the flow, and its positivity, established independently by Proposition 8.31, is preserved at every scale. The condensation threshold and the lower bound formula depend on λ only through its positivity. They are therefore unaffected by the running of λ. $\square$

Theorem 8.18 (Scale evolution and consistent family)**.** *The evolution of the family of effective potentials along the scale flow is driven by $\beta(\sigma) = -2C\sigma^{-5}$ and admits a unique global solution on the interval $\sigma \in [M_P^{-1}, \sigma_C]$, where σ_C is the evaluation endpoint of Definition 8.1, lying in the condensed phase beyond the Gaussian breakdown scale of Section 8.6. The flow is a two-parameter evolution family satisfying the cocycle property*

$$\Phi_{\sigma_3,\sigma_2}\,\Phi_{\sigma_2,\sigma_1} = \Phi_{\sigma_3,\sigma_1}, \qquad M_P^{-1} \le \sigma_1 \le \sigma_2 \le \sigma_3 \le \sigma_C,$$

where Φ_{σ_2,σ_1} carries the effective potential from σ_1 to σ_2. The additive one-parameter semigroup is reserved for the Gaussian heat flow C_t (Definition 8.4); the effective-potential flow is a cocycle, not a semigroup. The flow preserves the $SO(4)$-invariance of the spectral data.

Proof. The function $\beta(\sigma) = -2C\sigma^{-5}$ is analytic and Lipschitz continuous on the interval. The ordinary differential equation $dV_{\text{eff}}/d\sigma = -\beta(\sigma)$ is non-autonomous, so its flow is a two-parameter family. By the Picard–Lindelöf theorem the initial value problem has a unique solution on the interval, and uniqueness gives the cocycle identity $\Phi_{\sigma_3,\sigma_2}\Phi_{\sigma_2,\sigma_1} = \Phi_{\sigma_3,\sigma_1}$ by concatenation of solutions. Because β depends only on $k = 1/\sigma$, it is isotropic, and therefore $SO(4)$-invariance of the initial spectral data is preserved at all scales. $\square$

Lemma 8.19 (Convergence of spectral sums)**.** *The physical quantities, including the effective potential and the quartic coupling λ, are sums over the spectrum of RP^3, with $d_l = (l+1)^2$ and $\varepsilon_l = l(l+2)/L^2$ for even l. For $p > 3/2$ the series $\sum_l d_l(\varepsilon_l + m^2)^{-p}$ converges absolutely. The two-point function is not the multiplicity-only sum. It is the full spectral kernel of Section 10, the resolvent of $L_V + \Delta_{RP^3} + m_\delta^2$, a tempered distribution on $\mathbb{R}^4 \times RP^3$. The multiplicity-only sum, the case $p = 1$, diverges linearly and is the divergent diagonal of the RP^3 Green kernel, not itself a distribution.*

Proof. Since $\varepsilon_l \sim l^2/L^2$ and $d_l \sim l^2$, the general term behaves as $L^{2p}l^{2-2p}$, which converges absolutely when $2p - 2 > 1$, that is, $p > 3/2$. For $p = 1$ the multiplicity sum diverges linearly. It is the diagonal $G(y,y) = \sum_l d_l(\varepsilon_l + m^2)^{-1}$ of the RP^3 Green kernel $G(y,y') = \sum_l K_l(y,y')(\varepsilon_l + m^2)^{-1}$, where K_l is the spectral projector kernel of the l-th eigenspace. The kernel is a distribution with a $1/r$ short-distance singularity, while its diagonal is divergent. The distributional objects used by the construction are the kernel itself and the product-space covariance of Section 10. A positive lower bound on the spectral mass removes possible infrared singularities. $\square$

Lemma 8.20 (Reflection positivity as a closed condition)**.** *If a family of measures $\{\mu_a\}$ satisfies reflection positivity and converges weakly to a measure μ, then μ also satisfies reflection positivity.*

Proof. Weak convergence means $\int G\, d\mu_a \to \int G\, d\mu$ for every bounded continuous G. For reflection positivity one takes $G = F(\theta F)$. For bounded continuous F the non-negativity passes to the limit directly. For the unbounded polynomial functionals of the present construction, the closure is the standard result for measures with uniform moment bounds. □

8.5 Spectral-sum-definition conditions and reconstruction

The Osterwalder–Schrader axioms provide sufficient conditions for a Euclidean quantum field theory to yield a relativistic quantum field theory satisfying the Wightman axioms. The present section records the spectral-sum-definition analogues of these conditions for the measure $d\mu$. The clustering analogue is a consequence of the positive scalar mass established in Section 8.6.

Theorem 8.21 (Reflection positivity). *Let $\theta : \tau \mapsto -\tau$ denote Euclidean time reflection. For the matter measure the associated Schwinger functions satisfy reflection positivity unconditionally:*

$$\mathbb{E}[F(\theta F)] \geq 0$$

for every polynomial F generated by matter fields $\phi(f)$ with $\operatorname{supp} f \subset \{\tau > 0\}$. For the gauge measure, which contains the nonlinear F^2 term and whose continuum construction is the open problem of Section 10, reflection positivity holds at finite lattice spacing and is preserved by any weak limit that exists, by Lemma 8.20; the statement for the gauge sector is therefore conditional on the existence of the weak limit.

Proof. The total measure splits into a matter part and a gauge part. The matter measure is a continuous Grassmann Gaussian measure. Its reflection positivity is a standard result. For the gauge measure, which contains the nonlinear F^2 term, the lattice regularisation described in the technical prerequisites above is used. At finite lattice spacing, reflection positivity for lattice gauge theory is a rigorous theorem. The F^2 and minimal coupling terms are real and positive definite under time reflection. If a continuum limit is taken, reflection positivity is preserved by Lemma 8.20. □

Remark 8.22 (Boundary of reflection positivity). Reflection positivity is established for the regularised gauge-sector proxy: the lattice measure at finite spacing, with closure under weak limits. The spectral-sum interacting content does not constitute a standard Osterwalder–Schrader measure construction.

Theorem 8.23 (Spectral-sum-definition conditions). *The measure $d\mu$ satisfies the spectral-sum-definition analogue of regularity, reflection positivity, and $SO(4)$-invariance at the level of spectral data.*

Proof. Regularity corresponds to the temperedness of the spectral kernel and the analyticity of the β-function of Lemma 8.11. Reflection positivity is Theorem 8.21. Symmetry corresponds to the Euclidean covariance of the matter and gauge field actions. The $SO(4)$-invariance of spectral data follows from Theorem 8.18, which shows that the scale flow preserves the $SO(4)$-equivariance of the spectral quantities. For the free part the distribution property and reflection positivity hold exactly. For the gauge part they hold at finite lattice spacing, and closure under weak limits preserves them in any limit taken. The clustering analogue is Lemma 8.38. The boundary of the standard formulation for the interacting content, which admits no measure-theoretic analogue of these conditions, is the triviality theorem recorded in Section 10. □

8.6 Positive scalar mass

This section records the mechanism by which the colour-singlet scalar sector acquires a positive lower bound. The mechanism has three steps. First, the spectral value of the long-root mode decreases as the internal space radius grows. Beyond a critical value it becomes tachyonic and triggers condensation of the glueball field. Second, the strict positivity of the conformal Laplacian on RP^3 guarantees that the quartic coupling of the effective potential is positive, so condensation occurs at a non-zero vacuum expectation value. Third, the explicit RP^3 spectrum excludes massless modes in the colour-singlet sector.

Lemma 8.24 (Long-root mode)**.** *The long-root mode is the $J = 2$ deformation mode of the R-sector connection, transforming in the representation $(2, 1)$ of* Spin(4)*. Its internal eigenvalue is*

$$\varepsilon_{\text{long}} = \frac{C_2(2,1)}{L^2} = \frac{16}{L^2},$$

where L is the radius of the internal space RP^3 and $C_2(j_L, j_R) = 2[j_L(j_L+1)+j_R(j_R+1)]$ is the quadratic Casimir.

Proof. The quantum numbers $(2, 1)$ follow from the harmonic analysis of the R-sector connection modes on RP^3 in Appendix A. The $(2, 1)$ mode carries charge $q = 2m_R = 2$ under the unbroken torus $\mathfrak{u}(1)_R$. The first non-zero internal mode of the R-sector connection carrying this charge is the $l = 3$ harmonic $(j_L, j_R) = (2, 1)$. The $l = 1$ sector is the Killing zero-mode sector, and the 1-form sector of RP^3 retains odd l. With the normalisation used here, the internal eigenvalue is the Casimir value C_2/L^2, and for $(2, 1)$ one has $C_2 = 2[2 \cdot 3 + 1 \cdot 2] = 16$. The mode survives in the even antipodal sector of RP^3. □

The onset radius of condensation $L_c = \sqrt{\pi}$ is determined by Proposition 8.27. The corresponding curvature scalar is $R = 6/L^2$, and the critical curvature is $R_c = 6/L_c^2 = 6/\pi$. Condensation sets in at the Gaussian breakdown width $b_c = 1/\sqrt{2}$, equivalently $L = L_c$. The condensation condition $R < R_c$, equivalently the internal-space radius exceeding the critical value $L_c = \sqrt{\pi}$, gives the closed form of Theorem 8.32, $m_\delta^2 > 0$. Whether the bound is realised, however, depends on where the flow is evaluated. The endpoint value σ_C and the inequality $R(\sigma_C) < R_c$ (equivalently $L(\sigma_C) > L_c$) are fixed only after solving the scale flow. The present section records the formal mechanism, not the numerical endpoint.

Remark 8.25 (Status of the endpoint). σ_C is a physical evaluation scale of the scale flow, not a regulator cutoff. Its numerical value, and the verification of the condensation condition $R(\sigma_C) < R_c$ there, are determined by the numerical solution of the scale flow. No cosmological identification of σ_C is required in this paper.

Theorem 8.26 (Long-root tachyon). *The squared mass of the long-root mode is*

$$m_{\text{long}}^2(L) = \varepsilon_{\text{long}} - \varepsilon_{\text{long}}(L_c) = \frac{8}{3}(R - R_c).$$

When $R < R_c$, $m_{\text{long}}^2 < 0$. The long-root mode is a tachyon and triggers condensation of the glueball field.

Proof. Here $\varepsilon_{\text{long}} = 16/L^2$, and at $L_c = \sqrt{\pi}$ one has $\varepsilon_{\text{long}}(L_c) = 16/\pi$. Taking the difference and rewriting in terms of the curvature scalar $R = 6/L^2$ and its critical value $R_c = 6/\pi$ yields the stated formula. □

The origin of the tachyon lies in the breakdown of the Gaussian statistical description at the scale σ_c. Beyond this scale the theory evolves in the condensed phase toward the evaluation endpoint σ_C.

Proposition 8.27 (Threshold in the self-dual Gaussian normalisation). *In the self-dual Gaussian normalisation of the scale field, the breakdown width is $b_c = 1/\sqrt{2}$, equivalently the onset radius of condensation is $L_c = \sqrt{\pi}$ and the critical curvature is $R_c = 6/\pi$.*

Proof. At the Gaussian breakdown scale of the scale flow the unbiased Gaussian structure of (A2) saturates the entropic uncertainty bound. For a one-dimensional Gaussian the Beckner–Hirschman inequality [26] reads $H_x + H_p \geq 1 + \ln \pi$, with equality if and only if the distribution is Gaussian. The unbiased structure of (A2) is Gaussian, so equality holds. The scale-invariant two-point structure of Theorem 4.30 is self-dual under Fourier transform, so the Gaussian assigns equal spread to position and momentum, $H_x = H_p$; hence $H_x = \frac{1}{2}(1 + \ln \pi)$. For a Gaussian $N(0, b^2)$ the position entropy is $H_x = \frac{1}{2}\ln(2\pi e b^2)$. Equating gives $2\pi e b^2 = \pi e$, so $b^2 = 1/2$, and the breakdown width is $b_c = 1/\sqrt{2}$. The internal radius is defined as $L = \sqrt{2\pi}\, b$, giving $L_c = \sqrt{2\pi}/\sqrt{2} = \sqrt{\pi}$. The constant curvature of RP^3 then gives $R_c = 6/L^2 = 6/\pi$. The entropic content of the threshold is the breakdown width $b_c = 1/\sqrt{2}$, the minimum-uncertainty Gaussian; the value $L_c = \sqrt{\pi}$ is its transcription into radius units through the convention $L = \sqrt{2\pi}\, b$, rather than an independent prediction. □

Lemma 8.28 (Strict positivity of the conformal Laplacian). *On RP^3 with curvature scalar $R = 6/L^2$, the conformal Laplacian $\mathcal{L} = \Delta + \frac{1}{8}R$ has spectrum*

$$\text{spec}(\mathcal{L}) = \left\{ \frac{l(l+2) + 3/4}{L^2} : l = 0, 2, 4, \dots \right\} \subset \left[\frac{3}{4L^2}, +\infty \right).$$

Hence $\mathcal{L} \geq 3/(4L^2) > 0$, and it is strictly positive.

Proof. The scalar Laplacian spectrum on RP^3 is $\varepsilon_l = l(l+2)/L^2$ for even l. The three-dimensional conformal coupling is $\xi = 1/8$. The eigenvalues of $\mathcal{L}$ are $\varepsilon_l + \xi R$, and the minimum at $l = 0$ is $3/(4L^2)$. □

The strict positivity of the conformal Laplacian is the central analytic input for the positive lower bound. It guarantees that the quartic coupling of the glueball effective potential is non-perturbatively strictly positive.

Theorem 8.29 (Effective potential from spectral sums)**.** *Let E be the colour-singlet scalar, the glueball field, coupled to the spectral data through a quadratic coupling entering the spectrum of the conformal Laplacian, $\varepsilon_l + \xi R \mapsto \varepsilon_l + \xi R + gE^2$, where $g > 0$ is the quadratic coupling (not fixed in absolute value in this paper) and ξR the conformal shift of Proposition 8.31. The effective potential of E is the flow-endpoint spectral sum*

$$V(E) = \sum_l d_l f(\varepsilon_l + \xi R + gE^2),$$

with f the one-loop function used in the spectral-sum definition. Expanding in E to fourth order gives

$$V(E) = V_0 + \frac{1}{2} m_{\text{long}}^2 E^2 + \frac{\lambda}{4} E^4 + O(E^6),$$

with spectral-sum coefficients

$$m_{\text{long}}^2 = 2g \sum_l d_l f'(\varepsilon_l + \xi R), \qquad \lambda = 2g^2 \sum_l d_l f''(\varepsilon_l + \xi R).$$

The quadratic coefficient is the $p = 1$ spectral diagonal, divergent in its bare form; its renormalised value is the geometric closed form of Theorem 8.26. The quartic coefficient converges absolutely by Lemma 8.19. With f the conformal-Laplacian logarithm the coefficient λ reproduces the quartic coupling of Proposition 8.31, as shown in the proof. The Landau form of the effective potential is the fourth-order truncation of the spectral-sum expansion.

Proof. The Taylor expansion of the spectral sum in E is

$$f(\varepsilon_l + \xi R + gE^2) = f(\varepsilon_l + \xi R) + gE^2 f'(\varepsilon_l + \xi R) + \frac{1}{2} g^2 E^4 f''(\varepsilon_l + \xi R) + O(E^6).$$

Summing with multiplicities d_l gives $V_0 = \sum_l d_l f(\varepsilon_l + \xi R)$, $m_{\text{long}}^2 = 2g \sum_l d_l f'(\varepsilon_l + \xi R)$, and, comparing with the Landau coefficient $(\lambda/4)E^4$,

$$\frac{\lambda}{4} = \frac{1}{2} g^2 \sum_l d_l f''(\varepsilon_l + \xi R), \qquad \lambda = 2g^2 \sum_l d_l f''(\varepsilon_l + \xi R).$$

For the conformal-Laplacian logarithm $f(\lambda) = -\ln \lambda$, one has $f''(\varepsilon_l + \xi R) = (\varepsilon_l + \xi R)^{-2}$, so that

$$\lambda = 2g^2 \sum_l d_l (\varepsilon_l + \xi R)^{-2},$$

the quartic coupling of Proposition 8.31 with normalisation $\kappa = 2g^2$. The sign is positive by the analytic positivity of Proposition 8.31. The higher-order terms $O(E^6)$ are part of the full spectral-sum potential. The Landau form is the truncation at fourth order. □

The long-root mass m_{long}^2 is determined twice, geometrically and spectrally. Theorem 8.26 gives the explicit geometric closed form $\frac{8}{3}(R - R_c)$ from the Casimir eigenvalue of the long-root mode. The spectral-sum coefficient $m_{\text{long}}^2 = 2g \sum_l d_l f'(\varepsilon_l + \xi R)$ of Theorem 8.29 is the general form; with f the conformal-Laplacian logarithm it is the $p = 1$

divergent diagonal $-2g\sum_l d_l(\varepsilon_l+\xi R)^{-1}$ of Lemma 8.19, whose renormalised value is identified with the geometric closed form. The two derivations therefore determine the same long-root mass, the geometric form supplying the explicit value used in Theorem 8.32.

The $p=1$ spectral diagonal $m^2_{\text{long}} = -2g\sum_l d_l(\varepsilon_l+\xi R)^{-1}$ is divergent in its bare form. Its renormalisation is fixed by the following prescription. (i) *Regularisation.* The divergence is a large-l short-distance divergence of the spectral sum; it is regularised by the geometric closed form, the Casimir eigenvalue $\varepsilon_{\text{long}} = 16/L^2$ of the long-root mode (Appendix A), which supplies the finite value $\frac{8}{3}(R-R_c)$ of Theorem 8.26. (ii) *Renormalisation condition.* The renormalised long-root mass is identified with this geometric closed form; the identification is the matching of the spectral-sum computation to the geometric (Casimir) computation. (iii) *Scheme independence.* The renormalised value is a Casimir eigenvalue of the internal geometry, hence independent of the regularisation scheme; no scheme-dependent subtraction constant appears. (iv) *Normalisation.* The coupling normalisation is $\kappa = 2g^2$, with g the quadratic coupling of Theorem 8.29; the field E is the canonically normalised glueball field $\operatorname{Tr} F_{\mu\nu}F^{\mu\nu}$, and its normalisation is fixed by the two-point function of Section 10. The quartic coupling λ of Proposition 8.31 is then a convergent ($p=2$) spectral sum, requiring no subtraction.

Lemma 8.30 (Scheme independence of the geometric matching)**.** *The identification of the renormalised long-root mass with the geometric closed form $\frac{8}{3}(R-R_c)$ is scheme-independent: the matched datum is the internal Casimir eigenvalue $\varepsilon_{\text{long}} = 16/L^2$ together with the critical curvature $R_c = 6/\pi$, both fixed by the internal geometry alone, and no scheme-dependent subtraction constant enters. The value of m^2_{long} therefore does not depend on the regularisation used to tame the $p=1$ diagonal.*

Proposition 8.31 (Analytic positivity of the quartic coupling)**.** *Let E be the glueball field, a colour-singlet scalar. Its effective potential $V(E) = \frac{1}{2}m^2_{\text{long}}E^2 + \frac{\lambda}{4}E^4$ has the quartic coupling*

$$\lambda = \kappa \sum_{l=0,2,4,\dots} \frac{d_l}{(\varepsilon_l+\xi R)^2}, \qquad d_l = (l+1)^2, \qquad \kappa > 0.$$

The series converges and every term is strictly positive. Hence $\lambda > 0$ holds analytically.

Proof. In the spectral-sum definition the quartic coupling is defined by the spectral sum of the statement. The expression $\lambda \propto \operatorname{Tr}\mathcal{L}^{-2}$ is its one-loop form obtained from the effective potential of the glueball field. The normalisation constant is $\kappa = 2g^2 > 0$, where g is the quadratic coupling of Theorem 8.29; this identification is established in the proof of that theorem. Lemma 8.28 guarantees $\varepsilon_l + \xi R \geq 3/(4L^2) > 0$ termwise. The general term behaves asymptotically as L^4/l^2, so the series converges absolutely. Its closed form is given by the polygamma function. □

Theorem 8.32 (Conditional positive scalar mass in the Landau potential)**.** *Let E be the glueball field with an effective potential of Landau form and $\lambda > 0$. When $R < R_c$ the potential attains its minimum at*

$$\langle E\rangle^2 = \frac{8}{3}\frac{R_c - R}{\lambda} > 0,$$

so the glueball field condenses. The fluctuation $\delta E = E - \langle E \rangle$ has squared mass

$$m_\delta^2 = V''(\langle E \rangle) = 2\lambda \langle E \rangle^2 = \frac{16}{3}(R_c - R) > 0.$$

When the flow endpoint lies in the condensed phase ($R < R_c$), the lowest colour-singlet excitation has a positive scalar mass $m_\delta > 0$.

Proof. The Landau form is the lowest-order symmetric allowed potential for a colour-singlet scalar. From $m_{\text{long}}^2 < 0$ and $\lambda > 0$ the minimum lies at a non-zero field value. The glueball field $E = \operatorname{Tr} F_{\mu\nu} F^{\mu\nu}$ is a colour singlet. Its condensation breaks no continuous symmetry, so no Goldstone boson appears. The fluctuation mass follows from the second derivative evaluated at the minimum. $\square$

Remark 8.33 (Mechanism versus endpoint). The condensation mechanism of Theorem 8.32 is unconditional: whenever $R < R_c$ the Landau potential condenses and the fluctuation mass is positive. The only conditional element is the location of the flow endpoint, namely whether $R(\sigma_C) < R_c$ holds there; this numerical question does not affect the mechanism itself.

Theorem 8.34 (Structure of the effective potential)**.** *Let E be the glueball field. The one-loop effective potential in the spectral-sum definition has the form*

$$V(E) = \frac{1}{2} m_{\text{long}}^2 E^2 + \sum_{k \geq 2} c_k E^{2k}, \qquad c_k = \frac{(-1)^k g^k}{k} \zeta(k; \xi R),$$

where $\zeta(k; \xi R) = \sum_l d_l (\varepsilon_l + \xi R)^{-k}$ are convergent spectral sums for $k \geq 2$ by Definition 8.1 and Lemma 8.19. The quartic coefficient $c_2 = \lambda/4 > 0$ is strictly positive by Proposition 8.31, and the signs of the higher coefficients alternate with k. The Landau truncation of Theorem 8.32 keeps the E^2 and E^4 terms. For $m_{\text{long}}^2 < 0$ it has a unique minimum at a non-zero field value, and the fluctuation mass at the minimum is strictly positive. The higher-order coefficients are convergent spectral sums, and their contributions are controlled by powers of $g \langle E \rangle^2 / \varepsilon_l$.

Proof. The one-loop effective potential is the spectral sum over the glueball modes,

$$V(E) = \frac{1}{2} m_{\text{long}}^2 E^2 + \sum_l d_l h(\varepsilon_l + \xi R + g E^2),$$

where h is the one-loop function, $h(x) = -\ln x$ in the convention of Appendix B, and the spectral sum carries no factor $1/2$, the normalisation of Theorem 8.29. Expanding in E^2 gives $V(E) = \frac{1}{2} m_{\text{long}}^2 E^2 + \sum_{k \geq 2} c_k E^{2k}$ with

$$c_k = \frac{(-1)^k g^k}{k} \sum_l d_l (\varepsilon_l + \xi R)^{-k}.$$

The $k = 1$ term of this expansion is the divergent $p = 1$ spectral diagonal of Lemma 8.19; it is absorbed into the renormalised m_{long}^2 by the renormalisation prescription described above, leaving the convergent $k \geq 2$ sums. The quartic coefficient is $c_2 = g^2\zeta(2;\xi R)/2 = \lambda/4 > 0$. The sums converge for $k \geq 2$. For $m_{\mathrm{long}}^2 < 0$, the Landau truncation

$$V_2(E) = \frac{1}{2}m_{\mathrm{long}}^2 E^2 + \frac{\lambda}{4}E^4$$

has its unique minimum at $\langle E\rangle^2 = -m_{\mathrm{long}}^2/\lambda > 0$ with curvature $V_2''(\langle E\rangle) = -2m_{\mathrm{long}}^2 = m_\delta^2 > 0$ (Theorem 8.32). The higher-order coefficients c_k, $k \geq 3$, are convergent spectral sums whose contributions near the minimum are suppressed by powers of $g\langle E\rangle^2/\varepsilon_l$ relative to the quartic term. $\square$

Remark 8.35 (Controlled expansion and the physical mass). Theorem 8.32 is established in the Landau truncation $V_2(E)$, the fourth-order truncation of the full spectral-sum potential $V(E)$ of Theorem 8.34. The Landau truncation is the leading order of a controlled expansion: the higher coefficients $c_k = (-1)^k g^k \zeta(k;\xi R)/k$, $k \geq 3$, are convergent spectral sums (Definition 8.1, Lemma 8.19) whose contribution near the minimum $\langle E\rangle$ is suppressed by powers of $g\langle E\rangle^2/\varepsilon_l$ relative to the quartic term. The minimum of $V(E)$ is therefore a perturbation of the Landau minimum, and the fluctuation mass m_δ of Theorem 8.32 is the physical mass of the full potential's minimum to leading order in this expansion, with corrections bounded by the higher-order terms. Within this controlled expansion the higher-order corrections do not change the sign of m_δ^2.

Theorem 8.36 (Stability of the Landau minimum)**.** *Let $V(E) = \frac{1}{2}m_{\mathrm{long}}^2 E^2 + \sum_{k\geq 2} c_k E^{2k}$ be the full spectral-sum potential of Theorem 8.34, with $c_2 = \lambda/4 > 0$ and $c_k = (-1)^k g^k \zeta(k;\xi R)/k$ for $k \geq 3$. Near the Landau minimum $\langle E\rangle$ the higher-order terms are uniformly suppressed: for every $k \geq 3$,*

$$\left|\frac{c_k\langle E\rangle^{2k}}{c_2\langle E\rangle^4}\right| \leq C\left(\frac{g\langle E\rangle^2}{\varepsilon_1}\right)^{k-2},$$

with ε_1 the first non-zero internal eigenvalue and C a constant. Hence for $g\langle E\rangle^2/\varepsilon_1 < 1$ the full potential has a unique non-trivial minimum $E_* = \langle E\rangle + O(g\langle E\rangle^3/\varepsilon_1)$, and

$$V''(E_*) = 2\lambda\langle E\rangle^2\bigl(1 + O(g\langle E\rangle^2/\varepsilon_1)\bigr) > 0.$$

In particular the higher-order coefficients do not change the sign of the squared fluctuation mass.

Proof. Each spectral sum $\zeta(k;\xi R) = \sum_l d_l(\varepsilon_l + \xi R)^{-k}$ converges for $k \geq 2$ (Lemma 8.19) and is dominated by its leading mode, $\zeta(k;\xi R) \leq C'\varepsilon_1^{-k}$. Hence $|c_k| \leq C'' g^k \varepsilon_1^{-k}$, and at the minimum the k-th term obeys $|c_k E^{2k}| \leq C''(gE^2/\varepsilon_1)^k$. The ratio to the quartic term is $(g\langle E\rangle^2/\varepsilon_1)^{k-2}$ as stated. For $g\langle E\rangle^2/\varepsilon_1 < 1$ the tail $\sum_{k\geq 3} c_k E^{2k}$ is a convergent geometric series, so by the implicit function theorem the minimum of V is a small perturbation of the Landau minimum and the second derivative is positive, with the correction of order $g\langle E\rangle^2/\varepsilon_1$. $\square$

The existence of the positive lower bound is insensitive to the precise value of the long-root spectral coefficient. The condensation condition $m^2_{\rm long} < 0$ is the inequality $L > L_c$, and $\varepsilon_{\rm long}(L) = 16/L^2$ decreases with L. The absolute scale of the lower bound is set by the internal radius L. Within the present construction L is determined only up to the condensation threshold and the evaluation endpoint σ_C, so the numerical value of m_δ in physical units is not fixed here. The scalar glueball masses obtained in lattice QCD simulations [40] provide a phenomenological reference scale. Fixing the absolute scale of L is not part of the present construction. The lower bound m_δ is a statement about the spectral-sum definition. The relation of this bound to the standard continuum formulation is analysed in Section 10. The standard-formulation realisation of the colour-singlet data carries the same lower bound m_δ.

Theorem 8.37 (Absence of massless modes)**.** *On RP^3 the scalar spectrum contains a zero mode that carries the glueball field. The first de Rham cohomology vanishes,*

$$H^1_{dR}(RP^3;\mathbb{R}) = 0,$$

so there are no harmonic 1-forms. The Laplacian on transverse-traceless symmetric tensors obeys the Lichnerowicz formula

$$\Delta_L\omega = \nabla^*\nabla\omega + \mathrm{Ric}\cdot\omega - 2\,\mathrm{Rm}\cdot\omega,$$

with $\mathrm{Ric} = \frac{2}{L^2}g$*, and its lowest eigenvalue on RP^3 is $6/L^2 = R > 0$, hence it admits no massless TT modes. Together with $m_\delta^2 > 0$ from Theorem 8.32, the colour-singlet spectrum contains no massless mode. The positive lower bound is consistent within the spectral-sum definition.*

Proof. The scalar Laplacian spectrum on RP^3 is explicit, with $\varepsilon_l = l(l+2)/L^2$ and multiplicities $d_l = (l+1)^2$ for even l. The $l = 0$ mode is the constant function, which carries the glueball field E. The vanishing of the first cohomology and the strict positivity of the TT Laplacian are standard results in spectral geometry [4, 41]. The Kaluza–Klein zero modes of the gauge fields belong to the 1-form sector and lie outside the colour-singlet spectrum. The term zero mode here refers to modes with vanishing internal momentum. The sole scalar zero mode carries the glueball field, and its fluctuation mass $m_\delta > 0$ is supplied by Theorem 8.32. □

A positive lower bound $m_\delta > 0$ implies exponential clustering of the Schwinger functions, which is the spectral-sum-definition analogue of the clustering axiom OS4 in Theorem 2.2. The precise statement is the following lemma.

Lemma 8.38 (Exponential clustering)**.** *Let $C(\tau)$ be the $p = 0$ temporal correlator of the reconstructed colour-singlet scalar, and $S_2(d)$ its two-point function at Euclidean separation $d > 0$ in the $\mathbb{R}^4$ factor:*

$$C(\tau) = \sum_{l\ even} \frac{d_l}{2M_l}\,e^{-M_l\tau}, \qquad S_2(d) = \sum_{l\ even} d_l\,G_{M_l}(d),$$

$$M_l^2 = \varepsilon_l + m_\delta^2, \qquad G_M(d) = \frac{M\,K_1(Md)}{4\pi^2 d},$$

where K_1 is the modified Bessel function and $\varepsilon_l = l(l+2)/L^2$, $d_l = (l+1)^2$ is the Z_2-even scalar spectrum of RP^3. With $m_\delta > 0$ the following hold.

(i) Sharp temporal bound. *For every $\tau_0 > 0$ and all $\tau \geq \tau_0$,*

$$C(\tau) \leq C(\tau_0)\, e^{-m_\delta(\tau-\tau_0)}.$$

(ii) Exponential rate. *$S_2(d)/G_{m_\delta}(d) \to 1$ as $d \to \infty$; equivalently $d^{-1}\log S_2(d) \to -m_\delta$. The $l \geq 2$ tower decays with the larger rate $M_2 - m_\delta = (m_\delta^2 + 8/L^2)^{1/2} - m_\delta > 0$.*

(iii) Spatial bound. *For every $\varepsilon > 0$ and every $d_0 > 0$ there is a constant $C(d_0, \varepsilon) < \infty$ such that $S_2(d) \leq C(d_0,\varepsilon)\, e^{-(m_\delta-\varepsilon)d}$ for all $d \geq d_0$.*

This is the spectral-sum-definition analogue of the clustering axiom. It holds on the free sector and on the effective sector (the fluctuation field of Proposition 7.9 is the free scalar of mass m_δ), and the vacuum is unique. The boundary of the standard formulation for the interacting content, which admits no measure-theoretic analogue of this statement, is the triviality theorem recorded in Section 10.

Proof. For (i), each internal mode has $M_l \geq m_\delta$, hence $e^{-M_l\tau} \leq e^{-M_l\tau_0} e^{-m_\delta(\tau-\tau_0)}$ term by term. For (ii), the zero mode $l = 0$, $d_0 = 1$, has $M_0 = m_\delta$; the ratio of the $l \geq 2$ sum to the zero-mode term is

$$R(d) = \sum_{l\geq 2} d_l\, \frac{M_l K_1(M_l d)}{m_\delta K_1(m_\delta d)},$$

and the uniform bounds on K_1 ($c_- z^{-1/2} e^{-z} \leq K_1(z) \leq c_+ z^{-1/2} e^{-z}$ for $z \geq z_0$) give

$$R(d) \leq C \sum_{l\geq 2} d_l \Big(\frac{M_l}{m_\delta}\Big)^{1/2} e^{-(M_l - m_\delta)d} \leq C'\, e^{-(M_2-m_\delta)d} \to 0,$$

the middle sum being finite for $d \geq d_0$ by the Weyl asymptotics $d_l = (l+1)^2$, $M_l \simeq l/L$. Hence $S_2(d) = G_{m_\delta}(d)\,(1 + R(d))$, and $d^{-1}\log G_{m_\delta}(d) \to -m_\delta$. For (iii), the same bounds give $S_2(d) \leq C(d_0)\, d^{-3/2} e^{-m_\delta d}$ for $d \geq d_0$, and $d^{-3/2} \leq C_\varepsilon\, e^{\varepsilon d}$. □

8.7 Endpoint physical quantities

The evolution of the effective potential along the scale flow leads to condensation in the colour-singlet scalar sector and to a positive scalar mass parameter m_δ. It remains to record how the physical quantities at the flow endpoint are defined and to state their well-definedness.

In the present construction the basic object is the scale flow itself. The effective theory evolves as the scale parameter σ varies continuously. Physical quantities are evaluated at the endpoint σ_C. They are not defined by integrals over a measure. They are defined by spectral sums at the endpoint of the flow, as stated in Definition 8.1. These spectral sums are exact summations over the spectrum of the geometric Laplacian on the compact internal space RP^3. They do not rely on a perturbative expansion.

Theorem 8.39 (Endpoint physical quantities)**.** *In the spectral-sum definition the physical quantities at the flow endpoint σ_C are defined by spectral sums and closed analytic expressions. The effective potential and the squared masses are given by*

$$V_{\text{eff}}(k) = -\frac{C}{2}\left(k^4 - M_P^4\right), \qquad m_{\text{long}}^2 = \frac{8}{3}(R - R_c), \qquad m_\delta^2 = \frac{16}{3}(R_c - R).$$

The quartic coupling is given by the convergent spectral sum

$$\lambda = \kappa \sum_{l=0,2,4,\ldots} \frac{d_l}{(\varepsilon_l + \xi R)^2}, \qquad d_l = (l+1)^2, \qquad \kappa > 0.$$

Its closed form is a polygamma function. Every term is strictly positive and the series converges absolutely.

Proof. The effective potential V_{eff} is given by Lemma 8.11. The squared mass m_{long}^2 is given by Theorem 8.26, and m_δ^2 by Theorem 8.32. The expression for the quartic coupling λ is given by Proposition 8.31, and its absolute convergence is guaranteed by Lemma 8.19. Each formula is explicit and does not rely on any limiting procedure. □

The closed analytic expressions for the endpoint physical quantities are summarised in the following table.

Table 6. Endpoint physical quantities in the spectral-sum definition

Physical quantity	Closed analytic expression
$V_{\text{eff}}(k)$	$-\frac{C}{2}(k^4 - M_P^4)$
$\beta(\sigma)$	$-2C\sigma^{-5}$
m_{long}^2	$\frac{8}{3}(R - R_c)$
$\langle E \rangle^2$	$\frac{8}{3}(R_c - R)/\lambda$
m_δ^2	$\frac{16}{3}(R_c - R)$
λ	$\kappa \sum_l d_l(\varepsilon_l + \xi R)^{-2}$, polygamma closed form

The squared masses m_{long}^2 and m_δ^2 follow directly from the spectral data and are exact closed forms. The effective potential V_{eff} is obtained within the local potential approximation, and the vacuum expectation value $\langle E \rangle^2$ is obtained within the Landau form of the effective potential. These approximations do not affect the existence of the formulas. The formulas themselves exist explicitly, and their numerical accuracy is controlled by the respective truncations.

The endpoint physical quantities are therefore well defined within the spectral-sum definition. The relation of this construction to the standard path integral formulation is analysed in Section 10.

8.8 Vacuum, spin–statistics, and summary

The scale flow construction has supplied a Euclidean measure, the spectral-sum-definition conditions for the free and gauge sectors, a positive scalar mass, and closed-form expressions for the endpoint physical quantities. It remains to record the vacuum structure, the spin–statistics relation, and the overall status of the construction.

Theorem 8.40 (Vacuum)**.** *(i) In the free sector the free matter field measure $d\mu_0$, Gaussian or Grassmann Gaussian, possesses a unique vacuum state Ω_0, the Fock vacuum.*

(ii) Within the spectral-sum definition, the vacuum state Ω of the colour-singlet sector is the cyclic vector of the reconstruction of the spectral data. The condensation $\langle E \rangle \neq 0$ of Theorem 8.32 is a property of this state. In general Ω is not the free Fock vacuum.

Proof. Part (i) is standard. A Gaussian measure corresponds to a free field whose Hilbert space is a Fock space, and the Fock vacuum is unique. For part (ii), the reconstruction of the spectral data within the spectral-sum definition gives a cyclic vacuum vector Ω. Because the spectral data include the non-Gaussian content of the condensate and the quartic coupling, Ω is not the free Fock vacuum Ω_0. The non-vanishing vacuum expectation value $\langle E \rangle \neq 0$ is a property of Ω. In the free theory one has $\langle E \rangle_{\Omega_0} = 0$. No contradiction arises because the two statements refer to different Hilbert spaces. □

The positive scalar mass $m_\delta > 0$ implies exponential decay of the relevant correlation functions. This is the spectral-sum-definition analogue of clustering. It is consistent with uniqueness of the vacuum in the stated setting, but a full derivation of vacuum uniqueness would require additional assumptions. For the purposes of this paper, the vacuum of the free sector is unique, and the interacting vacuum is characterised as the cyclic vector of the reconstruction.

The spin–statistics relation of the chiral Weyl pair is the standard theorem of Section 6 (Theorem 6.9); it is not re-derived here.

Theorem 8.41 (Locality)**.** *Let x and y be spacelike separated points. Then*

$$[\varphi_i(f), \varphi_j(g)]_\mp = 0$$

whenever the supports of f and g lie in spacelike separated neighbourhoods of x and y.

Proof. The Osterwalder–Schrader reconstruction gives the locality property of the fields (Theorem 2.2). This part of the reconstruction does not require the clustering condition. □

Theorem 8.42 (Summary of results within the spectral-sum definition)**.** *Let A_μ be the gauge field of Proposition 4.21, equipped with the Euclidean measure $d\mu$ defined after emergence in Proposition 8.6. Then the following hold.*

(i) Within the spectral-sum definition, the physical quantities of the colour-singlet sector defined in Theorem 8.39 are well defined through spectral sums at the flow endpoint. The spectral data satisfy the spectral-sum-definition analogues of regularity and reflection positivity, as stated in Theorem 8.23, and are $SO(4)$-invariant by Theorem 8.18. The colour-singlet sector carries a positive scalar mass $m_\delta > 0$ by Theorems 8.32 and 8.37, conditional on the flow endpoint lying in the condensed phase (Section 8.6).

(ii) *The relation to the standard continuum formulation is as follows. The free part has a standard-formulation construction, given in Section 10. For the interacting scalar content, the boundary of the standard continuum formulation is the triviality of the four-dimensional elementary scalar models, recorded in Section 10.*

Proof. (i) The endpoint physical quantities are defined by Theorem 8.39 through closed analytic expressions whose convergence is guaranteed by Lemma 8.19. The positive scalar mass $m_\delta > 0$ follows from Theorems 8.32 and 8.37, which show that, when the condensation condition $R < R_c$ holds, i.e. conditional on the flow endpoint lying in the condensed phase, the colour-singlet scalar condenses without breaking any continuous symmetry and no massless modes survive in the colour-singlet spectrum. (ii) The standard-formulation realisation of the free part is the free scalar field on $\mathbb{R}^4 \times RP^3$ of Theorem 10.5. Its Schwinger functions satisfy the standard OS axioms, and the OS reconstruction theorem yields a Wightman field theory satisfying W1–W6, with the same scalar mass m_δ. The field A_μ itself is not gauge-invariant. Treating it as an operator-valued distribution would require a gauge fixing that spoils locality. The Wightman fields are therefore the gauge-invariant local observables of the colour-singlet sector. □

Within the spectral-sum definition the Wightman axioms are understood in the sense of Definition 8.1(iv). The spectral-sum-definition analogues of regularity, Euclidean invariance, reflection positivity, and clustering hold by Theorem 8.23, Theorem 8.18, and the positive scalar mass $m_\delta > 0$ (conditional on the flow endpoint lying in the condensed phase). The construction is internally consistent within the spectral-sum definition. On the free sector the standard Wightman axioms hold in the measure-theoretic sense; on the interacting sector the spectral-sum-definition analogues hold.

The construction of this section supplies a concrete instance of the spectral datum of Section 5: the internal space is RP^3 with its Laplace spectrum (Section 4), and the physical quantities at the flow endpoint are the corresponding spectral sums. The following theorem records the resulting correspondence between the two systems of this paper, the coarse-graining principle of Section 3 and the spectral formulation of Section 5.

Proposition 8.43 (The gauge sector satisfies the formulation conditions)**.** *The gauge field A_μ constructed in this section, the $G_{\rm SM} = SU(3) \times SU(2) \times U(1)$ connection of Proposition 4.21, on the internal space RP^3 with colour-singlet spectrum $\{\varepsilon_l = l(l+2)/L^2,\ d_l = (l+1)^2\}$ (Section 4), together with the endpoint physical quantities of the scale flow (Theorem 8.39) and the positive scalar mass m_δ (Theorem 8.32), constitutes a spectral datum satisfying Definitions 5.1–5.8 of Section 5. It therefore satisfies the formulation criterion of Theorem 5.11, carrying the positive scalar mass $m_\delta > 0$ conditional on the flow endpoint lying in the condensed phase.*

Proof. The conditions are verified item by item. (FSA-1) The spectral datum of the colour-singlet sector is the RP^3 spectrum with the flow-endpoint parameters, all constructed in Sections 4 and 8 (Definition 5.1). (FSA-2) The physical quantities are the endpoint spectral sums of Theorem 8.39, of the form of Definition 5.2. (FSA-3) The physical Hilbert space is the reconstruction space of Definition 6.2: on the free sector the Fock space (unconditional), and on the interacting sector the kinematic carrier and physical content established by the mapping relations (Theorems 7.8 and 9.5). (FSA-4) The Born rule is the kinematic rule of projective measurement (Theorems 6.3 and 6.16). (FSA-5) The dynamics is the reconstruction and Stone's theorem (Theorem 6.10), with the Fock Hamiltonian on the free sector. (FSA-6) The composite structure is the tensor product of the reconstruction (Section 6.3). (FSA-7) Locality holds on the free sector and on the effective sector (Theorem 6.8, Proposition 7.9). (FSA-8) The mapping holds: the free sector by the recovery theorem (Theorem C.2), the interacting sector by the analytic, structural, effective and non-perturbative mapping relations (Theorems 7.3, 7.7, 7.8, 9.5 and 9.3). The formulation criterion of Theorem 5.11 is therefore satisfied, item by item. The positive scalar mass is $m_\delta > 0$ (Theorem 8.32), with no massless mode in the colour-singlet spectrum (Theorem 8.37). □

Remark 8.44 (Status). Proposition 8.43 is the correspondence between the two systems of this paper: the coarse-graining principle of Section 3 and the spectral formulation of Section 5. Within the spectral-sum definition the gauge sector is closed as an internal construction by Proposition 8.43, carrying the positive scalar mass m_δ conditional on the flow endpoint lying in the condensed phase. The boundary of the standard continuum formulation for the colour-singlet scalar content, taken as an elementary scalar, is the triviality theorem (Section 10); the standard-formulation counterpart of the gauge-field sector is the standard problem of Section 10.

9 The formal correspondence and the non-perturbative content

The free-sector mapping of Appendix C proceeds through the Gaussian measure. Its covariance is the full spectral kernel, and Minlos' theorem together with the Osterwalder–Schrader reconstruction produces the Fock space. The interacting content of the formulation, the derived parameters $\lambda > 0$ and $\langle E \rangle \neq 0$, is carried by the spectral-sum definition; its correspondence to the physical structure follows the structure of the formulation rather than the Gaussian route. Section 10 records the boundary of the standard continuum formulation for elementary scalar interactions.

9.1 The formal correspondence without a measure

The physical quantities of the formulation are defined without a measure. The measure appears only as the statistical content of the unbiasedness clause and as the optional correspondence of the free sector. The formal correspondence of the interacting sector is therefore constructed directly from the algebraic relations of the spectral data.

Definition 9.1 (Mapping dictionary). *The mapping dictionary is the correspondence that assigns to each closed form of the spectral datum its structural role in the physical structure of Proposition 9.2. It consists of the following assignment rules.*

(i) The quartic coupling λ, a convergent spectral sum, is assigned to the e^4 vertex.

(ii) *The scalar mass parameter m_δ is assigned to the single-particle mass, located at the spectral pole $p^2 = m_\delta^2$.*

(iii) *The condensate $\langle E\rangle$ is assigned to the constant mode of the reconstructed colour-singlet field, the constant shift $\phi = e + \langle E\rangle$ of Section 10.3.1.*

(iv) *The mass tower $\{M_l\}$ is assigned to the single-particle states of the free sector.*

Proposition 9.2 (Formal correspondence without a measure)**.** *Let λ, m_δ, and $\langle E\rangle$ be the derived parameters of Sections 7 and 8. The following correspondence holds without the use of a measure.*

(a) Condensate as a constant mode. *The physical vacuum of the condensed sector is the vacuum of the shifted Gaussian structure of Section 10.3.1: the colour-singlet field is $\phi = e + \langle E\rangle$, with e the fluctuation field of mass m_δ. The constant $\langle E\rangle$ is not an element of the one-particle space $h_\delta = L^2(\mathbb{R}^3, d^3p/2\omega_\delta)$, so the shift is not a Weyl displacement within $\mathcal{F}(h_\delta)$; it adjoins a constant mode to the reconstruction. The vacuum of the condensed sector accordingly belongs to a representation inequivalent to the Fock representation, and its vacuum expectation value is $\langle\phi\rangle = \langle E\rangle$ (Theorem 8.40).*

(b) Mapping dictionary. *Each closed form of the spectral data corresponds to an element of the operator structure. The coupling λ, a convergent spectral sum, corresponds to the e^4 vertex. The parameter m_δ corresponds to the single-particle mass through the spectral pole. The condensate $\langle E\rangle$ corresponds to the constant mode of the reconstructed field. The mass tower $\{M_l\}$ corresponds to the single-particle states.*

(c) No measure is used. *Every element of the correspondence is constructed from the algebraic relations of the spectral data. No measure integral, no Minlos theorem, and no Osterwalder–Schrader reconstruction enters the interacting correspondence. This is consistent with the definition of physical quantities in Definition 5.2, which does not depend on a measure.*

Proof. (a) The constant shift $\phi = e + \langle E\rangle$ is the shift of Section 10.3.1, where the reconstructed structure of the shifted Gaussian measure is spelled out. The displacement parameter is the condensate, a closed expression of the spectral data. Since $\langle E\rangle \notin h_\delta$, the shifted vacuum lies outside the Fock representation, and the reconstruction adjoins the constant mode carrying the vacuum expectation value $\langle E\rangle$. (b) Each entry of the dictionary refers to the theorem that establishes the corresponding structure from the spectral data. (c) By construction, the dictionary and the shift invoke only the closed forms of the spectral data and the standard algebraic structures of the Fock space. None of these requires a measure. □

9.2 Non-perturbative content as spectral-sum representation

The non-perturbative content of the formulation does not require resummation of a perturbative series. The physical quantities are spectral sums and closed forms of the spectral data by Definition 5.2. In particular, the quartic coupling λ is itself a convergent spectral sum, not the leading order of an expansion. Every amplitude at any order of the effective expansion is therefore an explicit function of the spectral data.

Theorem 9.3 (Amplitudes by spectral-sum representation). *Let the effective theory of Theorem 7.8 be given, with the spectral data $(\lambda, m_\delta, \{M_l\})$. The following definitions are made within the spectral-sum definition. Then the following hold.*

(a) Every order. *Every order of the effective expansion of the scattering amplitude is an explicit function of the spectral data. The Feynman rules of the effective theory depend only on λ and m_δ. The renormalisation is the dispersion subtraction of the formulation. The one-loop bubble has the dispersion representation*

$$B(s) = \int_{4m_\delta^2}^{\infty} \frac{ds'}{\pi} \frac{\operatorname{Im} B(s')}{s' - s},$$

with the convention-independent spectral density

$$\operatorname{Im} B(s') = \frac{1}{16\pi} \sqrt{1 - \frac{4m_\delta^2}{s'}}\, \theta(s' - 4m_\delta^2).$$

The dispersion integral is subtracted at the subtraction point s_0, giving the convergent expression

$$B_{\text{ren}}(s) = \int_{4m_\delta^2}^{\infty} \frac{ds'}{\pi} \operatorname{Im} B(s') \left(\frac{1}{s' - s} - \frac{1}{s' - s_0} \right).$$

Higher orders are renormalised by the same dispersion subtraction of their spectral representations. Hence every order is a definite function of the spectral data. The tree-level $2 \to 2$ amplitude is $\mathcal{M}^{(0)} = -6\lambda$. The renormalised one-loop contribution, summed over the three channels, is

$$\mathcal{M}^{(1)}(s,t,u) = -18\lambda^2 \left(B_{\text{ren}}(s) + B_{\text{ren}}(t) + B_{\text{ren}}(u) \right),$$

the factor $18 = 6^2/2$ arising from the quartic vertex $-6i\lambda$ and the symmetry factor $1/2$ of the bubble.

(b) The non-perturbative character is carried by the coupling. *The coupling λ is a convergent spectral sum of the spectral data, independent of any expansion. The amplitude is therefore determined by the exact spectral data.*

(c) No resummation is required. *The physical quantities are spectral sums and closed forms directly. The effective expansion is an optional description of the effective theory, not the definition of the physical quantities.*

Proof. (a) The Feynman rules give the tree amplitude -6λ and the one-loop bubble

$$B(s) = \int \frac{d^4k}{(2\pi)^4} \frac{1}{(k^2 - m_\delta^2)((k-p)^2 - m_\delta^2)},$$

with $s = p^2$. Its imaginary part is the standard two-particle phase space, and its real part follows from the dispersion relation up to the subtraction convention. The bubble depends on the spectral data only through m_δ. (b) The coupling λ is a convergent spectral sum, not the leading order of an expansion. (c) This is the constitutive definition of the formulation, Definition 5.2. □

Remark 9.4. Within the formulation, non-perturbative means spectral-sum defined. A quantity is non-perturbative if it is a spectral sum or a closed form of the spectral data. This differs from the standard-formulation notion, where non-perturbative refers to the limit or resummation of a continuum construction. The spectral-sum representation is the formulation's way of encoding non-perturbative content. The physical quantities were never defined by a perturbative series, so no resummation is involved. This internal notion is not claimed to solve the standard non-perturbative construction problem of the measure-based formulation.

Theorem 9.5 (Interaction operator within the spectral-sum definition)**.** *Let the effective sector of Theorem 7.8 carry the interaction $\lambda : e^4 :$. Within the formulation the following hold.*

(a) Physical content. *The physical quantities considered here are exact functions of the spectral data. The amplitudes and loop corrections are given by Theorem 9.3 with the dispersion subtraction. The positive scalar mass m_δ and the condensate $\langle E \rangle$ are closed forms of the spectral data.*

(b) Structural status. *The operator is the mapping-dictionary entry $\lambda \mapsto e^4$ vertex of Proposition 9.2(b). Its structural role is fixed by the spectral-data closed form of the coupling.*

(c) Definitional scope. *Within the formulation the definition of the interaction operator is its physical content and formal correspondence. The strict Fock-space operator definition of $: e^4 :$ for an elementary scalar belongs to the standard formulation, whose continuum limits are Gaussian (Section 10). The formulation does not require that definition, since its physical quantities are spectral sums.*

(d) Closure. *The operator-level definition of $\lambda : e^4 :$ is therefore closed within the formulation. The physical content and structural status are fully determined by the spectral data. The boundary of the standard formulation for the elementary-scalar counterpart is recorded in Section 10.*

Proof. (a) The amplitudes are explicit spectral-data functions by Theorem 9.3. The positive scalar mass m_δ and the condensate $\langle E \rangle$ are closed forms. (b) This is the mapping dictionary of Proposition 9.2(b). (c) The definitional scope is the constitutive definition of the formulation, Definition 5.2. (d) Closure follows from (a)–(c). □

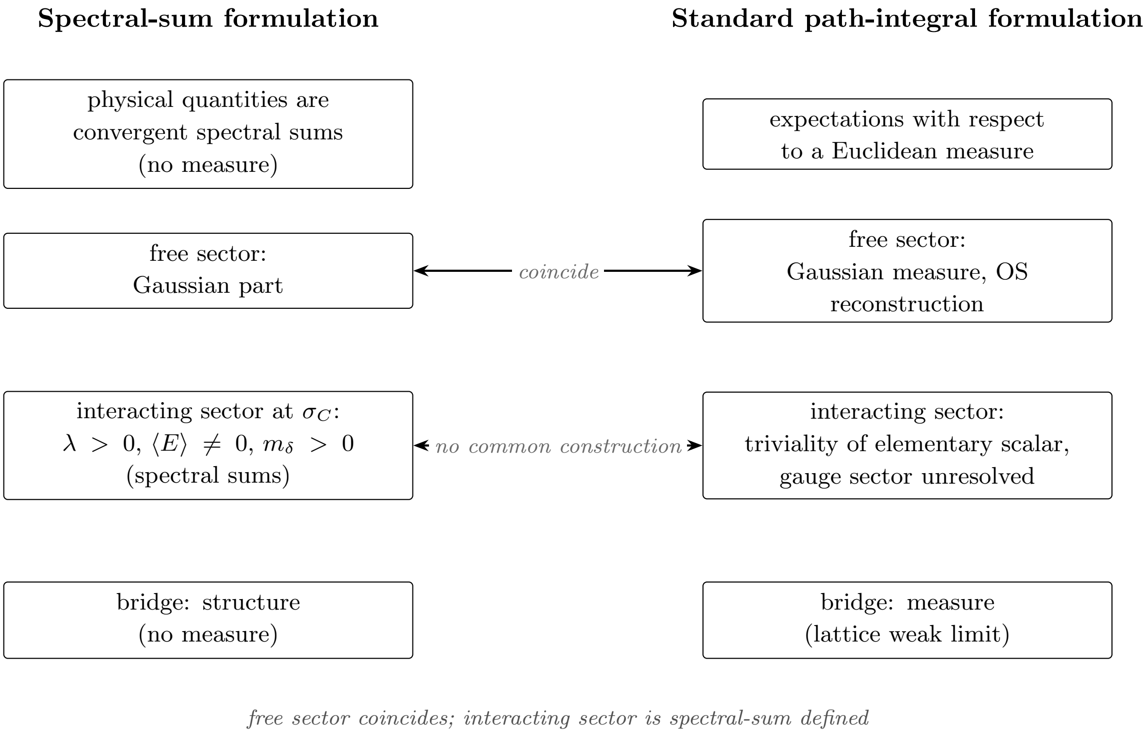

Figure 2. Relation between the spectral-sum formulation and the standard path-integral formulation. The free sector coincides. The interacting sector is carried by a spectral-sum construction rather than a standard measure-theoretic construction. The interacting-sector quantities are evaluated at the flow endpoint σ_C; $\langle E \rangle \neq 0$ and $m_\delta > 0$ hold when $R(\sigma_C) < R_c$ (Theorem 8.32), while $\lambda > 0$ is analytic (Proposition 8.31).

10 The spectral-sum definition and the standard path integral formulation

This section records the relation between the spectral-sum definition of this paper and the standard path integral formulation of quantum gauge theory. The analysis has two parts. Part 1 discusses the non-trivial content carried by the spectral sums. Part 2 constructs the standard-formulation realisation of the free part.

10.1 The two formulations

In the spectral-sum definition, physical quantities are defined directly by spectral sums over the spectrum of the internal space RP^3, evaluated at the endpoint of the scale flow. In the standard path integral formulation, physical quantities are defined as expectations with respect to a Euclidean measure. The measure is constructed in practice as the weak limit of lattice measures with the plaquette action [42]. Reflection positivity of the lattice theory is a rigorous theorem [43], and the continuum limit is the object of the standard constructive programme.

The correspondence between the standard requirements and the spectral-sum statements is summarised in Table 7.

Table 7. Dictionary between standard requirements and their spectral-sum-definition analogues

Standard requirement	Spectral-sum statement	Location
Wightman axioms W1–W6	Spectral-sum-definition analogue of the Wightman axioms	Definition 8.1, Theorem 8.42
Positive scalar scale	Positive colour-singlet scalar scale $m_\delta > 0$ within the spectral-sum definition	Theorems 8.32–8.37
Non-triviality	Spectral sums, $\lambda > 0$, condensate	Theorems 8.31, 8.32
Gauge group $G_{\rm SM}$	$SU(3) \times SU(2) \times U(1)$	Proposition 4.21
Spacetime dimension 4	Emergent $4 = 3 + 1$	Theorem 4.4, Corollary 4.25
Euclidean invariance	Gaussian part $E(4)$, flow $SO(4)$	Theorems 4.27, 8.18
Reflection positivity	Free field on $\mathbb{R}^4 \times RP^3$	Theorems 10.5, 10.6
Short-distance asymptotic freedom	β-function closed form, interaction stability	Lemma 8.11, Theorem 8.17

Each row of Table 7 records a spectral-sum-definition analogue of the corresponding standard requirement, not an equivalence (Definition 8.1). In particular, the positive colour-singlet scalar scale $m_\delta > 0$ is a clustering scale within the spectral-sum definition.

10.2 Part 1: The non-trivial content

10.2.1 The scalar sector in the standard continuum formulation

The colour-singlet sector is represented by a real scalar field with a quartic self-interaction. Its coupling is strictly positive by Proposition 8.31, and its potential develops a condensate by Theorem 8.32. This content is organised at the spectral level. The standard-formulation counterpart, taken as an elementary four-dimensional scalar field with quartic self-interaction, falls under the triviality theorem; taken as a composite field of the gauge sector, its standard-formulation status is tied to the gauge sector (Remark 10.3).

Theorem 10.1 (Triviality of the elementary scalar sector)**.** *Let $\{S^{(n)}\}$ be a continuum limit of four-dimensional lattice scalar models with quartic self-interaction, whose colour-singlet data match the spectral data of this construction with quartic coupling $\lambda > 0$ and condensate $\langle E \rangle \neq 0$. Then the limit is Gaussian: in the continuum limit the quartic coupling vanishes, and no non-trivial elementary scalar interaction survives.*

Proof. The colour-singlet sector is a four-dimensional real scalar field with quartic self-interaction. Four-dimensional scalar field theory of an elementary field with quartic interaction is trivial. The continuum limits of the four-dimensional lattice scalar models are Gaussian. The rigorous statement for the critical four-dimensional Ising and φ^4 lattice models is the scaling-limit theorem of Aizenman and Duminil-Copin [44]. The earlier analyses of Aizenman [45] and Fröhlich [46] established free and generalised-free scaling limits in five or more dimensions and, in four dimensions, the triviality of the families described by renormalised perturbation theory. Hence no standard-formulation continuum theory of this type can carry the non-trivial spectral-sum content $\lambda > 0$ as an elementary scalar interaction. $\square$

The difference between the two formulations is therefore structural. The spectral-sum definition carries the quartic coupling $\lambda > 0$ by construction through Proposition 8.31. A standard continuum formulation of an elementary scalar sector cannot.

10.2.2 The gauge sector and the standard formulation

Proposition 10.2. *The standard-formulation continuum limit of the non-Abelian gauge sector is the unresolved content of the standard formulation itself. The spectral-sum construction does not depend on it.*

Proof. The standard formulation requires the existence of a weak limit of the lattice measures [42, 43, 47] and the control of the continuum limit by multi-scale cluster expansion [48–50]. The completion of this programme is an open problem of the standard formulation. Recent progress includes direct decay estimates for finite lattice gauge theories at weak coupling [51]. The spectral-sum construction defines its physical quantities directly by spectral sums. It presupposes neither the lattice measures nor their limit. $\square$

Remark 10.3 (Scalar order parameter and the gauge sector). The quantity m_δ is the mass of the colour-singlet scalar order parameter, and within the spectral-sum definition it is the positive scalar mass of that sector. The scalar mass is closed internally within the spectral-sum construction.

The quantity m_δ can enter the standard programme's estimates only as an input improving the decay constants of the cluster expansion. It cannot replace the expansion itself. Defining Wilson loops by spectral sums would be a redefinition of the observable, not a translation of it. The question whether such a definition agrees with the standard Wilson loop is part of the standard programme.

10.2.3 Conclusion of Part 1

The relation between the two formulations is not a translation in the literal sense. The quartic coupling $\lambda > 0$, the condensate $\langle E \rangle \neq 0$, and the positive scalar mass $m_\delta > 0$ are defined in the spectral-sum definition. As elementary-scalar data of the standard continuum formulation they are excluded by Theorem 10.1.

Of the elementary scalar content, the standard continuum formulation can support at most the free part, which is the subject of Part 2. The two formulations are complementary in their methods. Both start from the same spectral data. The standard continuum formulation carries the measure construction, the Gaussian realisation of the free sector. The spectral-sum definition carries the formal correspondence and the definition of the physical quantities. The free sector is common to both constructions, and the quartic coupling and the condensate are carried by the spectral-sum definition.

10.3 Part 2: The free-sector realisation

The only content that admits a standard-formulation realisation is the free part of the colour-singlet spectral data. This includes the spectral masses M_l, the scalar mass parameter m_δ, and the condensate $\langle E\rangle$. This part constructs the standard-formulation realisation as the free scalar field on $\mathbb{R}^4 \times RP^3$ with a constant shift. The internal spectral modes of this field form the mass tower $\{M_l\}$, which is exactly the spectral data of the construction. The zero mode is the four-dimensional scalar field of mass m_δ. The construction uses only the spectral data and standard theorems. This is the concrete $M = RP^3$ instance of the abstract recovery of Appendix C (Theorems C.1 and C.2); the construction is spelled out in detail here because the standard-formulation comparison is made in this section.

The route is as follows. If a family of candidate Schwinger functions satisfies the standard OS axioms, then the Osterwalder–Schrader reconstruction yields the Hilbert space and the Wightman fields directly. For the Gaussian candidates constructed below, Minlos' theorem produces the Euclidean measure explicitly. No lattice regularisation, no cluster expansion, and no analysis of a continuum limit is required. The question is therefore reduced to constructing candidate Schwinger functions from the spectral data that satisfy OS0–OS4. For the free part this is possible. For the elementary-scalar interacting part it is impossible by Theorem 10.1.

10.3.1 Step 1: The candidate Schwinger functions

Let $\varepsilon_l = l(l+2)/L^2$ with multiplicities $d_l = (l+1)^2$ for even l be the scalar spectrum of RP^3, and set $M_l^2 = \varepsilon_l + m_\delta^2$, where $m_\delta^2 = \frac{16}{3}(R_c - R) > 0$ is the squared fluctuation mass of Theorem 8.32. Let $K_l(y, y')$ denote the spectral projector kernel of the l-th eigenspace of the scalar Laplacian on RP^3. With $\{Y_{lm}\}_{m=1}^{d_l}$ an orthonormal eigenbasis,

$$K_l(y, y') = \sum_{m=1}^{d_l} Y_{lm}(y) Y_{lm}(y').$$

Let G_M denote the free massive propagator of mass M in four dimensions. Define the two-point function of the colour-singlet sector by the full spectral kernel

$$S_2\big((x, y), (x', y')\big) = \sum_{l \text{ even}} K_l(y, y')\, G_{M_l}(x - x') + \langle E\rangle^2,$$

where $x, x' \in \mathbb{R}^4$ and $y, y' \in RP^3$, and the constant $\langle E\rangle \neq 0$ is the condensate of Theorem 8.32. The field is $\phi = \psi + \langle E\rangle$, where ψ is the free scalar field on $\mathbb{R}^4 \times RP^3$ with two-point function

$$S_{2,\psi}\big((x, y), (x', y')\big) = \sum_{l \text{ even}} K_l(y, y')\, G_{M_l}(x - x').$$

The higher Schwinger functions are defined by Wick's theorem applied to ψ, together with the constant shift $\langle E\rangle$.

The multiplicity-only sum $\sum_l d_l\, G_{M_l}(x-x')$ is not a distribution. Its momentum-space general term $d_l/(p^2+M_l^2)$ tends to the positive constant L^2 as $l \to \infty$, so the series diverges linearly, as stated in Lemma 8.19. Minlos' theorem cannot be applied to it as it stands. The kernel itself is well defined. By the completeness relation $\sum_l K_l(y,y') = \delta(y-y')$, the eigenfunction oscillation regularises the sum for smooth internal test functions. The kernel is the Green function of $L_V + \Delta_{RP^3} + m_\delta^2$ on $\mathbb{R}^4 \times RP^3$. It is the two-point function of the massive free scalar field on the product manifold. The field therefore lives on $\mathbb{R}^4 \times RP^3$. The factor $\mathbb{R}^4$ carries the Euclidean and reflection structures of spacetime, and the compact factor RP^3 is the internal space whose spectrum generates the mass tower $\{M_l\}$.

10.3.2 Step 2: The standard OS axioms

Lemma 10.4 (Reflection positivity of the shifted free field on the product space)**.** *Let ψ be the free scalar field on $\mathbb{R}^4 \times RP^3$ with two-point function*

$$S_{2,\psi}\big((x,y),(x',y')\big) = \sum_{l \text{ even}} K_l(y,y')\, G_{M_l}(x-x'),$$

where K_l is the spectral projector kernel of the l-th eigenspace of Δ_{RP^3}, a positive operator, and each G_{M_l} is the free massive propagator on $\mathbb{R}^4$ with $M_l \geq m_\delta > 0$. Then the Gaussian measure of ψ satisfies reflection positivity with respect to time reflection in the $\mathbb{R}^4$ factor, and the constant shift $\phi = \psi + c$ with $c \in \mathbb{R}$ preserves reflection positivity.

Proof. Decompose a test function $f \in \mathcal{S}(\mathbb{R}^4 \times RP^3)$ into internal spectral modes, $f(x,y) = \sum_l f_l(x)$, where f_l is the projection onto the l-th eigenspace. Then

$$\begin{aligned}\iint \overline{f(x,y)}\, S_{2,\psi}\big((x,y),(\theta x',y')\big)\, f(x',y')\, dx\, dy\, dx'\, dy' \\ = \sum_l \iint \overline{f_l(x)}\, G_{M_l}(x-\theta x')\, f_l(x')\, dx\, dx' \;\geq\; 0. \quad (10.1)\end{aligned}$$

The inequality is reflection positivity of the massive free field of mass $M_l \geq m_\delta > 0$ on $\mathbb{R}^4$, applied to f_l supported in $\{\tau > 0\}$. Each term is non-negative and the sum converges. For the shift, let F be a polynomial in the fields $\phi(f_i)$ with $\operatorname{supp} f_i \subset \{\tau > 0\}$. Writing $\psi(f) = \phi(f) - c\int f$, the polynomial $G(\psi) = F(\psi + c)$ is again a polynomial in the $\tau > 0$ fields $\psi(f_i)$, and

$$\mathbb{E}[F(\theta F)] = \mathbb{E}_0[G(\theta G)] \geq 0$$

by reflection positivity of the zero-mean Gaussian, since the constant c is reflection-invariant. The shift therefore preserves reflection positivity. □

The candidate Schwinger functions of Step 1 satisfy the standard OS axioms as follows.

(OS0) Regularity. The two-point function is the resolvent kernel of $L_V + \Delta_{RP^3} + m_\delta^2$ on $\mathbb{R}^4 \times RP^3$. It is a positive-definite tempered distribution, continuous as a bilinear form on $\mathcal{S}(\mathbb{R}^4 \times RP^3) \times \mathcal{S}(\mathbb{R}^4 \times RP^3)$. Every higher moment is a finite polynomial in it, hence tempered.

(OS1) Euclidean invariance. Each massive propagator is $O(4)$-invariant in the $\mathbb{R}^4$ factor. Each K_l is invariant under the isometries of RP^3. The shift is constant.

(OS2) Reflection positivity. This is Lemma 10.4.

(OS3) Symmetry. Wick's theorem is symmetric.

(OS4) Clustering. The truncated two-point function decays exponentially in $|x - x'|$ with rate at least m_δ, the lightest spectral mass, attained by the zero mode $l = 0$. The truncated functions of the free field vanish.

10.3.3 Step 3: The Euclidean measure

Theorem 10.5 (Standard-formulation realisation of the colour-singlet data)**.** *The candidate Schwinger functions of Step 1 are the moments of a unique Euclidean measure μ_∞ on $\mathcal{S}'(\mathbb{R}^4 \times RP^3)$. The measure is Gaussian with mean $\langle E \rangle$ and covariance $S_{2,\psi}$. It satisfies the standard OS axioms OS0–OS4 with scalar mass $m_\delta > 0$.*

Proof. The covariance $S_{2,\psi}$ is the Green function of $L_V + \Delta_{RP^3} + m_\delta^2$. It is the kernel of the resolvent of a positive elliptic operator at the negative spectral value $-m_\delta^2$, which maps $\mathcal{S}(\mathbb{R}^4 \times RP^3)$ to itself and is positive definite as a bilinear form. Hence $S_{2,\psi}$ is a positive-definite tempered distribution, and by Minlos' theorem [29] there exists a unique Gaussian measure on $\mathcal{S}'(\mathbb{R}^4 \times RP^3)$ with this covariance and mean $\langle E \rangle$. Its moments are the candidate Schwinger functions. The standard OS axioms hold by Step 2. Clustering follows from the scalar mass $m_\delta > 0$. The bottom of the spectrum of $L_V + \Delta_{RP^3} + m_\delta^2$ is m_δ^2, attained by the zero mode at $p = 0$. $\square$

10.3.4 Step 4: OS reconstruction

By the standard Osterwalder–Schrader reconstruction theorem, the measure μ_∞ of Theorem 10.5 defines a Wightman field theory of the colour-singlet sector satisfying the standard Wightman axioms W1–W6, with a unique vacuum, a positive scalar mass $m_\delta > 0$ above the vacuum, and vacuum expectation value $\langle E \rangle \neq 0$ for the scalar field. The reconstructed fields are the internal spectral modes. The projection ψ_l of ψ onto the l-th eigenspace of Δ_{RP^3} is a Wightman field on $\mathbb{R}^4$ of mass M_l. The zero mode $l = 0$ is the four-dimensional field of mass m_δ. The truncated functions of the theory vanish, and its S-matrix is trivial. The field $\phi = \psi + \langle E \rangle$ is the free field shifted by the constant $\langle E \rangle$. The constant lies outside the one-particle space of the massive free field, so the reconstruction adjoins a constant mode, and the shifted theory is not unitarily equivalent to the unshifted free field; the difference between the two representations is the constant background $\langle E \rangle$. In the spectral-sum definition, by contrast, the spectral data include the quartic coupling $\lambda > 0$, and the vacuum of the colour-singlet sector is not unitarily equivalent to the Fock vacuum by Theorem 8.40. This difference is the content of Theorem 10.1. The standard continuum formulation carries the colour-singlet data in their free form; the quartic coupling is carried by the spectral-sum definition.

10.3.5 The recovery theorem

Theorem 10.6 (Recovery of a standard quantum field theory)**.** *The colour-singlet spectral data $\{\varepsilon_l, d_l, m_\delta\}$ determine a unique standard-formulation quantum field theory. It is the free scalar field of mass m_δ on $\mathbb{R}^4 \times RP^3$ with product metric and RP^3 of radius L. Its two-point function is the full spectral kernel of Step 1. The correspondence preserves the scalar* *mass m_δ, the two-point function, the vacuum, and the clustering rate. The reconstructed Wightman theory is the family $\{\psi_l\}$ of free fields on $\mathbb{R}^4$ with masses $M_l = \sqrt{\varepsilon_l + m_\delta^2}$. The zero mode is the four-dimensional colour-singlet field of mass m_δ. The correspondence is an isomorphism of spectral data. The Kaluza–Klein decomposition of the free field on the product space is exactly the spectrum used in the construction.*

Proof. By Theorem 10.5 the spectral data determine the Gaussian measure μ_∞ on $\mathcal{S}'(\mathbb{R}^4 \times RP^3)$ with covariance $S_{2,\psi}$, the two-point function of the massive free field of mass m_δ on the product manifold. The scalar mass is the bottom of the spectrum, m_δ, the zero mode at $p = 0$. Clustering at rate m_δ follows from OS4. The Wightman reconstruction of Step 4 gives the family $\{\psi_l\}$ with masses $\{M_l\}$. The spectral data are exactly the Kaluza–Klein data of this field by Lemma 8.19. □

10.3.6 What is not translated

Wilson loops and the non-singlet sector are not translated. Their standard-formulation construction belongs to the standard formulation's own problem, as stated in Proposition 10.2. The construction of the colour-singlet sector is defined within the spectral-sum definition. Of the elementary scalar content, the standard continuum formulation can support at most the free content constructed here; the quartic coupling and the condensate are carried by the spectral-sum definition. The complementary correspondence is the product-space formulation of Appendix B. It realises the spectral data as the internal-mode content of an explicit action on $\mathbb{R}^4 \times RP^3$. Its one-loop content in the internal-trace order reproduces the spectral sums, without entering the standard four-dimensional continuum formulation. Its Kaluza–Klein decoupling reproduces the four-dimensional effective structure.

10.4 Relation to the standard formulation

This subsection states how the formulation of Sections 7–9 stands in relation to the standard formulation of quantum field theory. On the free sector the two formulations coincide through the recovery theorem. On the interacting sector they are complementary, and the precise locus of the complementarity is the triviality theorem.

10.4.1 Where the two formulations coincide

On the free sector the formulation and the standard formulation coincide. The recovery theorem of Appendix C constructs the free sector of the formulation by the standard Osterwalder–Schrader reconstruction applied to the spectral kernel. The result is the free scalar field of mass m_δ on $\mathbb{R}^4 \times M$, a standard quantum field theory in the sense of Wightman and Osterwalder–Schrader. There is no difference of content here. The same spectral datum produces the same free field, the same mass tower, the same two-point function, and the same clustering.

10.4.2 The triviality theorem and the interacting sector

The interacting content of the formulation, namely the quartic coupling $\lambda > 0$ and the condensate $\langle E \rangle \neq 0$, has no realisation as an elementary four-dimensional scalar field theory in the standard continuum formulation. This is the content of the triviality theorem. The continuum limits of the four-dimensional lattice ϕ^4 models are Gaussian, so the elementary interacting scalar sector is trivial in the standard continuum formulation. The statement is a theorem of the standard formulation. It fixes the boundary at which the standard formulation stops for elementary scalar interactions. Within the spectral-sum definition the corresponding content satisfies its own axioms in the sense of Definition 8.1, including the clustering statement of Lemma 8.38 (Section 8), which needs no measure.

Remark 10.7 (The measure-theoretic locus of triviality). The triviality theorem is a statement about the standard formulation, whose physical content is carried by a measure. What the theorem excludes is the existence of a non-trivial measure with an interacting continuum limit. The only measures that survive the continuum limit are Gaussian. The formulation's interacting content does not proceed through such a measure. The theorem therefore does not touch the formulation's construction, but it does fix the boundary of the standard formulation against which the formulation's interacting content is compared.

10.4.3 Two constructions from the same spectral data

The two formulations share the free-sector spectral content. On the interacting sector the standard formulation has no measure-theoretic realisation (the triviality theorem), and this paper defines a spectral-sum counterpart instead.

Both formulations begin from the same spectral datum. The standard formulation maps the spectral data to the physical structure by a measure: the covariance defines a Gaussian measure, and the physical structure is reconstructed from that measure. The formulation maps the same spectral data to the physical structure by a formal correspondence, the mapping dictionary and the constant-mode shift of Proposition 9.2, which invoke only the closed forms of the spectral data and the standard algebra of the Fock space, without a measure. The two constructions coincide on the free sector, where the measure-theoretic construction exists and the formal correspondence reduces to it; they diverge exactly where the measure-theoretic construction stops, at the elementary-scalar interacting content excluded by triviality. The spectral datum therefore fixes a common free-sector realisation and an internal interacting correspondence: the free sector represented by the Gaussian measure and its reconstruction, the interacting sector by the formal correspondence, sharing the same free content and differing only in the construction.

10.4.4 The classification of the remaining items

The boundaries of the formulation fall into three classes.

(a) *Excluded by theorem.* Triviality rules out the standard continuum realisation of the elementary interacting scalar sector. This is a theorem of the standard formulation; it does not touch the interacting content, which is defined by spectral sums.

(b) *Closed by definition.* The spectral-sum representation (Theorem 9.3) and the mapping dictionary (Proposition 9.2) are constructions introduced here and closed by their own definitions and theorems.

(c) *Outside the definition.* The operator-level ultraviolet completion of the four-dimensional fields, beyond the spectral-sum definition of the physical quantities, is not needed here, since the physical quantities are spectral sums from the outset.

The relation of the formulation to the standard formulation is as follows. On the free sector the two coincide. On the interacting sector the formulation supplies a formal correspondence where the standard formulation has no measure-theoretic construction for elementary scalar interactions, because triviality cuts the latter at the free sector. The interacting content of the formulation is defined within the spectral-sum definition and is non-perturbative in the spectral-sum sense. The formulation shares the free-sector spectral content with the standard formulation; on the interacting sector it defines a spectral-sum counterpart rather than a standard measure-theoretic realisation.

11 Conclusion

This paper has analysed a spectral formulation of gauge structure and fermion content. The central datum is a discrete spectrum with multiplicities. The mass tower, the quartic coupling, and the condensate are obtained from convergent spectral sums and closed expressions associated with this datum.

The structural construction begins with the disorder-dominated coarse-graining principle. Its chiral spinor unit leads to the real projective three-space obtained from the three-sphere by antipodal identification. The associated Clifford structure gives a four-dimensional carrier. The scale field resolves this carrier into three directions tangent to its level sets and one flow direction. The internal isometry algebra decomposes into two SU(2) factors. Condensation in the right sector leaves the left SU(2) factor and a compact U(1) factor that provides the electroweak structure. The chiral carrier organises the fermion representations used in the formulation. A distinct colour carrier is subjected to the same minimality principle, which selects the SU(3) colour algebra.

The spectral datum, the definition of physical quantities, the state space, the mapping to field theoretic structures, and the criterion for a datum to define a quantum field theory are stated explicitly. Encoding relations specify how the spectral data give rise to fields, observables, and correlation functions. Standard quantum kinematics is incorporated within this setting. In the free sector, the Gaussian structure leads through Osterwalder–Schrader reconstruction to the usual free field construction. In the interacting sector, the mapping dictionary and the constant mode shift relate the spectral quantities to the corresponding field theoretic structures.

The triviality theorem provides the comparison with the standard continuum description. The continuum limit of the elementary four-dimensional scalar sector in the standard construction is Gaussian. The spectral sum formulation retains the quartic coupling and the condensate as finite quantities associated with the compact internal spectrum. The two descriptions share the free reconstruction while assigning different roles to the interacting scalar sector.

Taken together, these results give a spectral organisation in which compact internal geometry supports gauge symmetry, matter structure, and scalar mass formation. The distinction between theorem, definition, convention, and assumption is maintained throughout the construction. The structural results obtained here also provide the basis for a numerical evaluation of the spectral quantities.

A Spectral analysis: long-root mode, Dirac spectrum, and the emergence window

This appendix collects the spectral analysis used in the main text. It records the quantum numbers of the long-root mode, the condensation threshold, the spin-2 content, and the emergence window.

A.1 Spin(4) harmonics on S^3 and RP^3

Let S^3 carry the round metric of radius L, and let $RP^3 = S^3/\mathbb{Z}_2$ be the quotient by the antipodal map. The scalar Laplacian spectrum of S^3 is

$$\varepsilon_l = \frac{l(l+2)}{L^2}, \qquad \text{multiplicity } (l+1)^2, \qquad l = 0, 1, 2, \ldots$$

This is a standard result. The eigenfunctions of degree l transform in the representation $(l/2, l/2)$ of $\mathrm{Spin}(4) \cong SU(2)_L \times SU(2)_R$. The quadratic Casimir of the representation (j_L, j_R), in the normalisation

$$C_2(j_L, j_R) = 2[j_L(j_L+1) + j_R(j_R+1)],$$

satisfies $C_2(l/2, l/2) = l(l+2)$. Hence the Laplacian eigenvalue of a Spin(4) harmonic is C_2/L^2 in this normalisation. This convention is used throughout the paper.

On RP^3 the antipodal map acts on the harmonic of degree l as $(-1)^l$. The even sector therefore survives. This gives the scalar spectrum used in Lemma 8.19,

$$\varepsilon_l = \frac{l(l+2)}{L^2}, \qquad d_l = (l+1)^2, \qquad l = 0, 2, 4, \ldots$$

A.2 The long-root mode

In the decomposition $\mathfrak{so}(4) = \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ and the breaking $\mathfrak{su}(2)_R \to \mathfrak{u}(1)_R$, the unbroken torus assigns to a state of $SU(2)_R$ weight m_R the charge $q = 2m_R$. The long-root mode, the $J = 2$ squash deformation of the R-sector connection, carries charge $q = 2$, hence $j_R = 1$.

The modes of the R-sector connection are $\mathfrak{su}(2)_R$-valued 1-forms on RP^3. The harmonic decomposition of 1-forms on S^3 is

$$\Omega^1(S^3) = \bigoplus_{l \geq 1} \left[\left(\frac{l}{2} + \frac{1}{2}, \frac{l}{2} - \frac{1}{2} \right) \oplus \left(\frac{l}{2} - \frac{1}{2}, \frac{l}{2} + \frac{1}{2} \right) \right].$$

On the quotient RP^3 the antipodal map acts on a 1-form of degree l with an additional sign from the differential. The retained sector therefore has odd l. The $l = 1$ sector is the Killing zero-mode sector, which carries the gauge structure itself.

For the non-zero modes, the first one with $j_R = 1$ is the $l = 3$ harmonic $(j_L, j_R) = (2, 1)$, since $j_R = l/2 - 1/2 = 1$ and $j_L = l/2 + 1/2 = 2$. Its Casimir is

$$C_2(2, 1) = 2[2 \cdot 3 + 1 \cdot 2] = 16.$$

By the normalisation of Section A, the internal eigenvalue is

$$\varepsilon_{\mathrm{long}} = \frac{C_2(2, 1)}{L^2} = \frac{16}{L^2}.$$

This is the content of Lemma 8.24.

A.3 The Dirac spectrum of RP^3

The Dirac operator on the round S^3 of radius L has spectrum

$$\pm\frac{j + 1}{L}, \qquad j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$$

with the standard multiplicities (for the round spheres see [52]; for lens spaces see [53]). The manifold $RP^3 = S^3/\mathbb{Z}_2$ admits two spin structures. On each, the Dirac spectrum is the restriction of the S^3 spectrum to the sector of definite antipodal parity. For the spin structure selected by the chiral structure of clause (A1), the retained spectrum is

$$\pm\frac{j + 1}{L}, \qquad j = \frac{1}{2}, \frac{5}{2}, \frac{9}{2}, \dots$$

Equivalently, in the extrusion index n of the generation ladder, $\varepsilon_n = (n + 3/2)/L$ with $n = 0, 2, 4, \dots$

A.4 The emergence window and the retained spinor modes

The heat semigroup C_t of Section 8.4 exponentially suppresses modes whose momentum exceeds $1/\sigma$ (weight $e^{-\sigma^2|k|^2/2}$); it does not delete them. The sharp projection Π_σ of Definition 8.4 only designates the slow-mode sector and is an auxiliary device, not the actual dynamics. The internal spectrum is resolved by a different scale. The internal geometry is a statistical object defined by the ensemble of level sets of the Gaussian scale field. Its coherence scale is the internal radius L, defined as the Gaussian integral width of the scale field. The coarse graining at the breakdown scale retains internal modes whose wavelength exceeds this coherence scale. Shorter-wavelength modes oscillate within the coherence length of the statistics and are averaged out.

The retained window is fixed by the window capacity $(kL)^2$ of the scale flow, where k is the flow momentum scale and L is the internal radius. The retained spinor modes are those with dimensionless Dirac eigenvalue $n + 3/2$ below $(kL)^2$. The window capacity is fixed self-consistently by the spectral sums, and its value determines the generation count. The internal radius is the Gaussian integral width of the scale field, $L = \sqrt{2\pi}\, b$ (Section 4.7); with $k = 1/\sigma$ at the emergence scale this gives the Gaussian benchmark $c = (kL)^2 = 2\pi$. The actual window capacity is the self-consistent value at which the spectral sums close, fixed by the closure condition rather than by hand, and it need not

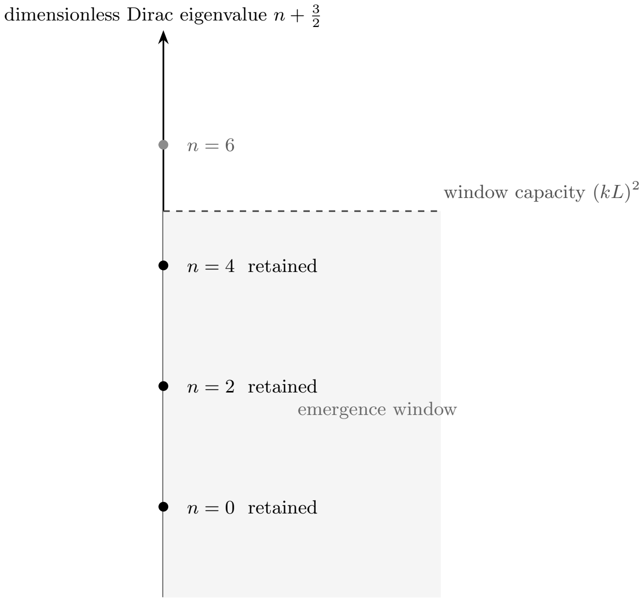

Figure 3. The emergence window. Illustrative mode-retention diagram for the RP^3 Dirac spectrum $n+\frac{3}{2}$ $(n=0,2,4,\dots)$: the retained sector is the set of modes below the window capacity $(kL)^2$. The window capacity is fixed self-consistently by the spectral sums, and its value determines how many modes are retained.

equal the benchmark 2π. The criterion $n+3/2<c$ retains $n=0,2,4$ and excludes $n=6$ for every capacity in the platform interval $(\frac{11}{2},\frac{15}{2})$, which contains the benchmark 2π (Propositions 8.14 and 8.15). The retained modes, ordered by their extrusion order along the scale flow, form the generation ladder, shown in Figure 3. The identification of this ladder with the Standard Model generations is recorded in Proposition 8.14.

B The product-space action: one-loop matching and Kaluza–Klein decoupling

Section 10 records the boundary of the standard four-dimensional continuum formulation for the elementary interacting scalar sector. This appendix constructs a complementary correspondence at the level of an action, along the lines of the classical Kaluza–Klein programme [54]. The spectral data of the present construction are organised as the internal-mode content of an explicit action S_{prod} on the product space $\mathbb{R}^4\times RP^3$. The one-loop effective potential of S_{prod}, computed in the internal-trace order, matches the spectral sums term by term. The Kaluza–Klein decoupling of the massive internal modes gives a four-dimensional effective theory.

The purpose of this appendix is accordingly limited. The product-space action is not presented as a renormalisable quantum field theory in seven dimensions; it is an auxiliary action whose internal-trace one-loop content reproduces the spectral sums. It is sufficient for this role even though it is power-counting non-renormalisable, because the physical quantities of the formulation are the spectral sums themselves, defined independently of any seven-dimensional loop expansion.

Two limitations are stated at the outset. First, S_{prod} is a seven-dimensional theory. The triviality theorem for four-dimensional scalar fields does not directly govern it, and no four-dimensional continuum limit is taken here. The internal radius L is the physical endpoint scale, not a removable cutoff. Second, this appendix does not construct the full renormalised theory of S_{prod} in seven-dimensional loop order. The matching theorem is proved in the internal-trace order, where every matched coefficient is a convergent spectral sum. The full loop-order renormalisation lies outside the spectral-sum definition.

A further boundary concerns the Kaluza–Klein reduction: the zero-mode truncation that yields the four-dimensional effective theory is not proved to be a consistent truncation, and the decoupling of the massive internal modes is established here only as a matching of the internal-trace one-loop content. The passage from the internal isometry algebra to the four-dimensional gauge symmetry is the zero-mode reduction recorded in Definition 4.14 of Section 4.3.

B.1 The product-space action

Throughout, $M = \mathbb{R}^4 \times RP^3$ carries the product metric, with RP^3 the round projective space of radius L and volume $\mathrm{Vol}(RP^3) = \pi^2 L^3$. The internal spectra are those of Appendix A. The gauge group is the emergent group $G_{\text{SM}} = SU(3) \times SU(2)_L \times U(1)_Y$ of Proposition 4.21.

Definition B.1 (The product-space action)**.** *The product-space action $S_{\text{prod}} = S_g + S_f + S_s$ on $M = \mathbb{R}^4 \times RP^3$ is defined by three sectors.*

1. Gauge sector. *For the G_{SM} connection one-form A on M,*

$$S_g[A] = \frac{1}{4g_7^2} \int_M d^7 z \, \sqrt{g} \, g^{MN} g^{PQ} \, \mathrm{Tr}(F_{MP} F_{NQ}),$$

with the seven-dimensional coupling g_7^2 of mass dimension -3.

2. Fermion sector. *For the spinor field ψ on M carrying the internal parity of clause (A1),*

$$S_f[\psi] = \int_M d^7 z \, \sqrt{g} \, \bar{\psi} \, (\gamma^\mu(\partial_\mu + A_\mu) + D_{\text{int}}) \, \psi,$$

where γ^μ are the $\mathbb{R}^4$ generators of Theorem 4.4 and D_{int} is the RP^3 Dirac operator of Appendix A.

3. Scalar sector. *For the colour-singlet scalar ϕ on M with zero-mode amplitude*

$$E(x) = \int_{RP^3} \phi(x, y) \, \mathrm{dvol}(y),$$

the action is

$$S_s[\phi] = \int_M d^7 z \, \sqrt{g} \left(\frac{1}{2} g^{MN} \partial_M \phi \, \partial_N \phi + \frac{1}{2}(m_s^2 + gE(x)^2)\phi^2 \right).$$

The scalar sector is the recovery-theorem tower of Theorem 10.6 together with the coupling $(g/2)E^2\phi^2$. This is the action-level form of the quadratic coupling entering the spectrum of Theorem 8.29, $\varepsilon_l + \xi R \mapsto \varepsilon_l + \xi R + gE^2$.

The seven-dimensional gauge sector has $[g_7^2] = L^3$, so the theory is power-counting non-renormalisable in seven dimensions. No seven-dimensional loop expansion beyond the internal-trace order is used here. The four-dimensional effective sector is obtained by the exact Kaluza–Klein projection of Section B.3, not by a continuum limit.

The internal radius L is the physical endpoint scale. No limit $L \to \infty$ and no lattice spacing are involved. The coupling g is fixed by the spectral-shift convention $\varepsilon_l \to \varepsilon_l + gE^2$ of Theorem 8.29. The Higgs sector is not part of this paper. The scalar content carried by $S_{\rm prod}$ is the colour-singlet sector, the glueball field E of Section 8. The electroweak Higgs is outside the present construction.

B.2 The one-loop matching theorem

Define the internal trace family

$$\zeta(s; m^2) = \sum_{l \text{ even}} d_l(\varepsilon_l + m^2)^{-s},$$

convergent for $\Re s > 3/2$, by Lemma 8.19. The internal-trace one-loop potential of the scalar sector around the background E_0 is the spectral sum

$$F(E_0) = \frac{1}{2} \sum_{l \text{ even}} d_l\, h(\varepsilon_l + m_s^2 + gE_0^2),$$

where h is the one-loop function. In the product-space computation the one-loop function is the logarithm, $h(\lambda) = -\ln \lambda$, with the sign convention used in Section 8.6.

Theorem B.2 (One-loop matching)**.** *The one-loop effective potential of the scalar sector of $S_{\rm prod}$, computed in the internal-trace order, matches the effective potential of Theorem 8.34 term by term, with the bare scalar mass m_s^2 in place of the conformal shift ξR. Its E_0^{2k} coefficient for $k \geq 2$ is the convergent spectral sum*

$$c_k = \frac{(-1)^k g^k}{2k} \zeta(k; m_s^2).$$

This matches the coefficient $c_k = (-1)^k g^k \zeta(k; \xi R)/k$ of Theorem 8.34 up to the stated sign and normalisation conventions, and up to the replacement of the conformal shift ξR by the bare scalar mass m_s^2: the product-space potential carries the standard factor $1/2$ of the one-loop determinant, while the spectral-sum potential of Theorems 8.29 and 8.34 carries the sum without that factor. Its E_0^2 coefficient is $-(g/2)\zeta(1; m_s^2)$, the $p = 1$ divergent diagonal of Lemma 8.19. Its renormalised value is the flow-defined long-mode mass $m_{\rm long}^2 = 2g \sum_l d_l f'(\varepsilon_l + \xi R)$ of Theorem 8.29.

Proof. Expand the one-loop function with $h = -\ln$. This gives

$$h(\varepsilon_l + m_s^2 + gE_0^2) = h(\varepsilon_l + m_s^2) + \sum_{k \geq 1} \frac{(-1)^k g^k}{k} \frac{E_0^{2k}}{(\varepsilon_l + m_s^2)^k},$$

a convergent expansion for $gE_0^2 < m_s^2$. Summing over l with the multiplicities d_l gives

$$F(E_0) = F(0) + \sum_{k\geq 1} \frac{(-1)^k g^k}{2k} \zeta(k; m_s^2)\, E_0^{2k}.$$

For $k \geq 2$ each sum $\zeta(k; m_s^2)$ converges absolutely by Lemma 8.19. The resulting coefficients agree with c_k up to the normalisation convention stated in the theorem. The $k = 2$ coefficient $c_2 = g^2\zeta(2; m_s^2)/4 > 0$ is a positive convergent quartic spectral sum; with the conformal shift ξR in place of the bare scalar mass m_s^2, and with the spectral-sum normalisation of Theorem 8.29, it reproduces the quartic coupling $\lambda = \kappa \sum_l d_l(\varepsilon_l + \xi R)^{-2}$ of Proposition 8.31 with $\kappa = 2g^2$. The $k = 1$ coefficient is the divergent diagonal. Its renormalised value is the long-mode mass of Theorem 8.29. □

The one-loop content of the gauge and fermion sectors is structurally the same. The internal-trace potential of the gauge sector is a spectral sum over the one-form and ghost spectra of Appendix A. The zero modes of Proposition B.4 are excluded from the fluctuation trace. The fermion sector contributes the spectral sum over the Dirac spectrum, which is bounded away from zero. The matching statement of Theorem B.2 concerns the colour-singlet scalar sector. The gauge and fermion sectors match at the level of their spectra and zero modes.

Remark B.3 (The full loop order). The full seven-dimensional one-loop computation requires completing the $\mathbb{R}^4$ momentum integration. This corresponds to the completed zeta function

$$\zeta_{\text{full}}(s; E_0) = \sum_{l \text{ even}} d_l \int \frac{d^4p}{(2\pi)^4} \left(p^2 + \varepsilon_l + m_s^2 + gE_0^2\right)^{-s},$$

with poles at $s = 2, 1, 0$. The E_0-dependence of the full loop order therefore requires the internal zeta family continued to the non-positive integers, together with the corresponding seven-dimensional counterterms. This continuation is standard-formulation renormalisation data. It is not part of the spectral-sum definition. This paper does not perform the full seven-dimensional loop-order renormalisation of S_{prod}. The matching theorem is exact within the internal-trace order.

B.3 Kaluza–Klein decoupling

Proposition B.4 (Zero-mode content)**.** *The zero-mode content of the product space $M = \mathbb{R}^4 \times RP^3$ matches the four-dimensional content of the construction.*

1. Scalars. *The unique zero mode is the constant mode, $l = 0$, $d_0 = 1$. It carries the colour-singlet field E.*
2. One-forms. *The first de Rham cohomology vanishes, $H^1_{dR}(RP^3; \mathbb{R}) = 0$. There are no harmonic one-forms. The gauge zero modes are the symmetry directions. The electroweak sector has four isometric Killing modes, and the colour sector has eight zero modes of $\mathfrak{su}(3)$. This gives twelve gauge bosons, the dimension of G_{SM}.*

3. Spinors. *The Dirac spectrum of Appendix A is bounded away from zero. There are no Dirac zero modes on RP^3. The four-dimensional chiral content is the emergence-window retention of Proposition 8.14.*

Proof. For scalars, the explicit spectrum of Appendix A gives $\varepsilon_0 = 0$ with multiplicity one. For one-forms, the vanishing of the first cohomology is Theorem 8.37. The gauge zero modes are modes of vanishing internal momentum, the Killing modes of the isometry algebra and the constant colour modes. They are not harmonic one-forms. For spinors, the spectrum is given in Appendix A, and the chiral content is the emergence-window convention of Proposition 8.14. □

Proposition B.5 (Zero-mode effective action)**.** *The zero-mode projection of S_{prod} is the four-dimensional effective action*

$$S_4^{(0)} = \int d^4x \left(\frac{1}{4g_4^2} \operatorname{Tr} F_{\mu\nu}F^{\mu\nu} + \bar{\psi}_{\text{win}}\gamma^\mu(\partial_\mu + A_\mu)\psi_{\text{win}} + \frac{1}{2}(\partial_\mu E)^2 + V(E) \right),$$

where ψ_{win} denotes the fermion content selected by the window criterion of Proposition 8.14, and $V(E)$ is the effective potential of Theorems 8.29 and 8.34. The four-dimensional gauge coupling is the metric-normalised Kaluza–Klein coupling

$$g_4^2 = \frac{g_7^2}{\operatorname{Vol}(RP^3)} = \frac{g_7^2}{\pi^2 L^3}.$$

Proof. Restrict the fields of Definition B.1 to the zero modes of Proposition B.4. In the gauge sector, write $A = A_\mu(x)dx^\mu + \omega(y)$, where A_μ is the four-dimensional connection. The zero-mode part of $F_{\mu\nu}F^{\mu\nu}$ integrates over the internal space with weight $\operatorname{Vol}(RP^3)$, giving the coupling relation. The fermion sector projects onto the window modes. The scalar sector projects the $l = 0$ mode onto the kinetic term of E, and the potential is $V(E)$. □

Proposition B.6 (Decoupling of the massive modes)**.** *Let*

$$M_{KK} = \sqrt{\frac{8}{L^2} + m_s^2}$$

be the lowest non-zero scalar mode, corresponding to $l = 2$. Integrating the massive internal modes with $l \geq 2$ in the internal-trace order modifies each four-dimensional coupling by a correction series whose coefficients are the restricted spectral sums

$$\sum_{l \geq 2} d_l(\varepsilon_l + m^2)^{-k}$$

and their gauge and Dirac analogues. Each correction to a coupling of mass dimension d is of order $(\Lambda/M_{KK})^{2k}$ for the low-energy scale Λ. The massive propagators cluster in coordinate space as $e^{-M_l|x-x'|}$ with $M_l \geq M_{KK}$. The decoupling is exact in the internal-trace order. The endpoint quantities are these spectral sums by Theorem B.2, so the four-dimensional effective theory matches the spectral content.

Proof. The restricted sums converge absolutely for $k \geq 2$ by Lemma 8.19. The suppression factors follow from the mass ratios. The zero mode is always lighter than the tower, $m_s^2 < 8/L^2 + m_s^2$. The colour-singlet decoupling is therefore structural, independent of the value of L. The identification of the correction sums with the spectral quantities is Theorem B.2. $\square$

B.4 The three technical conditions

The decoupling of Proposition B.6 rests on three technical conditions.

(a) *Coupling normalisation.* The relation $g_4^2 = g_7^2/\mathrm{Vol}(RP^3)$ is the metric-normalisation formula. The absolute value of the coupling is not fixed by this paper.

(b) *Scale separation.* The decoupling requires all low-energy scales Λ_{low} of the four-dimensional sector to satisfy $\Lambda_{\mathrm{low}} \ll 1/L$. The colour-singlet separation is structural. The quantitative separation for the gauge and fermion sectors is an open condition, tied to the absolute value of L, which this paper does not fix.

(c) *Chiral content.* The fermion content is the emergence-window convention of Proposition 8.14. It is not a zero-mode theorem. No Dirac zero modes exist on RP^3.

B.5 Status of the correspondence

The correspondence constructed in this appendix connects the spectral data to a product-space action on $\mathbb{R}^4 \times RP^3$, and then to a four-dimensional effective theory. The spectral data are the internal-mode content of the product-space action. The internal-trace one-loop potential matches the spectral sums. The zero-mode content matches the four-dimensional content. The decoupling corrections are convergent spectral sums.

This appendix does not claim the full seven-dimensional loop-order renormalisation. It does not claim a standard four-dimensional continuum realisation of the non-trivial scalar sector. It does not fix the absolute values of the couplings or of L. The product-space theory is a seven-dimensional theory whose low-energy sector matches the standard-model structure described in the construction. It complements the recovery theorem of Theorem 10.6, which realises the free sector in the standard formulation. The non-trivial sector is carried by the spectral-sum definition. The product-space action S_{prod} provides an action-level realisation of the spectral data that generates the effective structure at low energies.

C Recovery of the free sector

This appendix constructs the free sector of the formulation from the spectral datum. The result is a standard quantum field theory in the standard measure-theoretic sense. The construction is the standard Osterwalder–Schrader reconstruction applied to the spectral kernel. It is unconditional on the free sector and supplies the Fock-space structure used by the mapping statements of Section 7. The concrete $M = RP^3$ instance, with the explicit candidate Schwinger functions and the Osterwalder–Schrader verification carried out in detail, is given in Section 10 (Part 2). The reconstruction of this appendix applies to the free part of the formulation. The interacting content, namely the quartic coupling and the condensate, is carried by the spectral-sum definition of Definition 8.1, not by an Osterwalder–Schrader measure. The boundary of the standard formulation for that content is the triviality theorem recorded in Section 10.

C.1 The candidate Schwinger functions

Let the spectral datum of Definition 5.1 be given. It consists of a compact three-manifold M with spectrum $\{\varepsilon_l\}$ and multiplicities $\{d_l\}$. The positive scalar mass parameter $m_\delta > 0$ is derived from this datum through Theorem 8.32. The spectral kernel of the datum is the full two-point function on $\mathbb{R}^4 \times M$,

$$G_{\text{full}}((x, y), (x', y')) = \sum_l K_l(y, y')\, G_{M_l}(x - x'),$$

where K_l is the spectral projector of the l-th eigenspace of Δ_M, and G_{M_l} is the Euclidean two-point function of the free scalar field of mass

$$M_l = \sqrt{\varepsilon_l + m_\delta^2}$$

on $\mathbb{R}^4$. The candidate n-point Schwinger functions are the Gaussian Wick moments of G_{full},

$$S_n = \sum_{\text{pairings}} \prod G_{\text{full}}((x_i, y_i), (x_j, y_j)),$$

where the sum runs over all pairings of the n points. For odd n, one sets $S_n = 0$. Each S_n is determined by the spectral datum alone.

Theorem C.1 (Reconstruction of the free sector)**.** *Let the spectral datum of Definition 5.1 be given, and let the candidate Schwinger functions be formed from the spectral kernel. Then the following hold.*

(a) Osterwalder–Schrader axioms. *The candidate Schwinger functions satisfy the standard Osterwalder–Schrader axioms of Euclidean covariance, symmetry, reflection positivity, and clustering [21, 25].*

(b) Minlos theorem. *By Minlos' theorem [29], the covariance G_{full} determines a Gaussian measure μ on $\mathcal{S}'(\mathbb{R}^4 \times M)$ whose moments are the candidate Schwinger functions.*

(c) Osterwalder–Schrader reconstruction. *The reconstruction theorem produces from μ a Hilbert space $\mathcal{H}$, a field algebra $\mathcal{A}$ of operator-valued distributions $\phi(f \otimes g)$, a strongly continuous unitary dynamics $U(t)$ with self-adjoint generator H, and a cyclic vacuum vector Ω [21, 22]. The vacuum expectation values of the fields are the candidate Schwinger functions.*

(d) The mass tower. *The Hamiltonian H has spectrum*

$$\operatorname{spec}(H) = \{0\} \cup \{M_l\},$$

with multiplicities $\{d_l\}$, the mass tower of the datum. The Kaluza–Klein decomposition of the free field on $\mathbb{R}^4 \times M$ matches the mode decomposition of the spectrum.

Proof. (a) The spectral kernel is the two-point function of the free scalar field of mass m_δ on the product $\mathbb{R}^4 \times M$, with product metric and M of fixed volume. The candidate Schwinger functions are its Wick moments. A free-field two-point function is reflection-positive and Euclidean-covariant, and its Wick moments satisfy the Osterwalder–Schrader axioms [21, 25]. (b) A reflection-positive, continuous covariance is the covariance of a Gaussian measure on $\mathcal{S}'(\mathbb{R}^4 \times M)$ by Minlos' theorem [29]. The moments of that measure are the candidate Schwinger functions. (c) The Osterwalder–Schrader reconstruction theorem applied to the measure μ produces the physical structure $(\mathcal{H}, \mathcal{A}, U, \Omega)$ [21, 22]. (d) The spectral decomposition of the internal Laplacian,

$$\Delta_M = \sum_l \varepsilon_l K_l,$$

decomposes the free field into independent modes of mass M_l. Each mode contributes a single-particle state of mass M_l with multiplicity d_l. The spectrum of H is therefore the mass tower $\{M_l\}$ with the stated multiplicities. The lowest mode is $M_0 = m_\delta$. □

Theorem C.2 (The free sector is a standard quantum field theory)**.** *The free-sector spectral datum $S = (\{\varepsilon_l\}, \{d_l\}, m_\delta)$ determines a unique standard-formulation quantum field theory. It is the free scalar field of mass m_δ on $\mathbb{R}^4 \times M$, with product metric. Its two-point function is the full spectral kernel. The correspondence preserves the scalar mass m_δ, the two-point function, the vacuum, and the clustering rate. The Kaluza–Klein decomposition of the free field matches the mass tower of the datum.*

Proof. The reconstruction of Theorem C.1 produces the free scalar field of mass m_δ on $\mathbb{R}^4 \times M$. The two-point function is the covariance by construction, so the correspondence preserves the two-point function, the vacuum, and the clustering rate. The scalar mass m_δ is preserved by Theorem C.1(d). The Kaluza–Klein decomposition matches the mode decomposition of the datum by the same item. Uniqueness holds within the Gaussian class, since a quasi-free field is determined by its two-point function. □

This records the free-sector instance of the mapping relation stated in Definition 5.8.

The two-point function of the free sector carries the Källén–Lehmann representation [1, 2] with spectral measure

$$\rho(\mu^2) = \sum_l d_l\, \delta(\mu^2 - M_l^2),$$

the spectral measure of the mass tower. The atomic part of ρ is the single-particle spectrum of the theory by Theorem 7.14. The positivity of ρ is Theorem 7.4. The reconstruction of Theorem C.1 is the geometric realisation of this measure-theoretic content on the product $\mathbb{R}^4 \times M$. It is the carrier of the structural mapping recorded in Section 7.2.

Declarations

Conflict of interest. The author declares no competing interests.

Data and supplementary material availability. No new observational dataset is introduced in this work. Supplementary source material, symbolic checks, companion numerical scripts, and Lean 4 formalisation files are available in the software archived at Zenodo (DOI `https://doi.org/10.5281/zenodo.22067006`) [55]. This article is archived at Zenodo with DOI `https://doi.org/10.5281/zenodo.22067118`.

Scope of the formal checks. The Lean 4 formalisation covers selected finite algebraic identities used inside the construction, including counting relations, representation-content identities, normalisation factors, and closure equalities. These checks verify the corresponding statements as consequences of the stated definitions. They do not certify the physical interpretation of the construction, the uniqueness of that interpretation, or the empirical status of the companion numerical comparisons.

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